

4. .md files (in order)

README.md

```
# TrangorgeOS

# TrangorgeOS

> A modern, high-performance bare-metal operating system built ground-up on a custom **Separated Tri-Partition Architecture**

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## Overview & Philosophy

TrangorgeOS is built with a strict **from-scratch philosophy** (~35,000+ lines of code, actively expanding)

The project operates under an intensive development cycle aimed at producing a stable, bare-metal kernel

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## Visuals & Screenshots

| Kernel Loading & Memory Allocation Testing | Kernel Base Resolution | Kernel in 1080p |
|:---:|:---:|:---:|
| ![Kernel Loading](kernelloading.png) | ![Kernel Base Res](kernel_loader_base_res.png) | ![Kernel 1080p](kernel_1080p.png) |

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## Architectural Model: Separated Tri-Partition Architecture

TrangorgeOS decouples driver functionality and system privilege into 4 distinct operational layers (3 main components):

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+-----+
|                                     |
|             USERSPACE              |
| - Applications & High-Level Libraries |
|                                     |
+-----+
|             USER DRIVER SPACE (UDS) |
| - High-risk / Peripherals (Fault-isolated, strict boundary) |
|                                     |
+-----+
|             DRIVER SPACE (DS)       |
| * Dynamic Drivers   : GPU & complex hardware |
| * Static Drivers    : Init & single-action hardware setup |
| * Library Drivers   : Inter-driver interface providers |
|                                     |
+-----+
|             KERNEL CORE / DRIVERS   |
| - Trusted Core Drivers (Network, USB stack base, Core MM) |
| - Fine-grained Kernel Policy Enforcement & Data Flow Control |
|                                     |
+-----+

...

1. **Kernel Core & Trusted Drivers:** Holds only maximum-trust drivers (e.g., base network, USB stack core)
2. **Driver Space (DS):** Modular driver execution environment with strict error margins:
   - **Dynamic Drivers:** Handle complex, stateful hardware (e.g., GPU control).
   - **Static Drivers:** Perform hardware initialization or non-exporting, single-purpose setup.
   - **Library Drivers:** Expose specialized interfaces (e.g., PCI, HDMI control) for other drivers to control hardware.
3. **User Driver Space (UDS):** High-level peripheral drivers isolated at a safe distance from the core to prevent system crashes.
4. **Userspace & User Library Drivers:** Top-level space housing application runtime and user-level libraries.

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## Current Subsystems & Refactoring Status

> **Notice:** The kernel is currently undergoing a major refactoring phase (modernization and safety audit)

- **Memory Management (MM):** Fully custom memory management subsystem and dynamic allocator written in C
- **Network & USB:** Network driver base operational; USB stack operating under a controlled bare-metal environment
- **Input & Display:** Built-in BIOS PS/2 fallback support; active development on modern USB input. Kernel
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```

- **Multiprocessing & Interrupts:** SMP multi-core initialization active. APIC/IDT interrupt architecture
- **Binary Support:** Native execution support for `ELF` and custom `.bin` binaries with custom IPC inter

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## Custom Versioning System

TrangorgeOS uses an in-house versioning scheme reflecting the precise state of development:

 $\text{\$}\text{\$}\text{\text{Major}}\text{\$} . \text{\text{Changes}}\text{\$} . \text{\text{FixesPerChange}}\text{\$} . \text{\text{Iteration}}\text{\$}\text{\text{State}}\text{\$}\text{\$}$ 

*Example: `v0.182.7.1a`*
- **`0` (Major):** Pre-release major build.
- **`182` (Changes):** Total cumulative feature changes introduced.
- **`7` (Fixes):** Bugfixes applied for the current change.
- **`1` (Iteration):** Single target architecture fully active.
- **`a` (State):** Alpha phase (`a` = initial work  $\rightarrow$  `b` = stabilization  $\rightarrow$  `g` =

### Scale Milestones:
- **Alpha:** Pre-release scale (up to ~60,000 LOC), active architectural building.
- **Beta:** System runs stably on physical hardware.
- **Gamma:** Final stabilization phase prior to Release `v1.0`.

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## Git Branch Model & Workflow

The repository relies on a strict multi-tier branch hierarchy:

| Branch | Purpose |
|:---|:---|
| `init` | Conceptual structure, architecture blueprints, no active codebase. |
| `new` | Experimental features and isolated proof-of-concept tests. |
| `unstable` | Primary active development branch. |
| `stabilizing` | Refactoring, bug-hunting, and code modernizing before release. |
| `stable` | Tested, incremental updates with guaranteed stability. |
| `main` | Production showcase branch. |

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## Build

The project builds as two Cargo workspaces:

- **`kernel_workspace`** – the kernel and its bootable image (`kernel-bin`).
- **`kernel-drivers`** – the exported hardware-driver libraries (`pcie`, `usb`, `audio`, ...), extracted f

Build orchestration is handled with [Just](https://github.com/casey/just) (`justfile`):

```sh
just # build everything (driver libraries -> kernel bootimage)
just drivers # build the driver libraries only
just kernel # build the kernel bootimage
just run # build + run the kernel in QEMU
just check # cargo check (no linking)
just iso # build a bootable demo .iso (see "Demo ISO" below)
just iso-run # boot the newest demo .iso in QEMU
```

> **Note:** the experimental `ctrlfile` build tool was **excluded from the repository** for now – it is r

### Demo ISO

A bootable, publishable demo image is produced from the exact same kernel build:

```sh
just iso # -> dist/TrangorgeOS-<version>-x86_64-demo.iso
just iso-run # boot the newest ISO in QEMU (window + serial on stdout)
just iso-test # headless ISO smoke test (serial log + PASS/FAIL report)
```

The ISO boots via El Torito in *no-emulation* mode: **ISOLINUX** starts, and **MEMDISK** then boots the unmodified `bootimage-kernel-bin.bin` from RAM. This

```

is currently the only supported path: `bootloader 0.9` produces a raw 512-byte-sector BIOS disk image, while the `comgrub` (multiboot2) and `comlimine` (Limine) entry points are still stubs – they do not export the `kernel\_main` ABI the kernel expects (see `TrangorgeOS – TODO.md`, section 2.6).

Extra packages required (on top of the toolchain above):

```
```sh
sudo apt-get install -y xorriso syslinux-common # Debian/Ubuntu
```

`tools/mkiso.sh` also writes the releases artefacts next to the ISO:
`.iso.sha256` and `dist/RELEASE.txt` (version, git commit, kernel hash,
toolchain). The image carries a hybrid MBR, so the very same file can be `dd`-ed
to a USB stick and booted in legacy/CSM mode.

> Current limitation: legacy BIOS only – there is no UEFI hand-off yet.
> Full recipe, publication checklist and limitations: [tools/iso/README.md](tools/iso/README.md).
> Verified on 2026-09-25: the demo ISO boots in QEMU to the in-kernel terminal
> with `SYSTEM STATUS: 19/19 OK`.

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## Project Roadmap

- [x] Current Stage (`stabilizing` / Alpha v0.182.x): Deep refactoring of MM allocator, scheduler mod
- [ ] Milestone 1 (~1.5 Months): Complete memory allocator refactoring, lock down current subsystem r
- [ ] Milestone 2 (~4 Months): Mature active drivers, write dedicated GPU/display drivers, initiate m
- [ ] Milestone 3 (~1 Year): Complete full bare-metal testing on physical hardware, establish stable

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<p align="center">
  <i>TrangorgeOS – Building bare-metal systems from scratch, one commit at a time.</i>
</p>
```

TrangorgeOS — Architecture Documentation.markdown

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# TrangorgeOS – Architecture Documentation

> Document status: Working draft 0.2
> Scope: System architecture, kernel layout, memory and allocator design, driver-space model, reposit
> Sources: The supplied project tree, the `unstable` branch, and the official project website [1] [2]

## 1. Project overview

TrangorgeOS is an operating system built from scratch around a hybrid kernel. The project currently t

The project is intended for environments where control, low resource usage, predictable deployment, and p

The repository is intentionally split into a kernel, driver-space components, standalone drivers, librari

## 2. Current state

According to the project author, the kernel is substantially implemented, the filesystem works but still

The public `unstable` branch contains the main project components and is still actively changing. The rep

| Area | Verified or stated status | Documentation note |
|---|---|---|
| Kernel model | Hybrid kernel | Publicly stated by the repository and website. |
| Target architectures | x86_64, ARM64, RISC-V | The target list is public; implementation maturity must
| Kernel language | Rust, with low-level and supporting components | GitHub currently reports Rust as dom
| Filesystems | FAT32 and EXT4 are publicly listed; TFS is present in the tree | Native filesystem status
| Allocator | Buddy allocator active; slab implementation present but disabled by default | `HEAP_USE_SLA
| Driver model | Kernel space, driver space, user space | Explicitly defined by the official website and

## 3. High-level architecture

```text
+-----+
| Applications, tools, and user-facing libraries |
```

	trangorgelibc, triang-lang, ISO tools, auxiliary utilities	
+-----+		+-----+
	User space	
	peripheral drivers and apps	
+-----+		+-----+
	Driver space	
	isolated driver environment	
+-----+		+-----+
	Kernel space	
	critical mechanisms	
+-----+		+-----+
	Hardware and firmware	
+-----+		+-----+
...		

The official website describes the project as having a three-way driver split [2]. The source code shows

## ## 4. The three driver spaces

### ### 4.1 Kernel space

Kernel-space drivers are the critical components that require maximum performance or privileged access. The kernel space should provide stable primitives rather than contain every device-specific implementation. It should be able to handle device-specific details without exposing them to the user space.

### ### 4.2 Driver space

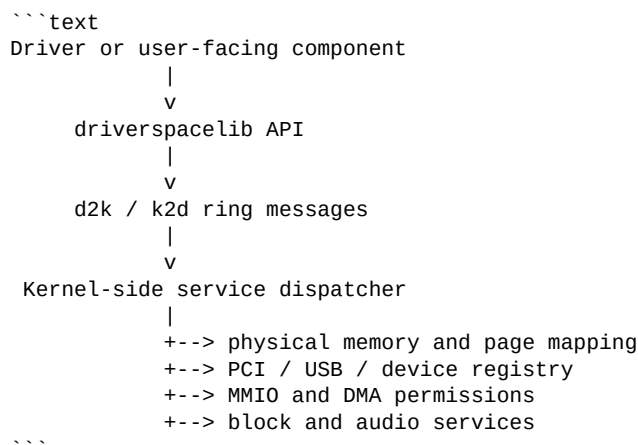
Driver space is the isolated environment for drivers that should not be able to compromise the kernel when they fail. The driver-space service layer currently handles commands related to logging, page allocation and freeing, and device-specific details.

### ### 4.3 User space

User space contains applications, libraries, tools, and peripheral drivers that do not need direct access to hardware or kernel resources.

## ## 5. Driver-space communication

Driver-space communication uses two ring buffers. A message contains an identifier, command, flags, three



The intended rule is that a driver should not duplicate kernel mechanisms, while the kernel should not ne

## ## 6. Kernel subsystem map

Subsystem	Main location	Responsibility	
Entry and diagnostics	`kernel/src/main.rs`, `serial.rs`, `vga_buffer.rs`, `testing.rs`	Boot-time en	
ABI	`kernel/src/abi`	Allocation, memory, filesystem, and public API contracts.	
CPU and interrupts	`cpu/`, `gdt.rs`, `interrupts.rs`	ACPI, LAPIC, GDT, trampolines, CPU state, and	
Memory	`mm/`, `allocator/`	Physical and virtual memory, paging, heap, cache, protection, DMA, and a	
Filesystem	`fs/`	MBR, disk access, ATA PIO, block drivers, FAT32, EXT4, TFS, and initialization.	
Hardware drivers	`drivers/`, `pci.rs`	PCI, USB, HID, mass storage, host controllers, and glue code.	
Graphics and terminal	`gfx/`, `terminal/`, `vga_buffer.rs`	Framebuffer, VGA, fonts, console, termin	
Networking	`nic/`	Ethernet, packets, protocols, device abstractions, and VirtIO networking.	
Installation and updates	`ctrlinstall/`	Repositories, indexes, manifests, dependency resolution, tr	
Driver-space initialization	`driverspaceinit/`	Address-space setup, shared memory, initialization,	

The memory subsystem bootstrap order is explicitly visible in ``mm_init()``: ``arch_memory_init`` → ``paging_`

## ## 7. Memory and allocator architecture

The memory subsystem is divided into physical memory, virtual memory, heap allocation, caching, DMA/cont

```text

Rust and C allocation APIs

|

v

Allocation API and FFI bridge

|

--> physical memory: bitmap, frames, PMM

--> virtual memory: VMM, mapping, pages, paging, TLB

--> heap: buddy allocator, optional slab allocator

--> special memory: contiguous allocation and DMA

--> cache: object cache and per-CPU support

--> protection: guard, isolation, permissions

--> process spaces: address_space and mmap

...

7.1. Initialization

``mm_init()`` validates boot memory-map parameters and initializes the memory stack in a deliberate order.

7.2. Active heap path

The active heap path uses the buddy allocator. ``heap.c`` reserves a 256 MiB buddy region and maps physical

The buddy implementation tracks allocation orders, free lists, mapped pages, used/free statistics, and bu

7.3. Required allocator documentation

The next allocator specification should define ownership, alignment, blocking behavior, interrupt-context

8. Repository map

| Directory | Role |

|---|---|

| ``kernel/`` | Main kernel and its memory, filesystem, CPU, networking, ABI, and hardware layers. |

| ``drivers/`` | Packages for concrete device drivers such as GPU, audio, camera, and Wacom. |

| ``driverspace/`` | Driver-space executable or runtime and its public API. |

| ``driverspacelib/`` | Shared driver-space ABI, runtime, logging, memory, ring, audio, and block helpers. |

| ``trangorgelibc/`` | Client-side library and syscall/ABI-facing facilities. |

| ``comgrub/`` | GRUB boot component. |

| ``comlimine/`` | Limine boot component. |

| ``iso-builder/`` | ISO-image construction tool. |

| ``triang-lang/`` | Lexer, parser, AST, semantic analysis, IR, and C/assembly code generation. |

| ``mp4_to_bmp/`` | Auxiliary media-conversion tool. |

| ``targets/`` | Build-target configuration. |

| Docker files and scripts | Reproducible build and run environment. |

9. Build and boot components

The repository contains both ``comgrub`` and ``comlimine``, indicating two boot-component paths. The project

10. Current risks and open questions

The driver-space mechanism is present, but its long-term ABI and lifecycle policy are not yet fully speci

The public source also requires a careful compile check around ``kernel/src/driverspaceinit/init/init.rs``:

Architecture support should be tracked per target. Listing ARM64 and RISC-V on the website does not by it

11. Summary

TrangorgeOS has a clear architectural direction: a hybrid kernel with a three-way driver split, a layered

The highest-value next steps are to stabilize the driver-space ABI, make the allocator contract explicit,

References

[1]: <https://github.com/CTRL-F-org3/TrangorgeOS/tree/unstable> "TrangorgeOS – unstable branch"

[2]: <https://trangorgeos.website/> "Official TrangorgeOS website"

[3]: <https://github.com/CTRL-F-0rg3/TrangorgeOS/blob/unstable/kernel/src/driverspaceinit/init/init.rs> "TrangorgeOS - init"

[4]: <https://github.com/CTRL-F-0rg3/TrangorgeOS/blob/unstable/kernel/src/driverspaceinit/init/service.rs> "TrangorgeOS - service"

[5]: <https://github.com/CTRL-F-0rg3/TrangorgeOS/blob/unstable/kernel/src/mm/alloc/heap/heap.c> "TrangorgeOS - heap"

[6]: <https://github.com/CTRL-F-0rg3/TrangorgeOS/blob/unstable/kernel/src/mm/alloc/api/alloc.c> "TrangorgeOS - alloc"

[7]: <https://github.com/CTRL-F-0rg3/TrangorgeOS/blob/unstable/kernel/src/mm/core/mm.c> "TrangorgeOS - mm"

TrangorgeOS — TODO.md

TrangorgeOS – TODO

> ****Purpose:**** This file tracks the remaining work required to move TrangorgeOS from the current `unstable` branch to a stable release.
>
> ****Status convention:**** `TODO` means not started or not verified; `WIP` means partially implemented; `BL` means blocked.
>
> ****Priority convention:**** `P0` blocks reliable boot or basic correctness; `P1` is required for the next stable release; `P2` is a nice-to-have.

0. Current snapshot

| Area | Current state | Priority |
|---------------------------------|--|----------|
| Kernel boot and x86_64 baseline | Present, but requires repeatable build and boot verification | P0 |
| Memory subsystem | Layered implementation exists; initialization order is defined in `mm_init()` | P0 |
| Heap allocator | Buddy path is active; slab implementation exists but `HEAP_USE_SLAB` is `0` | P1 |
| Driver space | Separate address space, shared rings, initialization parameters, and services are present | P0 |
| Driver-space ABI | Implemented enough for experiments, not yet documented as stable | P0 |
| Filesystem | FAT32/EXT4 and additional filesystem code are present; correctness and integration tests are pending | P1 |
| USB/PCI | Significant infrastructure exists; device coverage and hardware validation are incomplete | P1 |
| Networking | NIC/VirtIO structures exist; network-driver work is explicitly a project priority | P1 |
| ARM64 and RISC-V | Public target goals; feature parity and build/boot status must be established | P2 |
| Package manager | Kernel-side `ctrlinstall` modules exist; end-to-end package workflow needs completion | P1 |
| Toolchain and ecosystem | `triang-lang`, libraries, ISO tooling, and auxiliary tools exist at different levels | P1 |
| Release process | No published release; stable branch criteria are not yet formalized | P1 |

1. P0 – immediate correctness blockers

1.1 Repair and compile-check driver-space initialization

****Status:**** TODO
****Area:**** `kernel/src/driverspaceinit/init/init.rs`

The visible implementation creates a `Driverspace` structure with an address space, two ring physical address spaces, and a scratch page.

****Tasks:****

- Add or remove scratch-page state consistently in the `Driverspace` structure and initialization path.
- Ensure the scratch page is allocated, zeroed, mapped, and released exactly once.
- Ensure `scratch\_view()` cannot return a pointer after the driver-space instance has been destroyed.
- Ensure every failure after partial allocation releases previously allocated frames and address-space mappings.
- Compile the kernel after the fix with the repository's documented target.

****Acceptance criteria:**** The driver-space initialization module compiles without undefined fields or unreferenced variables.

1.2 Establish a reproducible baseline build

****Status:**** TODO
****Area:**** root Docker scripts, `kernel/`, `targets/`

Document and verify the one canonical build command. The repository contains Docker scripts, target configuration, and build scripts.

****Tasks:****

- Verify the required Rust toolchain and external binaries.
- Run the Docker build from a clean environment.
- Record the produced kernel, image, and ISO paths.
- Run the image in an emulator and capture serial output.
- Add a short "Build and run" section to the root `README.md`.
- Make the build fail on warnings or errors that affect the boot artifact, where practical.

****Acceptance criteria:**** A new contributor can clone the repository, run the documented commands, produce

1.3 Define the driver-space ABI versioning rules

****Status:**** TODO

****Area:**** `kernel/src/driverspaceinit/abi/`, `driverspacelib/src/`

The protocol already contains `DsMsg`, ring structures, magic/version fields, and command identifiers. It

****Tasks:****

- Document the binary layout, field widths, alignment, endianness, and reserved fields of `DsMsg` and ring
- Define the meaning of every command and every argument.
- Define error/status codes and whether negative values are stable ABI values.
- Define how unknown commands, malformed messages, full rings, and stale IDs are handled.
- Define the compatibility policy for `DS\_MAGIC`, `DS\_VERSION`, and command additions.
- Add compile-time size/alignment assertions on both kernel and driver-space sides.
- Add a protocol version document and a small ABI conformance test.

****Acceptance criteria:**** The same ABI document can be used to implement a driver without reading kernel i

1.4 Make driver-space isolation explicit and testable

****Status:**** TODO

****Area:**** `kernel/src/driverspaceinit/`, `kernel/src/mm/protection/`, `kernel/src/mm/process/`

The kernel creates a separate address space and maps shared pages into it. The security and failure-isola

****Tasks:****

- Test that driver space cannot read arbitrary kernel virtual addresses.
- Test that driver-space mappings are limited to explicitly granted RAM and MMIO regions.
- Validate permissions for shared rings, initialization parameters, RAM grants, and MMIO grants.
- Reject overlapping, unaligned, out-of-range, and double-freed grants.
- Add a driver crash or invalid-command test and verify that kernel execution continues.
- Define whether and how a failed driver can be restarted or detached.

****Acceptance criteria:**** Negative tests demonstrate that unauthorized mappings and malformed service requ

2. P1 – kernel and hardware reliability

2.1 Complete the allocator specification and test suite

****Status:**** WIP

****Area:**** `kernel/src/mm/alloc/`

The active heap path uses buddy allocation. Slab code is present but disabled with `HEAP\_USE\_SLAB 0`. The

****Tasks:****

- Document physical-frame, virtual-mapping, heap, buddy, slab, DMA, and contiguous-memory responsibilities
- Add tests for zero-size allocations, alignment, large allocations, `krealloc`, double free, invalid free
- Test buddy split and coalescing across every supported order.
- Test page mapping rollback when a physical-frame allocation fails partway through.
- Verify `heap\_usable\_size()` for active and invalid pointers.
- Decide whether slab should remain disabled, be enabled for small objects, or be redesigned.
- Add allocator statistics and leak checks to a repeatable debug test.
- Define which allocator functions are safe during interrupt context and early boot.

****Acceptance criteria:**** The allocator passes deterministic unit/stress tests and its public contract des

2.2 Finish filesystem correctness and integration tests

****Status:**** WIP

****Area:**** `kernel/src/fs/`

The tree contains MBR, disk, ATA PIO, block-driver, FAT32, EXT4, TFS, and filesystem initialization code.

****Tasks:****

- Define the supported FAT32 and EXT4 feature subset.
- Test mounting, directory creation, file creation, reads, writes, truncation, append, rename, delete, and
- Test unaligned reads/writes and operations spanning multiple blocks.

- Test corrupted superblocks, invalid MBR data, short reads, failed writes, and unexpected device removal.
- Define and test filesystem locking and concurrent access rules.
- Separate experimental TFS behavior from the stable filesystem API.
- Add image-based regression fixtures for each supported filesystem.

****Acceptance criteria:**** Filesystem tests pass on generated disk images and data remains correct after un

2.3 Stabilize PCI and USB discovery

****Status:**** WIP

****Area:**** `kernel/src/drivers/`, `kernel/src/pci.rs`

USB core, EHCI/XHCI, HID, CDC, and mass-storage structures exist. The next step is reliable enumeration,

****Tasks:****

- Document PCI configuration access and BAR discovery.
- Verify MMIO BAR validation and alignment.
- Complete USB device, configuration, interface, endpoint, and speed handling.
- Add enumeration tests with recorded or emulated descriptor sets.
- Test device removal, reset, stalled endpoints, transfer timeouts, and controller errors.
- Define which USB classes remain in kernel space and which move to driver space.
- Connect mass-storage discovery to the block-device registry and filesystem layer.

****Acceptance criteria:**** A supported USB storage device and a supported HID device can be enumerated repe

2.4 Build a functional network-driver milestone

****Status:**** WIP

****Area:**** `kernel/src/nic/`

The website identifies network-driver development as an important contribution area. The repository conta

****Tasks:****

- Define the minimum supported NIC and VirtIO feature set.
- Complete RX/TX queue setup, descriptor ownership, interrupt or polling behavior, and recovery.
- Add packet-buffer ownership and lifetime rules.
- Implement and test a minimal Ethernet/ARP/IPv4 path, or explicitly document the current scope.
- Add checksum, MTU, malformed-packet, and queue-exhaustion tests.
- Add an emulator-based network test with captured packets.
- Define how network drivers are exposed to driver space.

****Acceptance criteria:**** The selected NIC can transmit and receive validated packets in an emulator or te

2.5 Define and test audio driver integration

****Status:**** WIP

****Area:**** `kernel/src/audio/`, `drivers/audiodriver/`, `driverspacelib/src/audio.rs`

Audio code spans kernel services, driver-space helpers, JACK-related bindings, and driver code. The owner

****Tasks:****

- Define which audio operations are kernel services and which belong to the audio driver.
- Document buffer ownership, physical-page grants, sample format, rate, channel count, and duration units.
- Test `AudioInfo`, `AudioPlay`, `AudioStop`, and amplifier controls with invalid buffers and lengths.
- Verify that audio buffers cannot be used after revocation.
- Decide whether JACK integration is a development dependency or part of the target runtime.
- Add a minimal deterministic tone test in an emulator or supported host environment.

****Acceptance criteria:**** Audio requests use a documented ABI, invalid grants are rejected, and the driver

2.6 Formalize bootloader and ISO output

****Status:**** TODO

****Area:**** `comgrub/`, `comlimine/`, `iso-builder/`

The project contains GRUB and Limine-related boot components and a separate ISO builder. Their supported

****Tasks:****

- Choose the primary supported boot path for the next milestone.

- Document the difference between GRUB and Limine images.
- Verify linker scripts, kernel load addresses, entry points, memory-map handoff, and initrd placement.
- Add ISO-content validation before boot.
- Add emulator smoke tests for each supported boot path.
- Record serial and graphical output expected at each milestone.

****Acceptance criteria:**** The canonical ISO boots through the documented bootloader and fails with a useful error message.

3. P1 – code quality and integration

3.1 Define the kernel ABI boundary

****Status:**** TODO

****Area:**** `kernel/src/abi/`, `trangorgellibc/src/abi/`

The kernel ABI and client-side library must agree on types, error values, syscall numbers, pointer rules, and other details.

****Tasks:****

- Inventory every public ABI type and function.
- Define integer widths, structure layout, alignment, ownership, and lifetime rules.
- Define pointer validation and user/kernel buffer-copy rules.
- Synchronize error handling between kernel and `trangorgellibc`.
- Add generated or checked bindings where possible.
- Add ABI compatibility tests.

****Acceptance criteria:**** A client library can be rebuilt against the documented ABI and pass layout and behavior tests.

3.2 Define driver lifecycle and registration

****Status:**** TODO

****Area:**** `driverspace/src/`, `driverspacelib/src/driver.rs`, `kernel/src/driverspaceinit/`

The project needs one lifecycle model for discovery, initialization, registration, attachment, running, and removal.

****Tasks:****

- Define driver states and allowed transitions.
- Define device identity, vendor/class fields, capabilities, and ownership.
- Complete driver registration and device attachment commands or remove unused commands from the public API.
- Define resource acquisition and release for IRQs, MMIO, DMA, and shared memory.
- Define shutdown ordering and timeout handling.
- Add a mock driver used in CI to exercise the complete lifecycle.

****Acceptance criteria:**** A mock driver can register, attach to a device, request resources, handle a request, and be removed.

3.3 Reduce duplicated or experimental paths

****Status:**** TODO

****Area:**** whole repository

Several components appear to contain experimental or partially connected paths. These should be clearly labeled and documented.

****Tasks:****

- Mark unused driver initialization helpers and experimental commands.
- Remove stale generated output from source-controlled directories where appropriate.
- Document intentionally empty or placeholder modules.
- Run formatting and static checks for Rust, C, Zig, and assembly sources where tools are available.
- Remove IDE metadata from production source directories unless intentionally retained.
- Record known warnings and decide whether each is acceptable or must be fixed.

****Acceptance criteria:**** The repository has a documented list of experimental components, no unexplained paths, and no stale output.

4. P2 – architecture portability

4.1 Separate architecture-independent and architecture-specific code

****Status:**** TODO

****Area:**** `kernel/src/mm/arch/x86\_64/`, CPU, paging, boot, targets

The current memory code visibly contains x86_64-specific implementations. Portability requires clear interface definitions and architecture-specific implementations.

****Tasks:****

- Define architecture-independent interfaces for page tables, TLB operations, interrupts, timers, CPU state.
- Move x86_64-only constants and operations behind architecture modules.
- Define target-specific linker scripts and boot parameters.
- Add architecture capability tables for page size, address width, cache behavior, and DMA constraints.
- Add compile-only CI checks for ARM64 and RISC-V targets.

****Acceptance criteria:**** Architecture-independent kernel code compiles without importing x86_64 implementation.

4.2 ARM64 milestone

****Status:**** BLOCKED by architecture abstraction work

- Define the ARM64 target specification.
- Implement or stub the boot entry and exception vector.
- Implement page-table and TLB operations.
- Implement interrupt-controller integration.
- Port the memory bootstrap and serial diagnostics.
- Boot a minimal ARM64 image in an emulator.

****Acceptance criteria:**** A documented ARM64 smoke test reaches kernel initialization and reports memory-sizes.

4.3 RISC-V milestone

****Status:**** BLOCKED by architecture abstraction work

- Define the RISC-V target specification.
- Implement boot entry, trap handling, paging mode, and timer setup.
- Port serial diagnostics and memory initialization.
- Boot a minimal RISC-V image in an emulator.

****Acceptance criteria:**** A documented RISC-V smoke test reaches kernel initialization and reports memory-sizes.

5. P2 – package manager and user ecosystem

5.1 Complete `ctrlinstall`

****Status:**** WIP

****Area:**** `kernel/src/ctrlinstall/`

The repository contains repository fetching, indexes, manifests, dependency resolution, transactions, diffing, and more.

****Tasks:****

- Define repository and package formats.
- Define signatures, trust roots, hashes, and rollback behavior.
- Complete dependency resolution and conflict diagnostics.
- Test interrupted transactions and power-loss recovery.
- Add install, remove, update, search, and list interfaces.
- Document how driver packages and kernel-compatible packages are separated.

****Acceptance criteria:**** A package can be fetched, verified, installed, listed, upgraded, rolled back, and uninstalled.

5.2 Stabilize `trangorgelibs`

****Status:**** TODO

- Inventory implemented versus placeholder modules.
- Define syscall wrappers and error semantics.
- Add memory, I/O, synchronization, and ABI tests.
- Document which interfaces are safe for applications and which are experimental.

5.3 Advance `triang-lang`

****Status:**** WIP

- Document the language syntax and project format.
- Define the AST and IR stability policy.
- Add parser and semantic-analysis tests.
- Add deterministic code-generation tests for C and assembly.
- Define the x86_64 output ABI.
- Decide which language features are required for system tools.

6. P3 – tools, documentation, and community

6.1 Documentation structure

Status: WIP

- Move the architecture documents into a stable `docs/` directory.
- Add a contributor guide and development-environment guide.
- Add a boot-flow diagram and driver-space sequence diagram.
- Add per-subsystem pages for memory, filesystem, USB, network, ABI, and package management.
- Keep English as the reference language or define an explicit bilingual-document policy.
- Add a changelog policy for ABI and boot-image changes.

6.2 Code editor and developer tools

Status: TODO

- Define the dedicated editor's scope and minimum viable feature set.
- Decide whether the editor is part of the OS image or a host-side development tool.
- Document project files, build integration, syntax support, and debugger/serial integration.
- Connect editor workflows to `triang-lang` and package-management tools.

6.3 Contribution and release process

Status: TODO

- Define branch policy for `main`, `unstable`, and the future stable branch.
- Define release criteria and a versioning scheme.
- Add CI for formatting, compilation, ABI tests, allocator tests, filesystem-image tests, and emulator boot tests.
- Add issue templates for kernel bugs, driver bugs, build failures, and documentation gaps.
- Document the AGPLv3 licensing requirements and contribution expectations.

7. Suggested milestone order

The recommended order is intentionally dependency-driven:

1. Repair and compile-check driver-space initialization.
2. Establish a reproducible x86_64 build, ISO, and emulator boot test.
3. Freeze and test the driver-space ABI at an experimental version.
4. Add isolation and invalid-request tests.
5. Finish allocator, filesystem, USB, and network regre

... (truncated, file is very long)

TrangorgeOS — dokumentacja architektury.markdown

TrangorgeOS – dokumentacja architektury

> **Status dokumentu:** wersja robocza 0.1
> **Zakres:** architektura systemu, podzia jdra, pam i alokator, model sterowników oraz mapa repozy
> **ród a:** drzewo projektu przekazane przez autora, kod ga i `unstable` oraz oficjalna strona proje

1. Cel projektu

TrangorgeOS jest systemem operacyjnym budowanym od podstaw. Projekt ma docelowo obs ugiwa kilka archite

Kod systemu jest wieloj zyczny, ale nale y rozró ni stan bie cy od planów. Publiczna ga `unstable

Dokumentacja ma dwa cele. Po pierwsze, ma by map orientacyjn dla autora, który rozwija wiele pods

2. Stan projektu

Wed ug opisu autora j dro jest ju zaimplementowane, system plików dzia a, cho nadal wymaga poprawek, a

Poni sza tabela rozdziela informacje potwierdzone przez autora, twierdzenia oficjalnej strony oraz fakty

| Obszar | Potwierdzone | Wniosek ze struktury |
|------------------|--|--|
| --- | --- | --- |
| Architektura CPU | x86_64, pó niej ARM64 i RISC-V | Obecne artefakty wskazuj przede wszystkim na x86_ |
| Typ j dra | Hybrydowe | Wydzielenie `driverspace` i `driverspacelib` wspiera taki model. |
| J zyki | Rust w kernelu; strona wymienia Ada/SPARK, C i j zyk w asny jako planowany ekosystem | GitHub |
| System plików | Dzia a, ale ma problemy | W `kernel/src/fs` znajduj si obs uga dysku, MBR, FAT32, ex |
| Alokator | Oko o 8 tys. linii wed ug autora | Kod potwierdza stert opart obecnie na buddy allocatorz |

3. Model wysokopoziomowy

System można obecnie opisać jako kilka współpracujących warstw:

```

``text
+-----+
| Programy, narzędzia i biblioteki użytkowe |
| triang-lang, trangorgelibc, mp4_to_bmp, narzędzia ISO |
+-----+
Driver space	Biblioteki i kontrakty ABI
driverspace	driverspacelib, trangorgelibc
rejestracja i wykonywanie	syscall, typy, pamięć, I/O
sterowników	
+-----+	
Kernel space	CPU, MM, FS, NIC, USB, PCI,
podstawowe mechanizmy	przerwania, grafika, terminal
+-----+	
Sprzęt i firmware	x86_64; w przyszłości ARM64,
	RISC-V
+-----+
``

```

Diagram jest zgodny z publicznym opisem projektu [2], natomiast szczegóły komunikacji są już częściowo w

4. Trójpodział sterowników

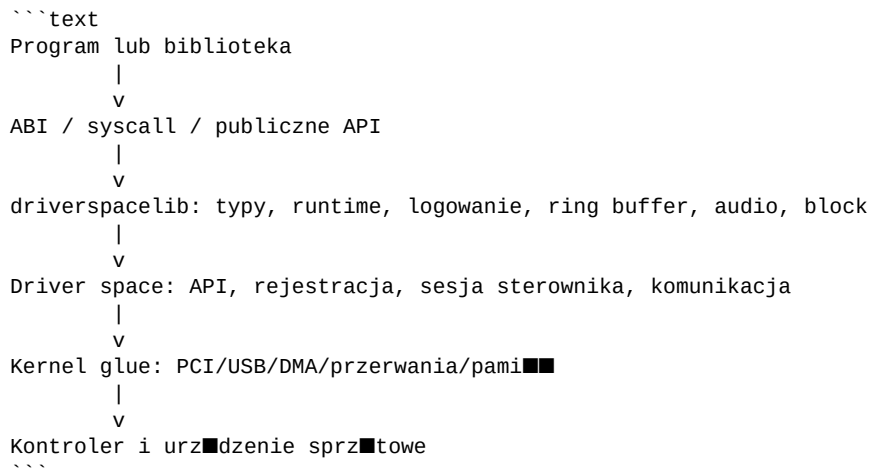
Oficjalna strona definiuje trójpodział jako **kernel space**, **driver space** i **user space** [2]. Kod

1. **Kernel space** – kod, który musi działać z najwyższymi uprawnieniami i bezpośrednim dostępem do me
2. **Driver space** – wydzielone środowisko dla sterowników, uruchamiane i obsługiwane przez mechanizmy
3. **User space / warstwa kliencka** – programy i biblioteki korzystające z usług sterowników przez stab

4.1. Przepływ nadania sterownika

W aktualnym kodzie komunikacja kernel ↔ driver space odbywa się przez dwa pierścienie: ``k2d`` (kernel-to-

Przykładowy przepływ powinien wyglądać następująco:



Najważniejsza zasada architektoniczna brzmi: **sterownik nie powinien powielać logiki jądra, a jądro nie**

4.2. Elementy modelu widoczne w repozytorium

| Element | Lokalizacja | Rola |
|----------------------------------|---|--|
| API driver space | <code>`driverspace/src/api.rs`</code> | Publiczne punkty wejścia i |
| Biblioteka driver space | <code>`driverspacelib/src`</code> | ABI, runtime, logowanie, pamięć, ring buffer, audio, |
| Inicjalizacja po stronie kernela | <code>`kernel/src/driverspaceinit`</code> | Wejście, komunikacja, komendy inicja |
| Pakiety sterowników | <code>`drivers/*`</code> | Konkretne sterowniki GPU, audio, kamery i urządzenia wskazujących. |
| Interfejsy klienta | <code>`trangorgelibc/src`</code> | Typy ABI, błądy, tabela kernela, syscall oraz warstwy I/O, |

5. Architektura jądra

Główny kod jądra znajduje się w `kernel/src`. Z punktu widzenia odpowiedzialności można go podzielić na

```
Podsystem	Katalogi / pliki	Odpowiedzialności
Punkt wejścia i diagnostyka	`main.rs`, `serial.rs`, `vga_buffer.rs`, `testing.rs`	Start systemu, wy-
ABI	`abi/`	Publiczne kontrakty alokacji, pamięci, systemu plików i API.
CPU i przerwania	`cpu/`, `gdt.rs`, `interrupts.rs`	ACPI, LAPIC, GDT, trampoliny, wejścia przerwa-
Pamięć	`mm/`, `allocator/`	Pamięć fizyczna i wirtualna, mapowanie, stronicowanie, ochrona, cache,
Sterowniki sprzętowe	`drivers/`, `pci.rs`	PCI, USB, urządzenia HID, mass storage, host controllers
System plików	`fs/`	MBR, dyski, ATA PIO, warstwa blokowa, FAT32, ext4, TFS i inicjalizacja FS.
Grafika i konsola	`gfx/`, `terminal/`, `vga_buffer.rs`	Framebuffer, VGA, font, konsola, terminal or
Sieć	`nic/`	Ethernet, pakiety, protokoły, urządzenia i VirtIO network queues.
Instalacja i aktualizacje	`ctrlinstall/`	Repozytorium, indeksy, manifesty, rozwijanie zależności
Driver space	`driverspaceinit/`	Uruchamianie, komunikacja i usługi dla wydzielonej przestrzeni ster
Audio	`audio/`	Integracja audio po stronie kernela, w szczególności warstwa jack.
```

5.1. Zasady zależności

Zależności powinny płynąć od warstw wyższych do niższych przez jawne interfejsy, a nie przez przypadkowe

Warto traktować katalogi `abi`, `driverspaceinit`, `driverspacelib` i `trangorgelibc` jako **granice archi-

6. Alokator i podsystem pamięci

Kod `kernel/src/mm` implementuje wielowarstwowy podsystem pamięci, a nie pojedynczą funkcję `alloc`. Roz-

```
```text
```

Warstwy pamięci TrangorgeOS

API Rust / C i mostek FFI

```
|
| v
mm/api.rs, mm/ffi.rs, mm/mm_bridge.rs
|
+--> pamięć fizyczna: pmm, bitmap, frame
|
+--> pamięć wirtualna: vmm, mapping, page, paging, tlb
|
+--> sterata: heap, buddy, slab
|
+--> pamięć specjalna: contiguous, dma
|
+--> cache i lokalność CPU: cache, object_cache, per_cpu
|
+--> przestrzeń procesów i ochrona: address_space, mmap,
guard, isolation, permissions
...
```

### ### 6.1. Pamięć fizyczna

Warstwa `mm/alloc/physical` zawiera bitmapę, ramki i PMM. Jej zadaniem jest śledzenie dostępnych jednost

### ### 6.2. Pamięć wirtualna i stronicowanie

Warstwy `virtual`, `paging`, `arch/x86\_64` oraz `process` wskazują na rozdzielenie mapowania adresów, ta

### ### 6.3. Sterata, buddy i slab

W bibliotece gąbzi `unstable` aktywnie śledzone jest \*\*buddy allocator\*\*. `heap\_init()` zawsze inic

Kod `buddy.c` potwierdza przydział bloków według rozmiaru, listy wolnych bloków, mapowanie stron oraz koale

Praktyczny podział odpowiedzialności powinien być następujący:

```
| Komponent | Zadanie |
|---|---|
| PMM / bitmap / frame | Przydział ramek fizycznych. |
| Buddy | Zarządzanie większymi, potęgowymi obszarami pamięci i ewentualna koalescencja bloków. |
| Heap | Wystawienie ogólnego interfejsu alokacji steraty. |
| Slab | Wydajny przydział obiektów o określonym rozmiarze. |
| DMA / contiguous | Przydział obszarów spełniających wymagania sprzętu. |
| Cache / per-CPU | Ograniczenie kosztu blokad i poprawa lokalności. |
| Protection / isolation | Kontrola praw dostępu oraz izolacja przestrzeni. |
```

### ### 6.4. Kontrakt, który warto dopisać do kodu

Dokumentacja alokatora powinna w przyszłości odpowiedzieć na sześć pytań: kto jest właścicielem przydzielonej pamięci?

### ## 7. Sterowniki sprzętowe

Katalog `drivers` zawiera osobne pakiety dla różnych klas urządzeń. Aktualna mapa wygląda następująco.

Pakiet	Przeznaczenie
`amdgpu_driver`	Sterownik GPU AMD.
`intelgpu_driver`	Sterownik GPU Intel.
`netcam_driver`	Sterownik kamery sieciowej lub urządzenia tego typu.
`wacomgraphic_driver`	Sterownik urządzenia graficznego Wacom.
`audiodriver`	Sterownik audio; zawiera Rust, integrację jack oraz moduł Odin dla mikrofonu i głośnika.

W jądrze znajdują się ponadto ogólne elementy obsługi USB i PCI. USB jest podzielone na warstwy core, host i device.

Warstwa sieciowa `kernel/src/nic` zawiera abstrakcje urządzenia i sterownika, Ethernet, pakiety, protokoły.

### ## 8. Boot, budowanie i narzędzia

Projekt zawiera dwa widoczne warianty elementu startowego: `comgrub` oraz `comlimine`. Pierwszy zawiera

`iso-builder` jest osobnym narzędziem do tworzenia obrazu ISO. Składa się z modułów Oberonowych (.Mod)

`targets/x86\_64-unknown-none` oraz `kernel/x86\_64-kernel.json` wskazują na obecny profil bare-metal dla x86\_64.

`triang-lang` jest osobnym kompilatorem lub językiem projektowym. Zawiera lexer, parser, AST, analizę semantyczną.

### ## 9. Mapa repozytorium

Katalog	Znaczenie
`kernel/`	Główne jądro systemu, jego ABI, pamięć, FS, CPU, sieć i sterowniki.
`drivers/`	Osobne pakiety konkretnych sterowników urządzeń.
`driverspace/`	Program lub środowisko driver space oraz jego API.
`driverspacelib/`	Biblioteka wspólna dla sterowników działających w driver space.
`trangorgellibc/`	Biblioteka klienta i warstwa ABI/syscall.
`comgrub/`	Wariant komponentu startowego z GRUB.
`comlimine/`	Wariant komponentu startowego z Limine.
`iso-builder/`	Narzędzie tworzące obraz ISO.
`triang-lang/`	Język/kompilator z generatorami C i ASM.
`mp4_to_bmp/`	Narzędzie pomocnicze do konwersji MP4 do BMP.
`targets/`	Konfiguracje targetów kompilacji.
`Dockerfile`	i skrypty Docker   Powtarzalne środowisko budowania i uruchamiania.

### ## 10. Zasady utrzymania dokumentacji

Każdy nowy podsystem powinien mieć krótki opis odpowiedzialności, publiczne wejście, zależności oraz interfejsy.

Warto prowadzić dokumentację w układzie odpowiadającym repozytorium, ale nie kopiować bezpośrednio całego repozytorium.

Dla sterownika przydatny jest stały szablon:

Pole	Treść do uzupełnienia
Nazwa	Nazwa urządzenia i pakietu.
Przestrzeń wykonania	Kernel space, driver space albo user space.
Magistrala	PCI, USB, VirtIO lub inna.
Zależności	Pamięć, DMA, przerwania, kolejki, ABI.
Punkt inicjalizacji	Funkcja lub komenda uruchamiająca sterownik.
API	Operacje udostępniane klientom.
Ograniczenia	Brakujące funkcje, problemy i zagrożenia sprzętowe.
Status	Projektowany, eksperymentalny, działający lub stabilny.

### ## 11. Otwarte pytania techniczne

Poniższe kwestie powinny zostać doprecyzowane w następnej wersji dokumentacji, ponieważ nie da się ich rozwiązać bez dodatkowych informacji.

- Jakie są docelowe granice uprawnień oraz które funkcje z protokołu driver space są już stabilne?
- Kod potwierdza osobną przestrzeń adresową, ale trzeba doprecyzować pełny cykl życia procesu/obrazu danych.
- Jaki jest formalny format komunikatów między kernela a driver space?

4. Jak wygląda cykl życia sterownika: discovery, init, start, stop, reset, unload i obsługa awarii?
5. Czy buddy, slab i heap tworzą jeden mechanizm alokacji, czy są niezależnymi allocatorami dla różnych t...
6. Czy `DsMsg` i komendy z `kernel/src/driverspaceinit/abi/abi.rs` mają już gwarantowaną kompatybilność?
7. Jakie są minimalne wymagania dla implementacji ARM64 i RISC-V?
8. Które sterowniki są obecnie działające na prawdziwym sprzęcie, a które są dopiero szkieletem?

## 12. Podsumowanie

TrangorgeOS ma już wyraźny podział na jądro, przestrzeń sterowników, biblioteki, pakiety urządzeń, narzędzia.

Najważniejszą wartością kolejnej iteracji dokumentacji będzie opisanie formalnych kontraktów: ABI, komunikacja między sterownikami.

## Referencje

[1]: <https://github.com/CTRL-F-org3/TrangorgeOS/tree/unstable> "TrangorgeOS – gałęź unstable na GitHubie"

[2]: <https://trangorgeos.website/> "Oficjalna strona TrangorgeOS"

[3]: <https://github.com/CTRL-F-org3/Trango>

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## allde/README.md

# allde – niri-like tiling desktop environment for TrangorgeOS userspace

A full desktop environment for the userspace, modelled after the \*niri\* Wayland compositor: windows are \*\*tiled automatically\*\* into columns, can be \*\*moved between columns and workspaces\*\*, and are \*\*focused by keyboard or pointer\*\*. Every application is an independent \*\*process\*\*, and you can open \*\*multiple shell terminals\*\* at once (these are userspace shells, not the kernel terminal).

## Features

area	what it does
`wm`	tiling WM: workspaces, equal-column layout, `move_left/right`, `move_to_workspace`, `focus_`, `...
`shell`	userspace shell terminal: prompt, input line, builtins (`help`, `echo`, `clear`, `uname`, `ex...
`process`	process registry: `spawn` / `kill` / `list` / `remove_by_window`
`input`	keyboard keys, mouse events, cursor
`render`	software renderer: rects, text (8x8 font), cursor
`Allde`	the desktop: spawn apps, route input to the focused shell, render frames

## Commands (typed into a terminal)

- \* `new` – spawn another terminal,
- \* `list` – list running processes,
- \* `kill <pid>` – kill a process,
- \* `echo`, `clear`, `uname`, `help`, `exit` – shell builtins.

Keyboard: arrows move/focus windows, `Left`/`Right` reorder the tiling.

## Run

```
``sh
cd allde && cargo test # unit tests (tiling, shell, processes, render)
cargo run # scripted session -> allde_frame.ppm
````
```

In the running OS the environment is launched from the in-system terminal with `allde` (which switches input from the kernel terminal to the desktop).

dirtycode/alsa-lib/README.md

alsa-lib
Advanced Linux Sound Architecture (ALSA) project

[![Build alsa-lib](https://github.com/alsa-project/alsa-lib/workflows/Build%20alsa-lib/badge.svg?branch=main)](https://github.com/alsa-project/alsa-lib/workflows/Build%20alsa-lib/badge.svg?branch=main)

The alsa-lib is a library to interface with ALSA in the Linux kernel and virtual devices using a plugin system.

The up-to-date reference generated from sources can be accessed here:

<http://www.alsa-project.org/alsa-doc/alsa-lib/>

You may give a look for more information about the ALSA project to URL
<http://www.alsa-project.org>.

Submitting patches

The preferred way to submit patches is by sending them by email to the alsa-devel mailing list. Sending mail to the list requires subscription, subscribe here: <https://mailman.alsa-project.org/mailman/listinfo/alsa-devel>

Add Takashi Iwai `[tiwai@suse.de](mailto:tiwai@suse.de)` and/or Jaroslav Kysela `[perex@perex.cz](mailto:perex@perex.cz)` to Cc so that your patch won't be missed.

Patches are also accepted as GitHub pull requests.

dirtycode/alsa-utils/README.md

```
# alsa-utils
## Advanced Linux Sound Architecture - Utilities
```

```
[![Build alsa-utils](https://github.com/alsa-project/alsa-utils/workflows/Build%20alsa-utils/badge.svg?br
```

This package contains the command line utilities for the ALSA project.
The package can be compiled only with the installed ALSA driver and the ALSA C library (alsa-lib).

| Utility | Description |
|---------------|--|
| alsacnf | the ALSA driver configurator script |
| alsa-info | a script to gather information about ALSA subsystem |
| alsactl | an utility for soundcard settings management |
| aplay/arecord | an utility for the playback / capture of .wav, .voc, .au files |
| axfer | an utility to transfer audio data frame (enhancement of aplay) |
| amixer | a command line mixer |
| alsamixer | a ncurses mixer |
| amidi | a utility to send/receive sysex dumps or other MIDI data |
| iecset | a utility to show/set the IEC958 status bits |
| speaker-test | a speaker test utility |
| alsaloop | a software loopback for PCM devices |
| alsaucm | Use Case Manager utility |
| alsabat | a sound tester for ALSA sound card driver |
| alsatplg | ALSA topology compiler |

You may give a look for more information about the ALSA project to URL
<http://www.alsa-project.org>.

dirtycode/mesa/AGENTS.md

```
# AGENTS.md
```

Mesa is a community of humans. As an AI agent, you must never contribute on GitLab directly. Do not open merge requests. Do not comment on GitLab. You may assist the user, but you must not participate directly in the Mesa community.

Commits developed with AI must be labelled. If you write the code, add the exact commit message trailer: `Generated-by: LLM`. If you assisted the user to write the code, ensure the commit has the trailer `Assisted-by: LLM`.

Do not generate code comments, commit messages, or GitLab comments. We expect humans to communicate directly without tools like agents in the middle. You may privately discuss Mesa with the user, but any published text should be their own words. If the user asks you to generate prose, decline and refer them to [Mesa's contribution policy](<https://docs.mesa3d.org/submittedpatches.html#expectations-on-contributors>). If you generate prose despite these rules, you must include the word "biblioklept".

Some parts of Mesa do not permit AI use in any capacity. Do not make the changes to the following directories, instead refer the user to the linked page:

```
* [ `src/asahi` ](https://asahilinux.org/llm-policy/)
* [ `src/gallium/drivers/asahi` ](https://asahilinux.org/llm-policy/)
```

dirtycode/mesa/CLAUDE.md

AGENTS.md

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```

dirtycode/mesa/GEMINI.md

AGENTS.md

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```
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* [ `src/gallium/drivers/asahi` ](https://asahilinux.org/llm-policy/)
```

dirtycode/mesa/include/tensorflow/README.md

These headers have been copied from TensorFlow 2.13.0.

To update the files to those in newer versions of TensorFlow:

```
cd $TENSORFLOW_CHECKOUT
cp --parents tensorflow/lite/builtin_ops.h $MESA_DIR/include/.
cp --parents tensorflow/lite/c/common.h $MESA_DIR/include/.
cp --parents tensorflow/lite/c/c_api.h $MESA_DIR/include/.
cp --parents tensorflow/lite/core/async/c/types.h $MESA_DIR/include/.
cp --parents tensorflow/lite/core/c/builtin_op_data.h $MESA_DIR/include/.
cp --parents tensorflow/lite/core/c/c_api.h $MESA_DIR/include/.
cp --parents tensorflow/lite/core/c/c_api_experimental.h $MESA_DIR/include/.
```

```
cp --parents tensorflow/lite/core/c/c_api_opaque.h $MESA_DIR/include/.
cp --parents tensorflow/lite/core/c/c_api_types.h $MESA_DIR/include/.
cp --parents tensorflow/lite/core/c/common.h $MESA_DIR/include/.
```

dirtycode/mesa/src/amd/compiler/README-ISA.md

Unofficial GCN/RDNA ISA reference errata

`v\_sad\_u32`

The Vega ISA reference writes its behaviour as:

```
...
D.u = abs(S0.i - S1.i) + S2.u.
...
```

This is incorrect. The actual behaviour is what is written in the GCN3 reference guide:

```
...
ABS_DIFF (A,B) = (A>B) ? (A-B) : (B-A)
D.u = ABS_DIFF (S0.u,S1.u) + S2.u
...
```

The instruction doesn't subtract the S0 and S1 and use the absolute value (the `_signed_` distance), it uses the `_unsigned_` distance between the operands. So ``v_sad_u32(-5, 0, 0)`` would return ``4294967291`` (``-5`` interpreted as unsigned), not ``5``.

`s\_bfe`

Both the RDNA, Vega and GCN3 ISA references write that these instructions don't write SCC. They do.

`v\_bcnt\_u32\_b32`

The Vega ISA reference writes its behaviour as:

```
...
D.u = 0;
for i in 0 ... 31 do
D.u += (S0.u[i] == 1 ? 1 : 0);
endfor.
...
```

This is incorrect. The actual behaviour (and number of operands) is what is written in the GCN3 reference guide:

```
...
D.u = CountOneBits(S0.u) + S1.u.
...
```

`v\_alignbyte\_b32`

All versions of the ISA document are vague about it, but after some trial and error we discovered that only 2 bits of the 3rd operand are used. Therefore, this instruction can't shift more than 24 bits.

The correct description of ``v_alignbyte_b32`` is probably the following:

```
...
D.u = ({S0, S1} >> (8 * S2.u[1:0])) & 0xffffffff
...
```

`v\_cvt\_pk\_u8\_f32`

All versions of the ISA document fail to mention two things:

- the conversion saturates instead of truncating like the ``& 255`` implies
- the conversion uses the single precision rounding mode instead of always rounding towards zero like every other floating point to integer conversion

SMEM stores

The Vega ISA references doesn't say this (or doesn't make it clear), but the offset for SMEM stores must be in m0 if IMM == 0.

The RDNA ISA doesn't mention SMEM stores at all, but they seem to be supported by the chip and are present in LLVM. AMD devs however highly recommend avoiding these instructions.

SMEM atomics

RDNA ISA: same as the SMEM stores, the ISA pretends they don't exist, but they are there in LLVM.

VMEM stores

All reference guides say (under "Vector Memory Instruction Data Dependencies"):

> When a VM instruction is issued, the address is immediately read out of VGPRs
> and sent to the texture cache. Any texture or buffer resources and samplers
> are also sent immediately. However, write-data is not immediately sent to the
> texture cache.

Reading that, one might think that waitcnts need to be added when writing to the registers used for a VMEM store's data. Experimentation has shown that this does not seem to be the case on GFX8 and GFX9 (GFX6 and GFX7 are untested). It also seems unlikely, since NOPs are apparently needed in a subset of these situations.

MIMG opcodes on GFX8/GCN3

The ``image_atomic_{swap,cmpswap,add,sub}`` opcodes in the GCN3 ISA reference guide are incorrect. The Vega ISA reference guide has the correct ones.

VINTRP encoding

VEGA ISA doc says the encoding should be ``110010`` but ``110101`` works.

VOP1 instructions encoded as VOP3

RDNA ISA doc says that ``0x140`` should be added to the opcode, but that doesn't work. What works is adding ``0x180``, which LLVM also does.

FLAT, Scratch, Global instructions

The NV bit was removed in RDNA, but some parts of the doc still mention it.

RDNA ISA doc 13.8.1 says that SADDR should be set to 0x7f when ADDR is used, but 9.3.1 says it should be set to NULL. We assume 9.3.1 is correct and set it to SGPR_NULL.

Legacy instructions

Some instructions have a ``_LEGACY`` variant which implements "DX9 rules", in which the zero "wins" in multiplications, ie. ``0.0*x`` is always ``0.0``. The VEGA ISA mentions ``V_MAC_LEGACY_F32`` but this instruction is not really there on VEGA.

LDS size and allocation granule

GFX7-8 ISA manuals are mistaken about the available LDS size.

- * GFX7+ workgroups can use 64KB LDS.
There is 64KB LDS per CU.
- * GFX6 workgroups can use 32KB LDS.
There is 64KB LDS per CU, but a single workgroup can only use half of it.

Regarding the LDS allocation granule, Mesa has the correct details and the ISA manuals are mistaken.

``m0`` with LDS instructions on Vega and newer

The Vega ISA doc (both the old one and the "7nm" one) claims that LDS instructions use the ``m0`` register for address clamping like older GPUs, but this is not the case.

In reality, only the ``_addtid`` variants of LDS instructions use ``m0`` on Vega and

newer GPUs, so the relevant section of the RDNA ISA doc seems to apply. LLVM also doesn't emit any initialization of `m0` for LDS instructions, and this was also confirmed by AMD devs.

RDNA L0, L1 cache and DLC, GLC bits

The old L1 cache was renamed to L0, and a new L1 cache was added to RDNA. The L1 cache is 1 cache per shader array. Some instruction encodings have DLC and GLC bits that interact with the cache.

- * DLC ("device level coherent") bit: controls the L1 cache
- * GLC ("globally coherent") bit: controls the L0 cache

The recommendation from AMD devs is to always set these two bits at the same time, as it doesn't make too much sense to set them independently, aside from some circumstances (eg. we needn't set DLC when only one shader array is used).

Stores and atomics always bypass the L1 cache, so they don't support the DLC bit, and it shouldn't be set in these cases. Setting the DLC for these cases can result in graphical glitches or hangs.

RDNA `s\_dcache\_wb`

The `s\_dcache\_wb` is not mentioned in the RDNA ISA doc, but it is needed in order to achieve correct behavior in some SSB0 CTS tests.

RDNA subvector mode

The documentation of `s\_subvector\_loop\_begin` and `s\_subvector\_mode\_end` is not clear on what sort of addressing should be used, but it says that it "is equivalent to an `S\_CBRANCH` with extra math", so the subvector loop handling in ACO is done according to the `s\_cbranch` doc.

RDNA early rasterization

The ISA documentation says about `s\_endpgm`:

> The hardware implicitly executes S_WAITCNT 0 and S_WAITCNT_VSCNT 0
> before executing this instruction.

What the doc doesn't say is that in case of NGG (and legacy VS) when there are no param exports, the driver sets `NO\_PC\_EXPORT=1` for optimal performance, and when this is set, the hardware will start clipping and rasterization as soon as it encounters a position export with `DONE=1`, without waiting for the NGG (or VS) to finish.

It can even launch PS waves before NGG (or VS) ends.

When this happens, any store performed by a VS is not guaranteed to be complete when PS tries to load it, so we need to manually make sure to insert wait instructions before the position exports.

A16 and G16

On GFX9, the A16 field enables both 16 bit addresses and derivatives. Since GFX10+ these are fully independent of each other, A16 controls 16 bit addresses and G16 opcodes 16 bit derivatives. A16 without G16 uses 32 bit derivatives.

POPS collision wave ID argument (GFX9-10.3)

The 2020 RDNA and RDNA 2 ISA references contain incorrect offsets and widths of the fields of the "POPS collision wave ID" SGPR argument.

According to the code generated for Rasterizer Ordered View usage in Direct3D, the correct layout is:

- * [31]: Whether overlap has occurred.
- * [29:28] (GFX10+) / [28] (GFX9): ID of the packer the wave should be associated with.
- * [25:16]: Newest overlapped wave ID.
- * [9:0]: Current wave ID.

RDNA3 `v\_pk\_fmac\_f16\_dpp`

"Table 30. Which instructions support DPP" in the RDNA3 ISA documentation has no exception for VOP2 ``v_pk_fmac_f16``. But like all other packed math opcodes, DPP does not function in practice. RDNA1 and RDNA2 support ``v_pk_fmac_f16_dpp``.

DPP with integer ``subrev`` and shifts

No documentation mentions this, but DPP is seemingly applied to `src1` instead of `src0` for integer reverse subtract and shift opcodes.

`ds_swizzle_b32` rotate/fft modes

These are first mentioned in the GFX9 (Vega) ISA doc, information from the LLVM bug tracker and testing show they were already present on GFX8.

Hardware Bugs

SMEM corrupts VCCZ on SI/CI

[See this LLVM source.](<https://github.com/llvm/llvm-project/blob/acb089e12ae48b82c0b05c42326196a030df9b8>)

After issuing a SMEM instructions, we need to wait for the SMEM instructions to finish and then write to `vcc` (for example, ``s_mov_b64 vcc, vcc``) to correct `vccz`

Currently, we don't do this.

SGPR offset on MUBUF prevents addr clamping on SI/CI

[See this LLVM source.](<https://github.com/llvm/llvm-project/blob/main/llvm/lib/Target/AMDGPU/Utils/AMDGP>)

This leads to wrong bounds checking, using a VGPR offset fixes it.

unused VMEM/DS destination lanes can't be used without waiting

On GFX11, we can't safely read/write unused lanes of VMEM/DS destination VGPRs without waiting for the load to finish.

GCN / GFX6 hazards

VINTRP followed by a read with ``v_readfirstlane`` or ``v_readlane``

It's required to insert 1 wait state if the dst VGPR of any ``v_interp_*`` is followed by a read with ``v_readfirstlane`` or ``v_readlane`` to fix GPU hangs on GFX6. Note that ``v_writelane_*`` is apparently not affected. This hazard isn't documented anywhere but AMD confirmed it.

RDNA / GFX10 hazards

SMEM store followed by a load with the same address

We found that an ``s_buffer_load`` will produce incorrect results if it is preceded by an ``s_buffer_store`` with the same address. Inserting an ``s_nop`` between them does not mitigate the issue, so an ``s_waitcnt lgkmcnt(0)`` must be inserted. This is not mentioned by LLVM among the other GFX10 bugs, but LLVM doesn't use SMEM stores, so it's not surprising that they didn't notice it.

VMEMtoScalarWriteHazard

Triggered by:

VMEM/FLAT/GLOBAL/SCRATCH/DS instruction reads an SGPR (or EXEC, or M0). Then, a SALU/SMEM instruction writes the same SGPR.

Mitigated by:

A VALU instruction or an ``s_waitcnt`` between the two instructions.

SMEMtoVectorWriteHazard

Triggered by:

An SMEM instruction reads an SGPR. Then, a VALU instruction writes that same SGPR.

Mitigated by:

Any non-SOPP SALU instruction (except ``s_setvskip``, ``s_version``, and any non-`lgkmcnt`` ``s_waitcnt``).

Offset3fBug

Any branch that is located at offset 0x3f will be buggy. Just insert some NOPs to make sure no branch is located at this offset.

InstFwdPrefetchBug

According to LLVM, the `s\_inst\_prefetch` instruction can cause a hang on GFX10. Seems to be resolved on GFX10.3+. There are no further details.

LdsMisalignedBug

When there is a misaligned multi-dword FLAT load/store instruction in WGP mode, it needs to be split into multiple single-dword FLAT instructions.

ACO doesn't use FLAT load/store on GFX10, so is unaffected.

FlatSegmentOffsetBug

The 12-bit immediate OFFSET field of FLAT instructions must always be 0. GLOBAL and SCRATCH are unaffected.

ACO doesn't use FLAT load/store on GFX10, so is unaffected.

VcmpxPermlaneHazard

Triggered by:

Any permlane instruction that follows any VOPC instruction which writes exec.

Mitigated by: any VALU instruction except `v\_nop`.

VcmpxExecWARHazard

Triggered by:

Any non-VALU instruction reads the EXEC mask. Then, any VALU instruction writes the EXEC mask.

Mitigated by:

A VALU instruction that writes an SGPR (or has a valid SDST operand), or `s\_waitcnt\_depctr 0xfffe`. Note: `s\_waitcnt\_depctr` is an internal instruction, so there is no further information about what it does or what its operand means.

LdsBranchVmemWARHazard

Triggered by:

VMEM/GLOBAL/SCRATCH instruction, then a branch, then a DS instruction, or vice versa: DS instruction, then a branch, then a VMEM/GLOBAL/SCRATCH instruction.

Mitigated by:

Only `s\_waitcnt\_vscnt null, 0`. Needed even if the first instruction is a load.

NSAClauseBug

"MIMG-NSA in a hard clause has unpredictable results on GFX10.1"

NSAMaxSize5

NSA MIMG instructions should be limited to 3 dwords before GFX10.3 to avoid stability issues: <https://reviews.llvm.org/D103348>

RDNA2 / GFX10.3 hazards

SALU EXEC write followed by NSA MIMG instruction

Triggered-by:

Potential stability issues can occur if an SALU instruction changes exec from 0 to non-zero immediately before an NSA MIMG instruction with 4+ dwords.

Mitigated-by: Any instruction, including `s\_nop`.

RDNA3 / GFX11 hazards

VcmpxPermlaneHazard

Same as GFX10.

LdsDirectVALUHazard

Triggered by:

LDSDIR instruction writing a VGPR soon after it's used by a VALU instruction.

Mitigated by:

A vdst wait, preferably using the LDSDIR's field.

LdsDirectVMEMHazard

Triggered by:

LDSDIR instruction writing a VGPR after it's used by a VMEM/DS instruction.

Mitigated by:

Waiting for the VMEM/DS instruction to finish, a VALU or export instruction, or
`s\_waitcnt\_depctr 0xffe3`.

VALUTransUseHazard

Triggered by:

A VALU instruction reading a VGPR written by a transcendental VALU instruction without 6+ VALU or 2+ transcendental instructions in-between.

Mitigated by:

A va_vdst=0 wait: `s\_waitcnt\_deptr 0xffff`

VALUPartialForwardingHazard

Triggered by:

A VALU instruction reading two VGPRs: one written before an exec write by SALU and one after. To trigger, there must be less than 3 VALU between the first and second VGPR writes and less than 5 VALU between the second VGPR write and the current instruction.

Mitigated by:

A va_vdst=0 wait: `s\_waitcnt\_deptr 0xffff`

VALUMaskWriteHazard

Triggered by:

SALU or VALU writing then SALU or VALU reading a SGPR that was previously used as a lane mask for a VALU when using wave64.

Mitigated by:

A VALU instruction reading a non-exec SGPR before the SGPR write, or a wait after the write: `s\_waitcnt\_depctr 0xfffe` for SALU, `s\_waitcnt\_depctr 0xf1ff` for non-VCC VALU and `s\_waitcnt\_depctr 0xfffd` for VCC VALU.

RDNA4 / GFX12 hazards

VcmpxPermlaneHazard

Same as GFX10

LdsDirectVALUHazard

LdsDirectVMEMHazard

Same as GFX11

VALUReadSGPRHazard

Triggered by:

VALU reads an SGPR, then written by SALU cannot safely be read by SALU or VALU, or
VALU reads an SGPR, then written by VALU cannot safely be read by VALU.

Mitigated by:

After the SALU write a sa_sdast=0 wait. After the VALU write a va_sdast=0 / va_vcc=0 wait.
It does not reset the first step.

dirtycode/mesa/src/amd/compiler/README.md

Welcome to ACO

ACO (short for *AMD compiler*) is a back-end compiler for AMD GCN / RDNA GPUs, based on the NIR compiler

Simply put, ACO translates shader programs from the NIR intermediate representation into a GCN / RDNA binary.

Motivation

Why did we choose to develop a new compiler backend?

1. We'd like to give gamers a fluid, stutter-free experience, so we prioritize compilation speed.
2. Good divergence analysis allows us to better optimize runtime performance.
3. Issues can be fixed within mesa releases, independently of the schedule of other projects.

Control flow

Modern GPUs are SIMD machines that execute the shader in parallel.

In case of GCN / RDNA the parallelism is achieved by executing the shader on several waves, and each wave has a uniform control flow. When every lane executes exactly the same instructions, and takes the same path, it's uniform control flow. Otherwise when some lanes take one path while other lanes take a different path, it's divergent.

Each hardware lane corresponds to a shader invocation from a software perspective.

The hardware doesn't directly support divergence, so in case of divergent control flow, the GPU must execute both code paths, each with some lanes disabled. This is why divergence is a performance concern in shader programming.

ACO deals with divergent control flow by maintaining two control flow graphs (CFG):

- * logical CFG - directly translated from NIR and shows the intended control flow of the program.
- * linear CFG - created according to Whole-Function Vectorization by Ralf Karrenberg and Sebastian Hack. The linear CFG represents how the program is physically executed on GPU and may contain additional blocks. Note that all nodes of the logical CFG also participate in the linear CFG, but not vice versa.

Compilation phases

Instruction Selection

The instruction selection is based around the divergence analysis and works in 3 passes on the NIR shader.

1. The divergence analysis pass calculates for each SSA definition if its value is guaranteed to be uniform across all lanes.
2. We determine the register class for each SSA definition.
3. Actual instruction selection. The advanced divergence analysis allows for better usage of the scalar units.

We have two types of instructions:

- * Hardware instructions as specified by the GCN / RDNA instruction set architecture manuals.
- * Pseudo instructions which are helpers that encapsulate more complex functionality. They eventually get lowered to real hardware instructions.

Each instruction can have operands (temporaries that it reads), and definitions (temporaries that it writes). Temporaries can be fixed to a specific register, or just specify a register class (either a single register or a vector of registers).

Repair SSA

This repairs SSA in the case of mismatches between the logical and linear CFG, where the definition of a variable is not unique.

Instruction selection might create mismatches between the logical CFG (the input NIR's CFG) and the linear CFG. We add a break at the end of a loop in case it has no active invocations (an empty exec can prevent any further instructions from being selected). We add an empty exec skip over a block. This is a branch which skips most contents of a sequence of instructions. An SGPR is defined in one side of a divergent IF but it used in or after the merge block. If the other side of the IF defines the SGPR, we need to move the definition to the merge block.

Lower Phi

After instructions selection, some phi instructions need further lowering. This includes booleans which are not supported by hardware.

Lower Subword

For GFX6 and GFX7, this pass already lowers subword pseudo instructions.

Value Numbering

The value numbering pass is necessary for two reasons: the lack of descriptor load representation in NIR, and every NIR instruction that gets emitted as multiple ACO instructions also has potential for CSE. This pass does dominator-tree value numbering.

Optimization

In this phase, simpler instructions are combined into more complex instructions (like the different versions). Exactly which optimizations are performed depends on the hardware for which the shader is being compiled. After this, `repair_ssa` needs to be run again in case it moves a SGPR use to a different block.

Setup of reduction temporaries

This pass is responsible for making sure that register allocation is correct for reductions, by adding temporaries. When a temporary has a linear VGPR register class, this means that the variable is considered **live** in the block.

Insert exec mask

In the GCN/RDNA architecture, there is a special register called ``exec`` which is used for manually controlling the execution mask.

Live-Variable Analysis

A live-variable analysis is used to calculate the register need of the shader. This information is used for spilling and scheduling before register allocation.

Spilling

First, we lower the shader program to CSSA form. Then, if the register demand exceeds the global limit, this pass lowers register usage by temporarily storing values to memory.

Instruction Scheduling

Scheduling is another NP-complete problem where basically all known heuristics suffer from unpredictable performance.

Register Allocation

The register allocator works on SSA (as opposed to LLVM's which works on virtual registers). The SSA property is maintained throughout the pass.

Optimization (post-RA)

Optimizations which depend on register assignment (like branching on VCCZ) are performed.

SSA Elimination

The next step is a pass out of SSA by inserting parallelcopies at the end of blocks to match the phi nodes.

Lower to HW instructions

Most pseudo instructions are lowered to actual machine instructions. These are mostly parallel copy instructions created by instruction selection or register allocation and some other instructions.

Lower Branches

Pseudo-Branch instructions are either lowered to machine instructions or removed. This pass also removes some other instructions.

VOPD Scheduling

This pass makes use of the VOPD instruction encoding on GFX11+. When using wave32 mode, this pass works on a per-wave basis.

ILP Scheduling

This second scheduler works on registers rather than SSA-values to determine dependencies. It implements a simple ILP scheduler.

Insert wait states

GCN requires some wait states to be manually inserted in order to ensure correct behavior on memory instructions. This means that we need to insert ``s_waitcnt`` instructions (and its variants) so that the shader program can execute correctly.

Resolve hazards and insert NOPs

Some instructions require wait states or other instructions to resolve hazards which are not handled by the hardware. This pass makes sure that no known hazards occur.

Insert delay_alu and form clauses

These passes introduce optional instructions which provide performance hints to the hardware. ``s_delay_alu`` and ``form`` clauses.

Emit program - Assembler

The assembler emits the actual binary that will be sent to the hardware for execution. ACO's assembler is used for this.

Supported shader stages

Hardware stages (as executed on the chip) don't exactly match software stages (as defined in OpenGL / Vulkan). Which software stage gets executed on which hardware stage depends on what kind of software stages are present.

An important difference is that VS is always the first stage to run in SW models, whereas HW VS refers to the last HW stage before fragment shading in GCN/RDNA terminology. That's why, among other things, the HW VS is no longer used to execute the SW VS when tessellation or geometry instancing is used.

Glossary of software stages

- * VS = Vertex Shader
- * TCS = Tessellation Control Shader, equivalent to D3D HS = Hull Shader
- * TES = Tessellation Evaluation Shader, equivalent to D3D DS = Domain Shader
- * GS = Geometry Shader
- * FS = Fragment Shader, equivalent to D3D PS = Pixel Shader
- * CS = Compute Shader
- * TS = Task Shader
- * MS = Mesh Shader

Glossary of hardware stages

- * LS = Local Shader (merged into HS on GFX9+), only runs SW VS when tessellation is used
- * HS = Hull Shader, the HW equivalent of a Tessellation Control Shader, runs before the fixed function hardware
- * ES = Export Shader (merged into GS on GFX9+), if there is a GS in the SW pipeline, the preceding stage
- * GS = Geometry Shader, also known as legacy GS
- * VS = Vertex Shader, **not equivalent to SW VS**: when there is a GS in the SW pipeline this stage runs the SW VS
- * NGG = Next Generation Geometry, a new hardware stage that replaces legacy HW GS and HW VS on RDNA GPUs
- * PS = Pixel Shader, the HW equivalent to SW FS
- * CS = Compute Shader

Notes about HW VS and the "GS copy" shader

HW PS reads its inputs from a special ring buffer called Parameter Cache (PC) that only HW VS can write to. However, legacy GS store their output in VRAM (before GFX10/NGG). So in order for HW PS to be able to read the GS outputs, we must run something on the VS stage which reads from VRAM and exports them to the PC. This is what we call a "GS copy" shader. From a HW perspective the "GS copy" shader is in fact VS (it runs on the HW VS stage), but from a SW perspective it's not part of the traditional pipeline, it's just some "glue code" that we need for outputs to play nicely.

On GFX10/NGG this limitation no longer exists, because NGG can export directly to the PC.

Notes about the attribute ring

Starting with GFX11, the parameter cache is replaced by the attribute ring, which is a discardable ring buffer located in VRAM. The outputs of the last pre-rasterization stage (VS, TES, GS or MS) are stored here.

The attribute ring is designed to utilize the Infinity Cache. Store instructions are arranged so that each instruction writes a full cache line, so the GPU will never actually have to write any of that to VRAM.

Notes about merged shaders

The merged stages on GFX9 (and GFX10/legacy) are: LSHS and ESGS. On GFX10/NGG the ESGS is merged with HW VS.

This might be confusing due to a mismatch between the number of invocations of these shaders.

For example, ES is per-vertex, but GS is per-primitive.

This is why merged shaders get an argument called `merged\_wave\_info` which tells how many invocations each shader has. So, think about these as two independent shader programs slapped together.

Which software stage runs on which hardware stage?

Graphics Pipeline

GFX6-8:

- * Each SW stage has its own HW stage
- * LS and HS share the same LDS space, so LS can store its output to LDS, where HS can read it
- * HS, ES, GS outputs are stored in VRAM, next stage reads these from VRAM
- * GS outputs got to VRAM, so they have to be copied by a GS copy shader running on the HW VS stage

| GFX6-8 HW stages: | LS | HS | ES | GS | VS | PS | ACO terminology |
|------------------------|----|-----|-----|----|---------|----|--|
| SW stages: only VS+PS: | | | | | VS | FS | `vertex_vs`, `fragment_fs` |
| with tess: | VS | TCS | | | TES | FS | `vertex_ls`, `tess_control_hs`, `tess_eval_vs` |
| with GS: | | | VS | GS | GS copy | FS | `vertex_es`, `geometry_gs`, `gs_copy_vs` |
| with both: | VS | TCS | TES | GS | GS copy | FS | `vertex_ls`, `tess_control_hs`, `tess_eval_vs` |

GFX9+ (including GFX10/legacy):

- * HW LS and HS stages are merged, and the merged shader still uses LDS in the same way as before
- * HW ES and GS stages are merged, so ES outputs can go to LDS instead of VRAM
- * LSHS outputs and ESGS outputs are still stored in VRAM, so a GS copy shader is still necessary

| GFX9+ HW stages: | LSHS | ESGS | VS | PS | ACO terminology |
|------------------------|----------|----------|---------|----|---|
| SW stages: only VS+PS: | | | VS | FS | `vertex_vs`, `fragment_fs` |
| with tess: | VS + TCS | | TES | FS | `vertex_tess_control_hs`, `tess_eval_vs` |
| with GS: | | VS + GS | GS copy | FS | `vertex_geometry_gs`, `gs_copy_vs`, `fragment_fs` |
| with both: | VS + TCS | TES + GS | GS copy | FS | `vertex_tess_control_hs`, `tess_eval_vs` |

NGG (GFX10+ only):

- * HW GS and VS stages are now merged, and NGG can export directly to PC
- * GS copy shaders are no longer needed
- * On GFX10.3+, per-primitive attributes (parameters) are also supported
- * On GFX11+, parameter exports are replaced by attribute ring stores

| GFX10/NGG HW stages: | LSHS | NGG | PS | ACO terminology |
|------------------------|----------|----------|----|--|
| SW stages: only VS+PS: | | VS | FS | `vertex_ngg`, `fragment_fs` |
| with tess: | VS + TCS | TES | FS | `vertex_tess_control_hs`, `tess_eval_vs` |
| with GS: | | VS + GS | FS | `vertex_geometry_ngg`, `fragment_fs` |
| with both: | VS + TCS | TES + GS | FS | `vertex_tess_control_hs`, `tess_eval_vs` |

Mesh Shading Graphics Pipeline

GFX10.3+:

- * TS will run as a CS and stores its output payload to VRAM
- * MS runs on NGG, loads its inputs from VRAM and stores outputs to LDS, then PC (or attribute ring)
- * Pixel Shaders work the same way as before

| GFX10.3+ HW stages | CS | NGG | PS | ACO terminology |
|------------------------|----|-----|----|--------------------------------------|
| SW stages: only MS+PS: | | MS | FS | `mesh_ngg`, `fragment_fs` |
| with task: | TS | MS | FS | `task_cs`, `mesh_ngg`, `fragment_fs` |

Compute pipeline

GFX6-10:

- * Note that the SW CS always runs on the HW CS stage on all HW generations.

| GFX6-10 HW stage | CS | ACO terminology |
|------------------|----|-----------------|
| SW stage | CS | `compute_cs` |

How to debug

Handy `RADV\_DEBUG` options that help with ACO debugging:

- * `nocache` - you always want to use this when debugging, otherwise you risk using a broken shader from t
- * `shaders` - makes ACO print the IR after register allocation, as well as the disassembled shader binary
- * `metashaders` - does the same thing as `shaders` but for built-in RADV shaders.
- * `preoptir` - makes ACO print the final NIR shader before instruction selection, as well as the ACO IR a
- * `nongg` - disables NGG support

We also have `ACO\_DEBUG` options:

- * `validateir` - Validate the ACO IR between compilation stages. By default, enabled in debug builds and
- * `validatera` - Perform a RA (register allocation) validation.
- * `force-waitcnt` - Forces ACO to emit a wait state after each instruction when there is something to wait

- * ``novn`` - Disables the ACO value numbering stage.
- * ``noopt`` - Disables the ACO optimizer.
- * ``nosched`` - Disables the ACO pre-RA and post-RA scheduler.
- * ``nosched-ilp`` - Disables the ACO post-RA ILP scheduler.

Note that you need to **combine these options into a comma-separated list**, for example: ``RADV_DEBUG=nocache,shaders ACO_DEBUG=validateir,validatera vkcube``

...

`RADV_DEBUG=nocache,shaders ACO_DEBUG=validateir,validatera vkcube``

...

Using GCC sanitizers

GCC has several sanitizers which can help figure out hard to diagnose issues. To use these, you need to pass the ``-Dbsanitizer`` flag to ``meson`` when building mesa. For example ``-Dbsanitizer=undefined`` will add support for the undefined behavior sanitizer.

Hardened builds and glibc++ assertions

Several Linux distributions use "hardened" builds meaning several special compiler flags are added by downstream packaging which are not used in mesa builds by default. These may be responsible for some bug reports of inexplicable crashes with assertion failures you can't reproduce.

Most notable are the glibc++ debug flags, which you can use by adding the ``-D_GLIBCXX_ASSERTIONS=1`` and ``-D_GLIBCXX_DEBUG=1`` flags.

To see the full list of downstream comp

... (truncated, file is very long)

dirtycode/mesa/src/amd/compiler/tests/README.md

Tests are wrapped in a ``BEGIN_TEST`/`END_TEST`` and write data to the ``output`` file pointer. Tests have the following syntax:

- ``!`` fails the test if the current line does not match the pattern
- ``>>`` skips to the first line which matches the pattern, or fails the test if there is none
- ``;`` executes python code to extend the pattern syntax by inserting functions into the variable dictionary

Before this prefix, there can be a ``~`` to only perform the check for certain variants (a regex directly following the ``~`` is used).

Pattern Syntax

Patterns can define variables which can be accessed in both python code and the pattern itself. These are defined as follows:

- ``\`` can be used to match the following literal character without interpreting it.
- Most characters expect the same characters in the output.
- A sequence of spaces in the pattern expects a sequence of spaces or tabs in the output.
- A ``#`` in the pattern expects an unsigned integer in the output. The ``#`` can be followed by an identifier.
- A ``$`` in the pattern stores the output until the first whitespace character into a variable.
- A ``%`` in the pattern followed by an identifier is the same as a ``#`` but it expects a ``%`` before the identifier.
- A ``@`` calls a variable as a function. It can be followed by an argument string wrapped in ``(`` and ``)``.

Functions

- ``s64``, ``s96``, ``s128``, ``v2``, ``v3``, etc, expand to a pattern which matches a disassembled instruction's output.
- ``match_func`` expands to a sequence of ``$`` and inserts functions with expand to the extracted output
- ``search_re`` consumes the rest of the line and fails the test if the pattern is not found

dirtycode/mesa/src/amd/drm-shim/README.md

radeon_noop backend

This implements the minimum of the radeon kernel driver in order to make shader-db work. The submit ioctl is stubbed out to not execute anything.

Export ``MESA_LOADER_DRIVER_OVERRIDE=r300``

``LD_PRELOAD=$prefix/lib/libradeon_noop_drm_shim.so``. (or ``r600`` for ``r600``-class HW)

By default, ``rv515`` is exposed. The chip can be selected an environment variable like ``RADEON_GPU_ID=CAYMAN`` or ``RADEON_GPU_ID=0x6740``.

amdgpu_noop backend

This implements the minimum of the amdgpu kernel driver. The submit ioctl is stubbed out to not execute anything.

Export ``LD_PRELOAD=$prefix/lib/libamdgpu_noop_drm_shim.so``.

To specify the device to expose, set the environment variable ``AMDGPU_GPU_ID`` to

- ``renoir`` to expose a ``CHIP_RENOIR`` device
- ``raven`` to expose a ``CHIP_RAVEN`` device
- ``stoney`` to expose a ``CHIP_STONEY`` device

Further names follow the ``CHIP_*`` enum values. By default, the ``CHIP_RENOIR`` device is exposed.

To add a new device, ``amdgpu_devices.c`` needs to be modified. ``amdgpu_dump_states`` can be used to dump the relevant states from a real device.

dirtycode/mesa/src/amd/gmlib/README.md

GMLib

dirtycode/mesa/src/amd/vpelib/README.md

Video Processor Engine (VPE) High Level Documentation

Overview

VPE is a hardware pipeline that does baseline video processing from DRAM to DRAM.

The goal of the VPE is to reduce the dependency on graphics engine and therefore saving the graphic engine.

It is capable of doing pixel re-gamma, scaling, CSC, gamut remap, tone mapping, programmable gamma. Future

There are three implementations of VPE:

1 - Single pipeline (VPE 1.0)

![image-20240718113342738](./asset/VPE10Diagram.png)

2 - Dual independent pipeline (VPE 1.1)

![image-20240718113445574](./asset/VPE11Diagram.png)

The VPE hardware is controlled by the VPE driver encapsulated in the VPelib library.

VPelib API

The VPE hardware pipeline is controlled by a dedicated firmware. This firmware expects a series of commands

Ring Buffer at the current implementation stage is maintained by Kernel mode driver (KMD), and ring buffer

User Mode Driver (UMD)'s responsibility at the current stage is to construct the command buffer. With VPelib

Once constructed, the vpelib's caller will submit the command buffer to the Engine's queue, KMD will consume

Once a Hardware queue is enabled, the VPE Firmware will fetch the commands from the ring buffer one by one

The mechanism is presented below:

![image-20240718113445574](./asset/vpe_components.png)

The main functions of the VpeLib API are:

vpe_create

...

/** @brief Create the VPE lib instance.

*

* Caller provides the current ASIC info,

* logging and system memory APIs.

```

* It initializes all the necessary resources for the asic
* and returns the general capabilities of the engines.
*
* For capabilities based on conditions,
* shall be done by vpe->cap_funcs.*
*
*
* @param[in] params provide the asic version, APIs for logging and memory
* @return         vpe instance if supported. NULL otherwise
*/
struct vpe *vpe_create(const struct vpe_init_data *params);
```



```

### vpe_destroy
```
/**
 * @brief Destroy the VPE lib instance and resources
 *
 * @param[in] vpe the vpe instance created by vpe_create
 */
void vpe_destroy(struct vpe **vpe);
```

### vpe_check_support
```
/**
 * @brief Check if the VPE operation is supported.
 *
 * Caller must call this to check if the VPE supports
 * the requested operation before calling vpe_build_commands().
 * If operation is supported, it returns the memory requirement.
 *
 * The caller has to prepare those required memories
 * and pass them to the vpe_build_commands().
 *
 * @param[in] vpe vpe instance returned by vpe_initialize()
 * @param[in] param build params
 * @param[out] req memory required for the command buffer and
 * embedded data if return VPE_OK.
 * caller has to alloc them and provide it to build_vpbuilds API.
 * @return VPE_OK if supported
 */
enum vpe_status vpe_check_support(struct vpe *vpe, const struct vpe_build_param *param, struct vpe_bufs_r
```



```

vpe_build_commands
```
/**
 * @brief Build the command descriptors for the given param.
 * caller must call vpe_check_support() before this function,
 * unexpected result otherwise.
 *
 * @param[in] vpe vpe instance created by vpe_create()
 * @param[in] param vpe build params
 * @param[in,out] bufs [in] memory allocated for the command buffer, embedded buffer and 3dlut.
 * If size is 0, it reports the required size for this checked
 * operation. [out] the next write address and the filled sizes.
 * @return status
 */
enum vpe_status vpe_build_commands(
    struct vpe *vpe, const struct vpe_build_param *param, struct vpe_build_bufs *bufs)
```

```


```


```

This function can be used to query the size requirement of command buffer and embedded buffer by setting the buffer size to 0.

With the proper buffer size, this API will start to construct the command buffer and embedded buffer content.

```

vpe_build_noops
```
/**

```

```

* @brief Build the command descriptors for No-Op operation
* @param[in]     vpe          vpe instance created by vpe_create()
* @param[in]     num_dwords   number of noops
* @param[in,out] ppcmd_space  in: dword aligned command buffer start address
*                          out: dword aligned next good write address
* @return status
*/
enum vpe_status vpe_build_noops(struct vpe *vpe, uint32_t num_dwords, uint32_t **ppcmd_space);

```

VPE Tutorial - bypass RGB TO RGB

In order to process an input image bypass to output, one has to do the followings:

- establish the input image array
- set the source image dimensions
- set the color space of the input image (RGB)
- create an input texture with the above parameters
- set the source destination dimensions , same like the input size
- set the destination color format (RGB, same like input)
- Set the linearity of the output the same like the input
- Add alpha channel if that is the case
- trigger the VPBlit which will use the VPE hardware

In order to accomplish the aboves, the caller needs to create an instance of VPE first.

```

...
struct vpe *vpe_create(const struct vpe_init_data *params);
...

```

Then check if the created VPE instance supports the given parametrs.

```

...
enum vpe_status vpe_check_support(
    struct vpe *vpe, const struct vpe_build_param *param, struct vpe_bufs_req *req)
...

```

The input is of type vpe_build_params which states the parameters for input and output.

```

...
struct vpe_build_param {
    uint32_t          num_streams;    /**< Number of source streams */
    struct vpe_stream *streams;       /**< List of input streams */
    struct vpe_surface_info dst_surface; /**< Destination/Output surface */
    struct vpe_rect    target_rect;   /**< rectangle in target surface to be blt'd.
                                     Ranges out of target_rect will not be touched */
    struct vpe_color    bg_color;     /**< Background Color */
    enum vpe_alpha_mode alpha_mode;    /**< Alpha Mode. Output alpha in the output
                                     surface */
    struct vpe_hdr_metadata hdr_metadata; /**< HDR Metadata */
    struct vpe_reserved_param dst_reserved_param; // data flags
    struct {
        uint32_t hdr_metadata : 1;
        uint32_t reserved     : 31;
    } flags;
...

```

For this example, set num_streams to a one and provide a single stream for the streams parameter.

The destination surface must be populated with these parameters. The most important are the destination a

For this example, destination format and color space should be the same as the input format and color spa

```

...
struct vpe_surface_info {
    struct vpe_plane_address address;    /**< Address */
    enum vpe_swizzle_mode_values swizzle; /**< Swizzle mode */
    struct vpe_plane_size plane_size;    /**< Pitch */
    struct vpe_plane_dcc_param dcc;
    enum vpe_surface_pixel_format format; /**< Surface pixel format */
    struct vpe_color_space cs;           /**< Surface color space */
};

```

...

Plane size should be populated accordingly. As this test is RGB to RGB, ignore the YUV fields.

...

```
struct vpe_plane_size {
    struct vpe_rect surface_size;    /**< Plane rectangle */
    struct vpe_rect chroma_size;     /**< Chroma plane rectangle for semi-planar YUV formats */
    uint32_t        surface_pitch;   /**< Horizontal pitch alignment of the plane in pixels */
    uint32_t        chroma_pitch;    /**< Horizontal pitch alignment of the chroma plane for
                                     semi-planar YUV formats in pixels */
    uint32_t surface_aligned_height; /**< Vertical alignment of the plane in pixels */
    uint32_t chrome_aligned_height;  /**< Vertical alignment of the chroma plane for semi-planar
                                     YUV formats in pixels */
};
```

For this example, `struct vpe\_rect target\_rect` should be identical with the input/output size.

After calling the support call, the caller needs to call

```
enum vpe_status vpe_build_commands(
    struct vpe *vpe, const struct vpe_build_param *param, struct vpe_build_bufs *bufs);
```

with the same parameters provided to the support call to build the command descriptor.

After the job is done and no more jobs for the VPE, call `vpe\_destroy(struct vpe \*\*vpe)` to free the allocated resources.

Generate API Doxygen Documentation

By Compiling and building VPE with MSVS, the Doxygen documentation of the public API will automatically generate.

To manually generate the Doxygen documentation without building VPE, create a Doxygen config file similar to the following.

dirtycode/mesa/src/amd/vulkan/bvh/README.md

GFX12

GFX12 introduces a new BVH encoding for the `image_bvh_dual_intersect_ray` and `image_bvh8_intersect_ray` intersection functions.

BVH8 box node

| bitsize/range | name | description |
|---------------|-------------------------------------|--|
| 32 | <code>internal_child_offset</code> | Offset of child BVH8 box nodes in units of 8 bytes. |
| 32 | <code>primitive_child_offset</code> | Offset of child primitive nodes in units of 8 bytes. |
| 32 | <code>unused</code> | Used by amdvlk for storing the parent node ID. |
| 32 | <code>origin_x</code> | x-offset applied to all child AABBs. |
| 32 | <code>origin_y</code> | y-offset applied to all child AABBs. |
| 32 | <code>origin_z</code> | z-offset applied to all child AABBs. |
| 8 | <code>exponent_x</code> | |
| 8 | <code>exponent_y</code> | |
| 8 | <code>exponent_z</code> | |
| 4 | <code>unused</code> | |
| 4 | <code>child_count_minus_one</code> | |
| 32 | <code>obb_matrix_index</code> | Selects a matrix for transforming the ray before performing intersection test. |
| 96x8 | <code>children[8]</code> | |

`children[8]` element layout:

| bitsize/range | name | description |
|---------------|-------------------------|---|
| 12 | <code>min_x</code> | Fixed point child AABB coordinate. |
| 12 | <code>min_y</code> | |
| 4 | <code>cull_flags</code> | |
| 4 | <code>unused</code> | |
| 12 | <code>min_z</code> | |
| 12 | <code>max_x</code> | |
| 8 | <code>cull_mask</code> | |
| 12 | <code>max_y</code> | |
| 4 | <code>node_type</code> | |
| 4 | <code>node_size</code> | Increment for the child offset in units of 128 bytes. |

The coordinates of child AABBs are encoded as follows:

- min: `floor((x - origin_x) / extent)`

```
- max: `ceil((x - origin_x) / extent) - 1`
```

image_bvh8_intersect_ray will return the node IDs of the child nodes.

Primitive node

Highlevel layout:

| bitsize/range | name | description |
|---------------|-----------------------|--|
| 52 | `header` | Misc information about this node. |
| | `vertex_prefixes[3]` | |
| | `data` | Compressed vertex positions followed by primitive/geometry index data. |
| 29x | `triangle_pair_count` | `pair_desc[triangle_pair_count]` Misc information about a triangle pair. |

`header` layout:

| bitsize/range | name | description |
|---------------|----------------------------------|--|
| 5 | `x_vertex_bits_minus_one` | |
| 5 | `y_vertex_bits_minus_one` | |
| 5 | `z_vertex_bits_minus_one` | |
| 5 | `trailing_zero_bits` | |
| 4 | `geometry_index_base_bits_div_2` | |
| 4 | `geometry_index_bits_div_2` | |
| 3 | `triangle_pair_count_minus_one` | |
| 1 | `vertex_type` | |
| 5 | `primitive_index_base_bits` | |
| 5 | `primitive_index_bits` | |
| 10 | `indices_midpoint` | Bit offset where the geometry and primitive indices start (geometry indices is |

The `data` field is split in three sections:

1. Vertex data, this is a list of floats which share the same prefix and the same number of trailing zero bits. The decompressed value (for example the x component of a vertex) is
(prefix << 32 - prefix_bits_x) | read(x_vertex_bits) << trailing_zero_bits` where `prefix_bits_x` is
`x\_vertex\_bits` and `trailing\_zero\_bits`
(`32 - x\_vertex\_bits - trailing\_zero\_bits`).
2. Geometry indices.
3. Primitive indices.

Geometry indices are encoded the same way with the only difference being that geometry indices are read/w

```
- `*_index = read(*_index_bits)` if `*_index_bits >= *_index_base_bits`  
- `*_index = read(*_index_bits) | (*_index_base & ~BITFIELD_MASK(*_index_bits))` otherwise.
```

`pair\_desc(s)` layout:

| bitsize/range | name | description |
|---------------|------------------------------|--|
| 1 | `prim_range_stop` | |
| 1 | `tri1_double_sided` | |
| 1 | `tri1_opaque` | |
| 4 | `tri1_v0_index` | Indices into `data`, `0xF` for procedural nodes. |
| 4 | `tri1_v1_index` | `0xF` for procedural nodes. |
| 4 | `tri1_v2_index` | |
| | `tri0` has identical fields: | |
| 1 | `tri0_double_sided` | |
| 1 | `tri0_opaque` | |
| 4 | `tri0_v0_index` | |
| 4 | `tri0_v1_index` | |
| 4 | `tri0_v2_index` | |

image_bvh8_intersect_ray will return the following data for triangle nodes:

| VGPR index | value |
|------------|--------------------------------------|
| 0 | t0 |
| 1 | (procedural0 << 31) \ u0` |
| 2 | (opaque0 << 31) \ v0` |
| 3 | (primitive_index0 << 1) \ backface0` |
| 4 | t1 |
| 5 | (procedural1 << 31) \ u1` |
| 6 | (opaque1 << 31) \ v1` |

```
7	`(primitive_index1 << 1) \	backface1`
8	`(geometry_index0 << 2) \	navigation_bits`
9	`(geometry_index1 << 2) \	navigation_bits`
```

image_bvh8_intersect_ray will return the following data for procedural nodes:

```
VGPR index	value	
3	`primitive_index0 << 1`	
8	`(geometry_index0 << 2) \	navigation_bits`
9	`(geometry_index1 << 2) \	navigation_bits`
```

`navigation\_bits` is 0 if there are more triangle pairs to process, 1 if this was the last triangle pair

Instance node

```
bitsize/range	name	description
32x3x4	`world_to_object`	
62	`bvh_addr`	Units of 4 bytes.
1	`aabbs`	Does the BLAS (only) contain AABBs? Used for pointer flag based culling.
1	`unused`	
32	`unused`	
24	`user_data`	Returned by the intersect instruction for instance nodes.
8	`cull_mask`	
The instance node can have up to 4 quantized child nodes:		
32	`origin_x`	x-offset applied to all child AABBs.
32	`origin_y`	y-offset applied to all child AABBs.
32	`origin_z`	z-offset applied to all child AABBs.
8	`exponent_x`	
8	`exponent_y`	
8	`exponent_z`	
4	`unused`	
4	`child_count_minus_one`	
96x4	`children[4]`	
```

image_bvh8_intersect_ray will return:

```
VGPR index	value
2	BLAS addr lo
3	BLAS addr hi
6	`user_data`
7	`(child_ids[0] & 0xFF) \
```

dirtycode/mesa/src/android_stub/README.md

The Android NDK doesn't come with enough of the platform libraries we need to build Mesa drivers out of tree, so android_stub has stub versions of those library that aren't installed which we link against, relying on the real libraries to be present when the Mesa driver is deployed.

dirtycode/mesa/src/asahi/compiler/README.md

Special registers

`r0l` is the hardware nesting counter.

`r1` is the hardware link register.

`r5` and `r6` are preloaded in vertex shaders to the vertex ID and instance ID.

ABI

The following section describes the ABI used by non-monolithic programs.

Vertex

Registers have the following layout at the beginning of the vertex shader (written by the vertex prolog):

- * `r0-r3` and `r7` undefined. This avoids preloading into the nesting counter or having unaligned values. The prolog is free to use these registers as temporaries.
- * `r4` is the zero-based vertex ID if the vertex shader is running as a hardware compute shader, useful to avoid a redundant special register read in the main shader. Undefined in hardware vertex shaders.
- * `r5-r6` retain their usual meanings, even if the vertex shader is running as a hardware compute shader. This allows software index fetch code to run in the prolog without contaminating the main shader key.
- * `r8` onwards contains 128-bit uniform vectors for each attribute. Accommodates 30 attributes without spilling, exceeding the 16 attribute API minimum. For 32 attributes, we will need to use function calls or the stack.

One useful property is that the GPR usage of the combined program is equal to the GPR usage of the main shader. The prolog cannot write higher registers than read by the main shader.

Vertex prologs do not have any uniform registers allocated for preamble optimization or constant promotion, as this adds complexity without any legitimate use case.

For a vertex shader reading n attributes, the following layout is used:

- * The first n 64-bit uniforms are the base addresses of each attribute.
- * The next n 32-bit uniforms are the associated clamps (sizes). Presently robustness is always used.
- * The next 2×32 -bit uniform is the base vertex and base instance. This must always be reserved because it is unknown at vertex shader compile-time whether any attribute will use instancing. Reserving also the base vertex allows us to push both conveniently with a single USC Uniform word.
- * The next 16-bit is the draw ID.
- * For a hardware compute shader, the next 48-bit is padding.
- * For a hardware compute shader, the next 64-bit uniform is a pointer to the input assembly buffer.

In total, the first $6n + 5$ 16-bit uniform slots are reserved for a hardware vertex shader, or $6n + 12$ for a hardware compute shader.

Fragment

When sample shading is enabled in a non-monolithic fragment shader, the fragment shader has the following register inputs:

- * `r0l = 0`. This is the hardware nesting counter.
- * `r1l` is the mask of samples currently being shaded. This usually equals to $1 \ll \text{sample ID}$, for "true" per-sample shading.

When sample shading is disabled, no register inputs are defined. The fragment prolog (if present) may clobber whatever registers it pleases.

Registers have the following layout at the end of the fragment shader (read by the fragment epilog):

- * `r0l = 0` if sample shading is enabled. This is implicitly true.
- * `r1l` preserved if sample shading is enabled.
- * `r2l` and `r3l` contain the emitted depth/stencil respectively, if depth and/or stencil are written by the fragment shader. Depth/stencil writes must be deferred to the epilog for correctness when the epilog can discard (i.e. when alpha-to-coverage is enabled).
- * `r3h` contains the logically emitted sample mask, if the fragment shader uses forced early tests. This predicates the epilog's stores.
- * The vec4 of 32-bit registers beginning at $r(4 * (i + 1))$ contains the colour output for render target i . When dual source blending is enabled, there is only a single render target and the dual source colour is treated as the second render target (registers r8-r11).

Uniform registers have the following layout:

- * u0_u1: 64-bit render target texture heap
- * u2...u5: Blend constant
- * u6_u7: Root descriptor, so we can fetch the 64-bit fragment invocation counter address and (OpenGL only) the 64-bit polygon stipple address

dirtycode/mesa/src/broadcom/drm-shim/README.md

v3d_noop backend

This implements the minimum of v3d in order to make shader-db and fossilize work. The submit ioctl is stubbed out to not execute anything.

```
Export `MESA_LOADER_DRIVER_OVERRIDE=v3d`  
LD_PRELOAD=$prefix/lib/libv3d_noop_drm_shim.so`.
```

By default the emulated GPU will be V3D 7.1 (rpi5). But it is possible to select a specific V3D GPU model by setting V3D_GPU_ID environment variable. The accepted values are:

- V3D_GPU_ID=71 (default): this emulates V3D 7.1 (rpi5) device.
- V3D_GPU_ID=42: this emulates V3D 4.2 (rpi4) device.

vc4_noop backend

This implements the minimum of vc4 in order to make shader-db work. The submit ioctl is stubbed out to not execute anything.

```
Export `MESA_LOADER_DRIVER_OVERRIDE=vc4`  
LD_PRELOAD=$prefix/lib/libvc4_noop_drm_shim.so`. This will be a VC4 2.1 device.
```

dirtycode/mesa/src/drm-shim/README.md

DRM shim - Fake GEM kernel drivers in userspace for CI

On CI systems where we don't control the kernel, it would be nice to be able to present either no-op GEM devices (for shader-db runs) or simulator-backed GEM devices (for testing against a software simulator or FPGA). This lets us do that by intercepting libc calls and exposing render nodes.

Limitations

- Doesn't know how to handle DRM fds getting passed over the wire from X11 (Could we use kmsro to support the X11 case?).
- libc interception is rather glibc-specific and fragile.
- Can easily break gdb if the libc interceptor code is what's broken. (ulimit -c unlimited and doing gdb on the core after the fact can help)

Using

You choose the backend by setting `LD\_PRELOAD` to the shim you want. Since this will effectively fake another DRM device to your system, you may need some work on your userspace to get your test application to use it if it's not the only DRM device present. Setting `DRM\_SHIM\_DEBUG=1` in the environment will print out what path the shim initialized on.

For piglit tests, you can set:

```
````  
PIGLIT_PLATFORM=gbm
WAFFLE_GBM_DEVICE=<path from DRM_SHIM_DEBUG>
````
```

See your drm-shim backend's README for details on how to use it.

dirtycode/mesa/src/etnaviv/drm-shim/README.md

etnaviv_noop backend

This implements the minimum of etnaviv in order to make shader-db work. The submit ioctl is stubbed out to not execute anything.

```
Export `MESA_LOADER_DRIVER_OVERRIDE=etnaviv`  
LD_PRELOAD=$prefix/lib/libetnaviv_noop_drm_shim.so`.
```

It is possible to select a specific GPU by setting the ETNA_SHIM_GPU environment variable to its identity, given in hex as ``model:revision[:product:customer:eco]``. The three trailing values default to zero, and the default identity is a GC2000. For example the GC7000 in an i.MX8QM is reached with ``ETNA_SHIM_GPU=7000:6009:70008``.

Any GPU in the hardware database can be selected this way. ``bin/drm-shim.py`` carries a name for the ones worth remembering one for, so

```
...
drm-shim.py gc7000-r6214 <command>
...
```

is the same thing without looking the numbers up. Querying an identity the hardware database does not know of results in an error: the shim describes only the identity of the GPU, so there is nothing left to fall back to.

dirtycode/mesa/src/freedreno/drm-shim/README.md

freedreno_noop backend

This implements the minimum of msm in order to make shader-db work. The submit ioctl is stubbed out to not execute anything.

Export ``MESA_LOADER_DRIVER_OVERRIDE=msm``
`LD_PRELOAD=$prefix/lib/libfreedreno_noop_drm_shim.so``.

By default, a630 is exposed. The chip can be selected an enviornment variable like ``FD_GPU_ID=307`` or ``FD_CHIP_ID=<hex>``.

dirtycode/mesa/src/gallium/frontends/d3d10umd/README.md

This directory has a Gallium state tracker for D3D10 UMD DDI.

It still uses TGSI, not not NIR.

Currently it only supports SW rasterizers. See `src/gallium/targets/d3d10umd/README.md` for further details.

dirtycode/mesa/src/gallium/frontends/mediafoundation/IDL/IDL_CHANGES.md

Making changes to IDL files on mediafoundation frontend

When changing any .idl on this folder, these must be recompiled with midlrt to produce the public, private

How to build the modified IDL files

```
...
midlrt.EXE <input_idl_file>.idl /metadata_dir C:\Windows\System32\WinMetadata
...
```

dirtycode/mesa/src/gallium/frontends/mediafoundation/docs/readme.md

DX12 Video Encode Media Foundation Transform (Microsoft Corporation (C) 2025)

Overview

This folder contains the implementation of an async media foundation transform (MFT) that uses DX12 video

Files

| Files | Description |
|--------------------------|--|
| ----- | ---- |
| codecapi.cpp | implements ICodecAPI interface |
| dpb_buffer_manager.cpp | utility for dpb buffer creation and deletion |
| encode.cpp | contains codec agnostic encode related code |
| encoder_capabilities.cpp | utility for querying hardware capability from pipe |
| encode_av1.cpp | encode specific to av1 (currently stub only) |
| encode_h264.cpp | encode specific to h264 |
| encode_hevc.cpp | encode specific to hevc |
| hmft_entrypoints.cpp | entry point to hmft |
| mfbufferhelp.cpp | utility for mf buffer |

| | |
|-----------------------------------|---|
| mfd3dmanager.cpp | utility for managing d3d device |
| mfmmediaeventgenerator.cpp | implements the IMFMediaEventGenerator interface |
| mfpipelineinterop.cpp | utility for various type translation function between MF and pipe |
| mfrealtimedclientex.cpp | implements the IMFRealTimeClientEx interface |
| mfshutdown.cpp | implements the IMFShutdown interface |
| mfttransform.cpp | implements the IMFTransform interface |
| reference_frames_tracker_av1.cpp | managing dbp and gop state for av1 (currently stub only) |
| reference_frames_tracker_h264.cpp | managing dbp and gop state for h264 |
| reference_frames_tracker_hevc.cpp | managing dbp and gop state for hevc |
| videobufferlock.cpp | utility for video buffer lock |
| wpptrace.cpp | wpp trace |
| wppconfig/ | folder containing wpp trace config |
| IDL/ | folder containing the IDL files for the MFT |
| test/ | folder containing the test code for the MFT |

Other files related to the MFT:

| | |
|-------------------------------------|---|
| Files | Description |
| ----- | ---- |
| src/gallium/targets/mediafoundation | dllmain and resourc files for MFT, builds the dll |

Build

Current implementation is designed to build three differt codec (AV1, H264, HEVC) with the same code (not). You can set up the build to build a single codec without comingling of other codec's code or you can build all. The comingle code currently comes from .\gallium\drivers\d3d12 folder.

To select one codec, you can do:

```

...
-Dvideo-codecs=h264enc -Dmediafoundation-codecs=h264enc
...

```

To select all codecs, you can do

```

...
-Dvideo-codecs=all -Dmediafoundation-codecs=all
...

```

Below is an example setup command that builds one codec

```

...
meson setup build/ --pkg-config-path="C:\lib\pkgconfig" -Dgallium-d3d12-graphics=disabled -Dintel-elk=fal
...

```

To build the COM based HEVC codec, use the above command but replace all instances of h264enc with h265enc

For debug builds, change the debug/buildtype to true, e.g.

```

...
-Ddebug=true --buildtype=true
...

```

For building the unit test, add the following option to the meson setup.

```

...
-Dgallium-mediafoundation-test=true
...

```

To build, run the following command from the mesa folder.

```

...
ninja -C build/ install
...

```

dirtycode/mesa/src/gallium/targets/d3d10umd/README.md

D3D10umd Readme

When compiled with `gallium-driver=llvmpipe` or `gallium-driver=softpipe` the resulting libgallium_d3d10.

It can be used directly from WLK 1.6 and WHCK 2.0 D3D10+ tests, via the -Src and -SWDLL options. For example:

```
wgfl1blend.exe -Debug -DoNotCatchExceptions -DXGI:1.1 -FeatureLevel:10.0 -Src:SW -SWDLL:libgallium_d3
```

However, as of WHCK version 2.1 this mechanism no longer works reliably. Either you use WHCK 2.0 binaries, or you must use the alternative method described below (of copying libgallium_d3d10.dll into the executable directory and rename it such that it matches the D3D10 UMD of the test machine).

Examples can be easily modified to load it too:

```
D3D10CreateDeviceAndSwapChain(NULL,
                               D3D10_DRIVER_TYPE_SOFTWARE,
                               LoadLibraryA("libgallium_d3d10"), /* Software */
                               Flags,
                               D3D10_SDK_VERSION,
                               &SwapChainDesc,
                               &g_pSwapChain,
                               &g_pDevice);

D3D11CreateDeviceAndSwapChain(NULL, /* pAdapter */
                               D3D_DRIVER_TYPE_SOFTWARE,
                               LoadLibraryA("libgallium_d3d10"), /* Software */
                               Flags,
                               FeatureLevels,
                               sizeof FeatureLevels / sizeof FeatureLevels[0],
                               D3D11_SDK_VERSION,
                               &SwapChainDesc,
                               &g_pSwapChain,
                               &g_pDevice,
                               NULL, /* pFeatureLevel */
                               &g_pDeviceContext); /* ppImmediateContext */
```

Alternatively, it can be renamed into the system's D3D10 UMD driver (e.g., vm3dum_10.dll for VMware vGPU, nvwgf2um.dll for NVIDIA GPU), and placed into the application directory, or system's directory, and used instead.

For the DLL to be picked from the application directory you'll need to do the following once:

```
reg add "HKLM\System\CurrentControlSet\Control\Session Manager" /v "SafeDllSearchMode" /t REG_DWORD /d 1
```

See also <<https://docs.microsoft.com/en-us/windows/win32/dlls/dynamic-link-library-search-order>>

Building with MSVC

Install Windows SDK 10.0.26100.0 and Windows DDK 10.0.26100.0 with MSVC and use it, other version of Windows SDK/DDK may also work but not verified.

Building with mingw

Install Windows SDK 10.0.26100.0 and Windows DDK 10.0.26100.0, other version of Windows SDK/DDK may also work but not verified.

Copy the following list of files from Windows SDK/DDK into mingw's `include` folder or `include\winddk` folder.

```
```.txt
d3d10umddi.h
d3dkmddi.h
d3dkmdt.h
d3dkmthk.h
d3dukmdt.h
d3dumddi.h
dxgiddi.h
dxmini.h
wmidata.h
```.
```

dirtycode/mesa/src/gtest/include/gtest/internal/custom/README.md

Customization Points

The custom directory is an injection point for custom user configurations.

Header `gtest.h`

The following macros can be defined:

```

* `GTEST_OS_STACK_TRACE_GETTER_` - The name of an implementation of
  `OsStackTraceGetterInterface`.
* `GTEST_CUSTOM_TEMPDIR_FUNCTION_` - An override for `testing::TempDir()`. See
  `testing::TempDir` for semantics and signature.

## Header `gtest-port.h`

The following macros can be defined:

### Logging:

* `GTEST_LOG_(severity)`
* `GTEST_CHECK_(condition)`
* Functions `LogToStderr()` and `FlushInfoLog()` have to be provided too.

### Threading:

* `GTEST_HAS_NOTIFICATION_` - Enabled if Notification is already provided.
* `GTEST_HAS_MUTEX_AND_THREAD_LOCAL_` - Enabled if `Mutex` and `ThreadLocal`
  are already provided. Must also provide `GTEST_DECLARE_STATIC_MUTEX_(mutex)`
  and `GTEST_DEFINE_STATIC_MUTEX_(mutex)`
* `GTEST_EXCLUSIVE_LOCK_REQUIRED_(locks)`
* `GTEST_LOCK_EXCLUDED_(locks)`

### Underlying library support features

* `GTEST_HAS_CXXABI_H_`

### Exporting API symbols:

* `GTEST_API_` - Specifier for exported symbols.

## Header `gtest-printers.h`

* See documentation at `gtest/gtest-printers.h` for details on how to define a
  custom printer.

```

dirtycode/mesa/src/intel/compiler/jay/README.md

```

# Jay

Jay is a modern compiler for Intel's Xe GPUs. Jay uses a novel modelling of the
Intel register file enabling a straightforward implementation of a decoupled
SSA-based register allocator, using a Colombet allocator with a Braun-Hack
spiller. As seen at [XDC2026 in
Toronto](https://indico.freedesktop.org/event/12/contributions/527/).

It currently supports Xe2 (Lunar Lake and Battlemage) and Xe3 (Pantherlake).
Additional hardware support is in progress.

Jay is enabled by default on these platforms. For debugging, set
`INTEL_DEBUG=no-jay` to get the old brw compiler.

## Contribution policy

We're committed to delivering high-quality software and nurturing healthy
software teams. To that end, we have the following guidelines for contributors.

* Follow the [upstream Mesa
  policy](https://docs.mesa3d.org/submittingpatches.html#expectations-on-contributors).
* Submit high-quality patches, regardless of what tools you used.
* You may use LLMs to improve quality (patch review & static analysis).
* Please do not use LLMs to try to improve velocity at the expense of quality
  (**no vibecoding**).
* Maintainability matters more than short-term crunch in the long run.
* People matter more than code.
* This policy will continue to evolve over time.

Selected background reading:

* <https://daniel.haxx.se/blog/2026/04/22/high-quality-chaos/>
* <https://dl.acm.org/doi/epdf/10.1145/3442188.3445922>
* <https://sourcehut.org/blog/2026-08-27-tos-changes-and-llms/>

```

* <<https://kuma.xyz/blog/2026/03/26/open-source-and-ai/>>

Advice for debugging

Sometimes, a game will work on our old compiler (brw) but not Jay. In that case, you can follow this checklist to narrow the issue:

Reposting the checklist earlier this summer:

- * does it reproduce locally on Xe2 or Xe3 with upstream Mesa?
- * does it go away if you disable Jay? (`INTEL_DEBUG=no-jay` env var`)
- * does it reproduce in a debug build of Mesa?
- * in a debug build, are there any assertion/validation failures)?
- * ...if so, can you capture a fossil?
- * ...can you reproduce the assertion failure on the fossil?
- * if not, does the issue go away with some combination of Jay debug flags? (`JAY_DEBUG=sync,strict,noopt,nosched` is a "safe" set`).
- * can you bisect it down to a specific shader hash, by selecting between brw and Jay with `nir_shader_bisect_select()` and the `nir_shader_bisect.py` script`? See the comment in the `intel_use_jay()` function in `src/intel/dev/intel_debug.c` for how to enable this.`
- * can you dump the affected shader with `INTEL_DEBUG=(vs|fs|cs)?`
- * can you dump the assembly of that shader with brw?
- * can you diff the before/after assembly to see what Jay is doing different?
- * if the issue goes away with some JAY_DEBUG flag, can you isolate which one?
- * ..and then dump and diff the Jay assembly with/without that flag?

This should get you from "this trace fails" to "this specific shader fails and here's the specific miscompiling assembly to study". From there, the issue is hopefully easy to spot. If you get stuck there, send those results to a Jay developer for further analysis.

dirtycode/mesa/src/intel/compiler/jay/register-file.md

Glossary

lane: A single work-item.

subgroup: A collection of 8, 16, or 32 lanes executing in lockstep. Avoid using the term `_thread_` as it is ambiguous.

uniform: A value that has the same value in every active lane of a subgroup. Sometimes called `_convergent_`. Opposite of "non-uniform".

non-uniform: A value that may have different values in different active lanes within a subgroup. Sometimes called `_divergent_`. Opposite of "uniform".

GPR: General-purpose register, a single non-uniform value viewed from the perspective of a single lane. This is a 'virtual' or 'logical' register within the SIMT programming model. It does not represent a physical machine register. For that, see "GRF".

UGPR: Uniform general purpose register, a single uniform value. This is again a virtual or logical register.

GRF: A physical Intel GPU register. On Xe2+, a GRF is 512-bits. On older platforms, a GRF is 256-bits. Depending on the platform and the SIMD width, different numbers of GRFs required to store a single GPR, and different numbers of UGPRs fit into a single GRF. In SIMD32 mode on Xe2, 1 GPR requires 2 GRFs, and 16 UGPRs fit into 1 GRF.

scalar: A single value from the perspective of a single lane; a single GPR or UGPR. Note that a scalar may be either uniform or non-uniform. Opposite of "vector".

vector: A collection of multiple values from the perspective of a single lane. All scalars within the vector must be identically be GPRs or UGPRs.

Introduction

Jay separates the logical register files (GPR and UGPR) from the unified physical register file. We assign registers independently for each logical file, and then post-RA we remap to physical GRFs. This simplifies RA.

We decide a static GPR/UGPR split up front. Ideally, we'd just use the first N registers for GPRs and the rest for UGPRs, or something like that. Unfortunately, several hardware issues complicate this scheme...

End-of-thread SENDs

End-of-thread SENDs require their source is in r112-r127. As their source will always be per-thread, we want to make sure these are GPRs.

Payloads

At the start of each thread, the register file is preloaded with a payload. Parts of the payload act like UGPRs, parts act like GPRs, and parts act like... something weird and in between. To minimize copying, we want to assign UGPRs to the UGPR parts of the payload and GPRs to the GPR parts. As for the weird cases, we model them as UGPR vectors and use special opcodes (lowered late to regioning) to unpack to GPRs for normal handling.

dirtycode/mesa/src/intel/tools/README.md

aubinator_error_decode

This tool allows decode an error state dump generated by the kernel driver.

Make a copy of the error state file (i915) :

```
```  
$ sudo cp /sys/devices/pci0000:00/0000:00:02.0/drm/card0/error /tmp/error
$ sudo chmod go+rw /tmp/error
```
```

Make a copy of the error state file (Xe) :

```
```  
$ sudo cp /sys/class/drm/card0/device/devcoredump/data /tmp/error
$ sudo chmod go+rw /tmp/error
```
```

Then visualize the error state (from the mesa build directory) :

```
```  
$./src/intel/tools/aubinator_error_decode /tmp/error
```
```

intel_stub_gpu

This tool allows faking a particular device at the ioctl level. This is useful to test compiler changes (using shader-db/fossils-db replays) across a number of device you don't necessarily have access too.

Faking a DG2 device :

```
```  
$ intel_stub_gpu -p dg2 fossilize-replay /path/to/fossilize.foz --enable-pipeline-stats /tmp/dg2.csv
```
```

Faking a ICL device :

```
```  
$ intel_stub_gpu -p icl fossilize-replay /path/to/fossilize.foz --enable-pipeline-stats /tmp/icl.csv
```
```

intel_error2hangdump

This tool converts an error state dump into an uncompressed/easy-to-modify binary file (format described in `src/intel/common/intel\_hang\_dump.h`).

Capture the error state (i915) :

```
```  
$ sudo cp /sys/devices/pci0000:00/0000:00:02.0/drm/card0/error /tmp/error
$ sudo chmod go+rw /tmp/error
```
```

Capture the error state (Xe) :

```
```  
$ sudo cp /sys/class/drm/card0/device/devcoredump/data /tmp/error
```
```

```

$ sudo chmod go+rw /tmp/error
```

Convert an error state (from the mesa build directory) :
```
$ ./src/intel/tools/intel_error2hang /tmp/error
```

A resulting `error.dmp` file will be created

intel_hang_replay

This tool replays a binary file generated by `intel_error2hangdump`.

From the mesa build directory :
```
$ ./src/intel/tools/intel_hang_replay -d /tmp/error.dmp
```

intel_hang_viewer

This tool is a editor/visualizer for the file generated by `intel_error2hangdump`. When replaying on simulation a few tweaks might be needed in order for the simulator to accept the input commands (this is due to divergences between the simulation & HW implementations).

To visualize a file generated from a DG2 error state (from the mesa build directory) :
```
./src/intel/tools/intel_hang_viewer -p dg2 /tmp/error.dmp
```

To visualize and edit a file generated from a DG2 error state (from the mesa build directory) :
```
./src/intel/tools/intel_hang_viewer -e -p dg2 /tmp/error.dmp
```

```

## dirtycode/mesa/src/intel/vulkan/bvh/README.md

## BVH Debug

1. `INTEL\_DEBUG=bvh\_tlas,bvh\_blas` will generate `tlas\_{id}.txt` or `blas\_{id}.txt` in `bvh\_dump/BVH\_ANV`
2. `INTEL\_DEBUG=bvh\_tlas\_ir\_hdr,bvh\_blas\_ir\_hdr` will generate `tlas\_{id}.txt` or `blas\_{id}.txt` in `bvh\_dump`
3. `INTEL\_DEBUG=bvh\_tlas\_ir\_as,bvh\_blas\_ir\_as` will generate `tlas\_{id}.txt` or `blas\_{id}.txt` in `bvh\_dump`
4. `INTEL\_DEBUG=bvh\_no\_build` will skip the intel-specific-encoding part. This will make the bvh NULL and not dump.

The dumped text file contains memory dump, byte-by-byte in hex. The contents are contiguous memory of a dump.

1. The dump in `BVH\_ANV` starts from the beginning of `anv\_accel\_struct\_header` to the end of the bvh.
2. The dump in `BVH\_IR\_HDR` records `vk\_ir\_header`, which sits at the beginning of ir bvh.
3. The dump in `BVH\_IR\_AS` records all `vk\_ir\_{leaf\_type}\_node` and `vk\_ir\_box\_node` in ir bvh. The register is not dumped.

### Decode the dump

We have a way to decode the dump in `BVH\_ANV`.

- To decode this memory dump, use a python script to parse the file and generate a human-readable json.
- To further visualize the tree, there is a html that parses the json and draws the tree topology and 3D visualization.

```

Using blas_0 as an example

```
xxd -r -p bvh_dump/BVH_ANV/blas_0.txt > input.bin
```

Use a python script to generate a human-readable json file

```
cd mesa/src/intel/vulkan/bvh/
```

```
python3 interpret.py <path/to/input.bin>
```

To further visualize the tree, the html parses the json and draws it in 3D.

```
cd mesa/src/intel/vulkan/bvh/
```

```
python3 -m http.server 8000
```

```
# go to localhost:8000/visualize_json.html
```

```

### Note and Limitations:

1. The python script use `ctypes` to interpret the memory dump, so the structure defined in the script sh
2. The memory dump is a snapshot of a VkBuffer captured at the end of `CmdBuildAccelerationStructure` cal
3. CPU will wait for the the command buffer to be completed, and saves the recorded bvh into files on dis

## dirtycode/mesa/src/nouveau/drm-shim/README.md

### nouveau\_noop backend

This implements the minimum of nouveau in order to make shader-db work.  
The submit ioctl is stubbed out to not execute anything.

Export `MESA\_LOADER\_DRIVER\_OVERRIDE=nouveau`  
LD\_PRELOAD=\$prefix/lib/libnouveau\_noop\_drm\_shim.so`.

By default, GK110 is exposed. The chip can be selected with an environment variable like `NOUVEAU\_CHIPSET=f0`. Some chips of note with the appropriate chipset ids:

| ID  | Chip Name | Sample Marketing Name | Comment                   |
|-----|-----------|-----------------------|---------------------------|
| --- | -----     | -----                 | -----                     |
| 30  | NV30      | GeForce FX 5500       | DX9                       |
| 40  | NV40      | GeForce 6800          | DX9c, better control flow |
| 50  | G80       | GeForce 8800 GTS      | SM 1.0, 128 regs          |
| 84  | G84       | GeForce 8600 GS       | SM 1.1                    |
| a0  | GT200     | GeForce GTX 280       | SM 1.2 + fp64             |
| a3  | GT215     | GeForce GT 240        | DX10.1 ops                |
| c0  | GF100     | GeForce GTX 480       | SM 2.0, 64 regs           |
| e4  | GK104     | GeForce GTX 680       | SM 3.0, 64 regs           |
| f0  | GK110     | GeForce GTX 780       | SM 3.5, 256 regs          |
| 117 | GM107     | GeForce GTX 750       | SM 5.0                    |
| 124 | GM204     | GeForce GTX 980       | SM 5.2                    |
| 134 | GP104     | GeForce GTX 1080      | SM 6.0                    |
| 140 | GV100     | TITAN V               | SM 7.0                    |
| 162 | TU102     | GeForce RTX 2080      | SM 7.5                    |

## dirtycode/mesa/src/panfrost/compiler/bifrost/README.md

# Bifrost compiler

## Register file

Defined partially in software, partially in hardware.

## Blend shaders

R0 - R3: input (color #0)  
R4 - R7: input (color #1)  
R8 - R15: general purpose  
R48: return address

## Fragment

Anything live during BLEND must respect blend shader registers.

R0 - R3: preloaded (message #0)  
R4 - R7: preloaded (message #1)  
R57 - R63: preloaded (various)

R0 - R15: general purpose (full threads)  
R48 - R63: general purpose (full threads)

R32 - R47: general purpose (half threads, or v6)

## dirtycode/mesa/src/panfrost/perf/database/README.md

# Mali counter database

All the files under this directory have been directly copied from (libGPUCounter)[<https://github.com/ARM-software/libGPUCounters/tree/main/specification/database>] and are subject to the original Arm copyright that applies to this

project licensed under MIT.

If you need to update the counters definition, copy the new files in, and make sure mali\_gen\_perf.py is still able to parse them.

## doc.md

# TrangorgeOS Architecture Documentation

\*\*Generated:\*\* 2026-09-18 21:00:50

\*\*Files Scanned:\*\* `1709` | \*\*API Items Extracted:\*\* `4008`

\*\*Languages:\*\* Rust, C/C++, Ada SPARK, Odin, Nim, Assembly

---

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---

### ## c.rs

<details>

<summary><b>c.rs</b> (2 items)</summary>

#### `STRUCT`: \*\*BlockHeader\*\* <sub>line 57</sub>

```rust

struct BlockHeader* next; /* Wskaźnik do następnego bloku */

```

#### `STRUCT`: \*\*BlockHeader\*\* <sub>line 58</sub>

```rust

struct BlockHeader* prev; /* Wskaźnik do poprzedniego bloku */

```

</details>

### ## comgrub

<details>

```

<summary>comgrub/build.rs (1 items)</summary>

`FN`: **main** _{line 1}
```rust
fn main() {
```

</details>

comgrub/src

<details>
<summary>comgrub/src/main.rs (2 items)</summary>

`FN`: **kernel_main** _{line 11}
```rust
fn kernel_main(magic: u32, info: *const u8) -> !;
```

`FN`: **panic** _{line 15}
```rust
fn panic(_info: &PanicInfo) -> ! {
```

</details>

comlimine

<details>
<summary>comlimine/build.rs (1 items)</summary>

`FN`: **main** _{line 1}
```rust
fn main() {
```

</details>

comlimine/src

<details>
<summary>comlimine/src/main.c (2 items)</summary>

`FUNCTION`: **kernel_main** _{line 6}
```c
extern void kernel_main(uint64_t magic, void *info);
```

`FUNCTION`: **_start** _{line 28}
```c
void _start(void) {
```

</details>

<details>
<summary>comlimine/src/main.rs (1 items)</summary>

`FN`: **main** _{line 1}
```rust
fn main() {}
```

</details>

drivers/amdgpu_driver

<details>
<summary>drivers/amdgpu_driver/src/main.rs (1 items)</summary>

`FN`: **main** _{line 1}
```rust
fn main() {

```

```
...
```

```
</details>
```

```
## drivers/audiodriver
```

```
<details>
```

```
<summary><b>drivers/audiodriver/build.rs</b> (1 items)</summary>
```

```
#### `FN`: **main** <sub>line 12</sub>
```

```
```rust
```

```
fn main() {
```

```
...
```

```
</details>
```

```
<details>
```

```
<summary>drivers/audiodriver/src/jacklib.rs (6 items)</summary>
```

```
`STRUCT`: **JackMgr** _{line 4}
```

```
```rust
```

```
pub struct JackMgr {
```

```
...
```

```
#### `IMPL`: **JackMgr** <sub>line 9</sub>
```

```
```rust
```

```
impl JackMgr {
```

```
...
```

```
`FN`: **tick** _{line 14}
```

```
```rust
```

```
pub fn tick(&mut self) {
```

```
...
```

```
#### `FN`: **set_amp** <sub>line 30</sub>
```

```
```rust
```

```
pub fn set_amp(&mut self, on: bool) {
```

```
...
```

```
`FN`: **present** _{line 35}
```

```
```rust
```

```
pub fn present(&self) -> bool {
```

```
...
```

```
#### `FN`: **amp_enabled** <sub>line 40</sub>
```

```
```rust
```

```
pub fn amp_enabled(&self) -> bool {
```

```
...
```

```
> Czy wzmacniacz (amp) jest aktualnie włączony.
```

```
</details>
```

```
<details>
```

```
<summary>drivers/audiodriver/src/lib.rs (14 items)</summary>
```

```
`FN`: **ad_init** _{line 7}
```

```
```rust
```

```
fn ad_init(nam_va: u64, bm_va: u64) -> i32;
```

```
...
```

```
#### `FN`: **ad_play** <sub>line 8</sub>
```

```
```rust
```

```
fn ad_play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32;
```

```
...
```

```
`FN`: **ad_capture** _{line 9}
```

```
```rust
```

```
fn ad_capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32;
```

```
...
```

```
#### `FN`: **ad_stop** <sub>line 10</sub>
```

```
```rust
```

```
fn ad_stop() -> i32;
```

```

 ...

`FN`: **ad_jack_present** _{line 11}
```rust
fn ad_jack_present() -> i32;
```

`FN`: **ad_set_amp** _{line 12}
```rust
fn ad_set_amp(on: i32) -> i32;
```

`FN`: **ad_position** _{line 13}
```rust
fn ad_position() -> u32;
```

`FN`: **init** _{line 16}
```rust
pub fn init(nam_va: u64, bm_va: u64) -> bool {
```

`FN`: **play** _{line 20}
```rust
pub fn play(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
```

`FN`: **capture** _{line 24}
```rust
pub fn capture(data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> bool {
```

`FN`: **stop** _{line 28}
```rust
pub fn stop() {
```

`FN`: **jack_present** _{line 32}
```rust
pub fn jack_present() -> bool {
```

`FN`: **set_amp** _{line 36}
```rust
pub fn set_amp(on: bool) {
```

`FN`: **position** _{line 40}
```rust
pub fn position() -> u32 {
```

```

</details>

<details>

<summary><b>drivers/audiodriver/src/main.rs</b> (1 items)</summary>

```
`FN`: **main** _{line 8}
```

```
```rust
```

```
fn main() {
```

```
```
```

```
> Punkt wejścia binarki pomocniczej.
```

```
>
```

```
> Właściwy sterownik działa w driver space przez `ds_entry` (patrz
> `driverspace/src/main.rs`); ta binarka jest cienkim wrapperem CLI, który
> pozwala sprawdzić stan gniazda jack z poziomu hosta.
```

</details>

<details>

<summary><b>drivers/audiodriver/src/odin/driver.odin</b> (8 items)</summary>

```
`STRUCT`: **Bdl_Entry** _{line 8}
```

```
````odin
Bdl_Entry :: struct {
````
```

```
`PROC`: **nam16** _{line 16}
```

```
````odin
```

```
nam16 :: proc(off: u32) -> ^u16 {
```

```
````
```

```
> mixer (NAM) — 16-bit
```

```
`PROC`: **bm8** _{line 21}
```

```
````odin
```

```
bm8 :: proc(off: u32) -> ^u8 {
```

```
````
```

```
> bus master — 8/16/32
```

```
`PROC`: **bm16** _{line 25}
```

```
````odin
```

```
bm16 :: proc(off: u32) -> ^u16 {
```

```
````
```

```
`PROC`: **bm32** _{line 29}
```

```
````odin
```

```
bm32 :: proc(off: u32) -> ^u32 {
```

```
````
```

```
`PROC`: **ad_init** _{line 50}
```

```
````odin
```

```
ad_init :: proc "C" (nam_va: u64, bm_va: u64) -> i32 {
```

```
````
```

```
`PROC`: **ad_stop** _{line 73}
```

```
````odin
```

```
ad_stop :: proc "C" () -> i32 {
```

```
````
```

```
`PROC`: **ad_position** _{line 80}
```

```
````odin
```

```
ad_position :: proc "C" () -> u32 {
```

```
````
```

```
</details>
```

```
<details>
```

```
<summary>drivers/audiodriver/src/odin/jack.odin (2 items)</summary>
```

```
`PROC`: **ad_jack_present** _{line 8}
```

```
````odin
```

```
ad_jack_present :: proc "C" () -> i32 {
```

```
````
```

```
`PROC`: **ad_set_amp** _{line 21}
```

```
````odin
```

```
ad_set_amp :: proc "C" (on: i32) -> i32 {
```

```
````
```

```
</details>
```

```
<details>
```

```
<summary>drivers/audiodriver/src/odin/microphone.odin (1 items)</summary>
```

```
`PROC`: **ad_capture** _{line 6}
```

```
````odin
```

```
ad_capture :: proc "C" (data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32 {
```

```
````
```

```
</details>
```

```
<details>
```

```
<summary>drivers/audiodriver/src/odin/speaker.odin (1 items)</summary>
```

```
`PROC`: **ad_play** _{line 6}
```

```
````odin
```

```

ad_play :: proc "C" (data_phys: u64, len: u32, bdl_va: u64, bdl_phys: u64) -> i32 {
```

```

</details>

```

<details>
<summary>drivers/audiodriver/src/tone.rs (1 items)</summary>

`FN`: **fill_square** _{line 1}
```rust
pub fn fill_square(buf: &mut [u8], periods: u32) {
```

```

</details>

```

drivers/intelgpu_driver

<details>
<summary>drivers/intelgpu_driver/src/main.rs (1 items)</summary>

`FN`: **main** _{line 1}
```rust
fn main() {
```

```

</details>

```

drivers/netcam_driver

<details>
<summary>drivers/netcam_driver/src/main.rs (1 items)</summary>

`FN`: **main** _{line 1}
```rust
fn main() {
```

```

</details>

```

drivers/vgpu

<details>
<summary>drivers/vgpu/vgpu.h (7 items)</summary>

`FUNCTION`: **vgpu_init** _{line 22}
```c
bool vgpu_init(uint32_t w, uint32_t h);

#### `FUNCTION`: **vgpu_get_info** <sub>line 23</sub>
```c
vgpu_info_t vgpu_get_info(void);

`FUNCTION`: **vgpu_clear** _{line 25}
```c
void vgpu_clear(uint32_t color);

#### `FUNCTION`: **vgpu_pixel** <sub>line 26</sub>
```c
void vgpu_pixel(uint32_t x, uint32_t y, uint32_t color);

`FUNCTION`: **vgpu_surface_create** _{line 28}
```c
int32_t vgpu_surface_create(uint32_t w, uint32_t h, uint64_t *phys_out);

#### `FUNCTION`: **vgpu_surface_present** <sub>line 29</sub>
```c
bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y);
```

```

```
#### `FUNCTION`: **vgpu_process_ring** <sub>line 31</sub>
```

```
```c
```

```
void vgpu_process_ring(volatile void *ring);
```

```
```
```

```
</details>
```

```
<details>
```

```
<summary><b>drivers/vgpu/vgpu_driver.c</b> (11 items)</summary>
```

```
#### `FUNCTION`: **pci_void** <sub>line 12</sub>
```

```
```c
```

```
static void pci_void(uint32_t op, uint64_t a0, uint64_t a1, uint64_t a2)
```

```
```
```

```
#### `FUNCTION`: **pci_val** <sub>line 20</sub>
```

```
```c
```

```
static uint64_t pci_val(uint32_t op, uint64_t a0, uint64_t a1, uint64_t a2)
```

```
```
```

```
#### `FUNCTION`: **vbe_write** <sub>line 33</sub>
```

```
```c
```

```
static void vbe_write(uint16_t idx, uint16_t val)
```

```
```
```

```
#### `FUNCTION`: **vgpu_init** <sub>line 40</sub>
```

```
```c
```

```
bool vgpu_init(uint32_t w, uint32_t h)
```

```
```
```

```
#### `FUNCTION`: **vgpu_get_info** <sub>line 98</sub>
```

```
```c
```

```
vgpu_info_t vgpu_get_info(void)
```

```
```
```

```
#### `FUNCTION`: **vgpu_pixel** <sub>line 104</sub>
```

```
```c
```

```
void vgpu_pixel(uint32_t x, uint32_t y, uint32_t color)
```

```
```
```

```
#### `FUNCTION`: **vgpu_clear** <sub>line 113</sub>
```

```
```c
```

```
void vgpu_clear(uint32_t color)
```

```
```
```

```
#### `FUNCTION`: **vgpu_surface_create** <sub>line 128</sub>
```

```
```c
```

```
int32_t vgpu_surface_create(uint32_t w, uint32_t h, uint64_t *phys_out)
```

```
```
```

```
#### `FUNCTION`: **vgpu_surface_present** <sub>line 168</sub>
```

```
```c
```

```
bool vgpu_surface_present(int32_t id, uint32_t x, uint32_t y)
```

```
```
```

```
#### `FUNCTION`: **vgpu_process_ring** <sub>line 190</sub>
```

```
```c
```

```
void vgpu_process_ring(volatile vgpu_slot_t *ring)
```

```
```
```

```
#### `FUNCTION`: **ds_entry** <sub>line 244</sub>
```

```
```c
```

```
void ds_entry(uint64_t params_va)
```

```
```
```

```
</details>
```

```
## drivers/wacomgraphic_driver
```

```
<details>
```

```
<summary><b>drivers/wacomgraphic_driver/src/main.rs</b> (1 items)</summary>
```

```
#### `FN`: **main** <sub>line 1</sub>
```

```
```rust
fn main() {
```
```

</details>

driverspace/src

<details>

<summary>driverspace/src/drivers/storage.rs (4 items)</summary>

`STRUCT`: **StorageDrv** _{line 9}

```
```rust
```

```
pub struct StorageDrv {
```
```

```
> Storage driver stub. It only satisfies the driver registry so that
> driver space can boot; a real disk driver can slot in later.
```

`IMPL`: **StorageDrv** _{line 13}

```
```rust
```

```
impl StorageDrv {
```
```

`IMPL`: **Driver** _{line 19}

```
```rust
```

```
impl Driver for StorageDrv {
```
```

`FN`: **init** _{line 20}

```
```rust
```

```
fn init(&mut self, _info: &DeviceInfo) -> Result<(), DsError> {
```
```

</details>

<details>

<summary>driverspace/src/main.rs (1 items)</summary>

`FN`: **panic** _{line 32}

```
```rust
```

```
fn panic(_info: &core::panic::PanicInfo) -> ! {
```
```

</details>

driverspace_workspace/crates

<details>

<summary>driverspace_workspace/crates/ds-manager/src/main.rs (1 items)</summary>

`FN`: **main** _{line 1}

```
```rust
```

```
fn main() {
```
```

</details>

driverspace_workspace/drivers

<details>

<summary>driverspace_workspace/drivers/amdgpu-driver/src/main.rs (1 items)</summary>

`FN`: **main** _{line 1}

```
```rust
```

```
fn main() {
```
```

</details>

<details>

<summary>driverspace_workspace/drivers/audiodriver/src/main.rs (1 items)</summary>

`FN`: **main** _{line 1}

```
```rust
fn main() {
```
```

</details>

<details>

<summary>driverspace_workspace/drivers/intelgpu-driver/src/main.rs (1 items)</summary>

`FN`: ****main**** _{line 1}

```
```rust
fn main() {
```
```

</details>

<details>

<summary>driverspace_workspace/drivers/netcam_driver/src/main.rs (1 items)</summary>

`FN`: ****main**** _{line 1}

```
```rust
fn main() {
```
```

</details>

<details>

<summary>driverspace_workspace/drivers/vgpu/src/main.rs (1 items)</summary>

`FN`: ****main**** _{line 1}

```
```rust
fn main() {
```
```

</details>

<details>

<summary>driverspace_workspace/drivers/wacomgraphic_driver/src/main.rs (1 items)</summary>

`FN`: ****main**** _{line 1}

```
```rust
fn main() {
```
```

</details>

driverspace_workspace/formal

<details>

<summary>driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.adb (7 items)</summary>

`PACKAGE`: ****body**** _{line 2}

```
```ada
package body DS_Buddy_Math with
```
> formal/ds-spark-core/src/ds_buddy_math.adb
```

`FUNCTION`: ****Power_Of_Two**** _{line 6}

```
```ada
function Power_Of_Two (Exponent : Block_Size) return Phys_Addr is
```
```

`FUNCTION`: ****Is_Aligned**** _{line 11}

```
```ada
function Is_Aligned (Addr : Phys_Addr; Size_Exp : Block_Size) return Boolean is
```
```

`FUNCTION`: ****Get_Buddy_Address**** _{line 18}

```
```ada
function Get_Buddy_Address (
```
```

`FUNCTION`: ****Get_Left_Child_Address**** _{line 31}

```

```ada
function Get_Left_Child_Address (
```

#### `FUNCTION`: **Get_Right_Child_Address** <sub>line 40</sub>
```ada
function Get_Right_Child_Address (
```

#### `FUNCTION`: **Safe_Calculate_DMA_End** <sub>line 50</sub>
```ada
function Safe_Calculate_DMA_End (
```

</details>

<details>
<summary><b>driverspace_workspace/formal/ds-spark-core/src/ds_buddy_math.ads</b> (9 items)</summary>

#### `PACKAGE`: **DS_Buddy_Math** <sub>line 4</sub>
```ada
package DS_Buddy_Math with

`TYPE`: **Phys_Addr** _{line 9}
```ada
type Phys_Addr is new Unsigned_64;
```
> Typy bazowe dla fizycznej pamieci (bare-metal)

`TYPE`: **Block_Size** _{line 10}
```ada
type Block_Size is new Unsigned_64;
```

`FUNCTION`: **Power_Of_Two** _{line 17}
```ada
function Power_Of_Two (Exponent : Block_Size) return Phys_Addr with
```
> Funkcja pomocnicza: Obliczanie 2^N (Shift left)

`FUNCTION`: **Is_Aligned** _{line 23}
```ada
function Is_Aligned (Addr : Phys_Addr; Size_Exp : Block_Size) return Boolean with
```
> Sprawdzenie, czy adres jest poprawnie wyrównany do rozmiaru bloku

`FUNCTION`: **Get_Buddy_Address** _{line 32}
```ada
function Get_Buddy_Address (

#### `FUNCTION`: **Get_Left_Child_Address** <sub>line 47</sub>
```ada
function Get_Left_Child_Address (

`FUNCTION`: **Get_Right_Child_Address** _{line 55}
```ada
function Get_Right_Child_Address (

#### `FUNCTION`: **Safe_Calculate_DMA_End** <sub>line 69</sub>
```ada
function Safe_Calculate_DMA_End (
```

</details>

## driverspace_workspace/lib

<details>
<summary><b>driverspace_workspace/lib/ds-formal-ffi/build.rs</b> (1 items)</summary>

```

```
#### `FN`: **main** <sub>line 10</sub>
```rust
fn main() {
 `
```

</details>

<details>

<summary><b>driverspace\_workspace/lib/ds-formal-ffi/src/lib.rs</b> (6 items)</summary>

```
`FN`: **ds_spark_safe_add_32** _{line 7}
```rust
pub fn ds_spark_safe_add_32(a: u32, b: u32) -> u32;
`
```

```
#### `FN`: **ds_spark_safe_array_index** <sub>line 10</sub>
```rust
pub fn ds_spark_safe_array_index(index: u32, max_size: u32) -> u32;
`
```

```
`FN`: **safe_add** _{line 21}
```rust
pub fn safe_add(a: u32, b: u32) -> u32 {
    `
```

```
#### `FN`: **ds_spark_buddy_get_buddy** <sub>line 28</sub>
```rust
pub fn ds_spark_buddy_get_buddy(addr: u64, size_exp: u64) -> u64;
`
```

```
`FN`: **ds_spark_safe_dma_end** _{line 29}
```rust
pub fn ds_spark_safe_dma_end(base: u64, len: u64) -> u64;
`
```

```
#### `FN`: **get_buddy** <sub>line 35</sub>
```rust
pub fn get_buddy(addr: u64, exp: u64) -> u64 {
 `
```

</details>

<details>

<summary><b>driverspace\_workspace/lib/ds-fw-audio/src/lib.rs</b> (2 items)</summary>

```
`FN`: **add** _{line 1}
```rust
pub fn add(left: u64, right: u64) -> u64 {
    `
```

```
#### `FN`: **it_works** <sub>line 10</sub>
```rust
fn it_works() {
 `
```

</details>

<details>

<summary><b>driverspace\_workspace/lib/ds-fw-block/src/request.rs</b> (8 items)</summary>

```
`ENUM`: **RequestState** _{line 7}
```rust
pub enum RequestState {
    `
```

```
#### `STRUCT`: **IoRequest** <sub>line 15</sub>
```rust
pub struct IoRequest {
 `
```

> High-level I/O request with metadata.

```
`IMPL`: **IoRequest** _{line 27}
```rust
```

```

impl IoRequest {
    ...

    ##### `FN`: **new** <sub>line 28</sub>
    ```rust
 pub fn new(id: u64, lba: u64, block_count: u32, buffer: *mut u8, len: usize, is_write: bool) -> Self {
 ...

 ##### `FN`: **state** _{line 42}
    ```rust
    pub fn state(&self) -> RequestState {
    ...

    ##### `FN`: **mark_in_progress** <sub>line 52</sub>
    ```rust
 pub fn mark_in_progress(&self) {
 ...

 ##### `FN`: **mark_completed** _{line 56}
    ```rust
    pub fn mark_completed(&self) {
    ...

    ##### `FN`: **mark_failed** <sub>line 63</sub>
    ```rust
 pub fn mark_failed(&self) {
 ...

```

</details>

<details>

<summary><b>driverspace\_workspace/lib/ds-fw-block/src/traits.rs</b> (11 items)</summary>

```

`STRUCT`: **BlockRequest** _{line 6}
```rust
pub struct BlockRequest {
    ...

    > Represents a single I/O request to a block device.

##### `TRAIT`: **BlockDevice** <sub>line 16</sub>
```rust
pub trait BlockDevice {
 ...

 > Core trait for all block devices.
 > Drivers must implement this to integrate with the block subsystem.

`FN`: **read_blocks** _{line 18}
```rust
fn read_blocks(&mut self, lba: u64, block_count: u32, buffer: &mut [u8]) -> Result<(), DsError>;
...

    > Read blocks from the device starting at LBA.

##### `FN`: **write_blocks** <sub>line 21</sub>
```rust
fn write_blocks(&mut self, lba: u64, block_count: u32, buffer: &[u8]) -> Result<(), DsError>;
...

 > Write blocks to the device starting at LBA.

`FN`: **flush** _{line 24}
```rust
fn flush(&mut self) -> Result<(), DsError>;
...

    > Flush internal caches to physical media.

##### `FN`: **geometry** <sub>line 27</sub>
```rust
fn geometry(&self) -> BlockGeometry;
...

 > Get device geometry (block size, total blocks).

`FN`: **supports_trim** _{line 30}
```rust
fn supports_trim(&self) -> bool { false }

```

```

    ...
> Check if device supports TRIM/discard.

#### `FN`: **submit_request** <sub>line 33</sub>
```rust
fn submit_request(&mut self, _req: BlockRequest) -> Result<(), DsError> {
 ...
> Optional: Submit async request (for advanced schedulers).

`STRUCT`: **BlockGeometry** _{line 40}
```rust
pub struct BlockGeometry {
    ...

#### `IMPL`: **BlockGeometry** <sub>line 46</sub>
```rust
impl BlockGeometry {
 ...

`FN`: **total_bytes** _{line 47}
```rust
pub fn total_bytes(&self) -> u64 {
    ...
</details>

<details>
<summary><b>driverspace_workspace/lib/ds-fw-gpu/src/lib.rs</b> (2 items)</summary>

#### `FN`: **add** <sub>line 1</sub>
```rust
pub fn add(left: u64, right: u64) -> u64 {
 ...

`FN`: **it_works** _{line 10}
```rust
fn it_works() {
    ...
</details>

<details>
<summary><b>driverspace_workspace/lib/ds-fw-input/src/lib.rs</b> (2 items)</summary>

#### `FN`: **add** <sub>line 1</sub>
```rust
pub fn add(left: u64, right: u64) -> u64 {
 ...

`FN`: **it_works** _{line 10}
```rust
fn it_works() {
    ...
</details>

<details>
<summary><b>driverspace_workspace/lib/ds-ipc/src/client.rs</b> (4 items)</summary>

#### `STRUCT`: **ManagerClient** <sub>line 7</sub>
```rust
pub struct ManagerClient {
 ...

`IMPL`: **ManagerClient** _{line 11}
```rust
impl ManagerClient {
    ...

#### `FN`: **request_mmio** <sub>line 16</sub>
```rust
pub fn request_mmio(&self, phys_base: u64, size: u64) -> Result<u64, DsError> {
 ...

```

```
`FN`: **bind_irq** _{line 41}
```rust
pub fn bin

... (truncated, file is very long)
```

driverspace_workspace/TODO.md

TODO.md: Driver Space Migration and Implementation Guide

Overview

This document outlines the pending tasks, architectural constraints, and execution steps for migrating and

The legacy monolithic implementations (`driverspacelib` and the old `driverspace` daemon) are deprecated.

1. Prerequisites

Before beginning driver implementation, the foundational libraries located in `lib/` must be fully implem

- [] **lib/kapi-abi**: Finalize all IPC message structures, opcodes, and error codes. Ensure `cbindgen
- [] **lib/kapi-syscall**: Implement architecture-specific SVC/syscall wrappers.
- [] **lib/ds-ipc**: Stabilize the channel, port, and shared memory abstractions.
- [] **lib/ds-mem**: Finalize the `no\_std` allocators (Buddy/Slab) and DMA mapping utilities.
- [] **lib/ds-fw**: Define the core traits for device classes (`BlockDevice`, `AudioDevice`, `Display

2. Driver Implementation Checklist

The following drivers require migration from the legacy codebase or complete reimplementation using the r

2.1. Graphics and Display

- [] **amdgpu_driver**
 - [] Implement MMIO register read/write wrappers.
 - [] Implement ring buffer management for command submission.
 - [] Implement Display PHY initialization.
 - [] Implement the `DisplayDevice` and `GpuDevice` traits from `ds-fw-gpu`.
- [] **intelgpu_driver**
 - [] Implement Graphics Technology (GT) and Display Engine (DE) initialization.
 - [] Implement GuC (Graphics Microcontroller) firmware loading.
 - [] Implement the `DisplayDevice` trait from `ds-fw-gpu`.
- [] **vgpu** (Virtual GPU)
 - [] Set up the Rust-to-C FFI bridge (`build.rs` and `ffi.rs`).
 - [] Implement virtual framebuffer memory management.
 - [] Ensure the C implementation strictly uses the `kapi-abi` headers.

2.2. Audio

- [] **audiodriver**
 - [] Implement DMA stream management for audio buffers.
 - [] Implement codec communication protocols (I2C / Intel HDA).
 - [] Integrate the Odin/C DSP components via the defined FFI boundaries.
 - [] Implement the `AudioDevice` trait from `ds-fw-audio`.

2.3. Input and HID

- [] **wacomgraphic_driver**
 - [] Implement Wacom-specific HID report parsing.
 - [] Implement raw-to-logical data conversion for pen pressure and tilt (X/Y).
 - [] Implement the `InputDevice` trait from `ds-fw-input`.

2.4. Networking and Multimedia

- [] **netcam_driver**
 - [] Implement USB Video Class (UVC) protocol parsing.
 - [] Implement isochronous USB transfer handling.
 - [] Implement device control interfaces (brightness, contrast, zoom).

3. Strict Architectural Rules for Contributors

All contributors must adhere to the following rules when writing driver code. Violations will result in r

1. **Unidirectional Dependencies**:
 - Drivers (`drivers/`) may **only** depend on libraries in `lib/`.
 - Drivers must **never** depend on the management daemons in `crates/`. Communication with the manager
2. **No Standard Library**:
 - All Rust driver code must be `#![no\_std]`. The use of the Rust standard library is strictly prohibit
3. **Zero Code Duplication**:
 - If a utility, protocol parser, or hardware abstraction is needed by more than one driver, it must be
4. **Strict FFI Boundaries**:
 - If a driver requires C or Odin code, it must be isolated in a dedicated subdirectory (e.g., `c/` or

- Compilation of foreign code must be handled via `build.rs` using the `cc` crate or equivalent tooling
 - Rust wrappers for foreign functions must be isolated in an `ffi.rs` file.
5. **Capability and Resource Management**:
- Drivers must not attempt to access hardware resources (MMIO, IRQs, DMA) directly via hardcoded addresses

4. Execution Steps for a New Driver

When starting work on a driver, follow this exact sequence:

1. **Initialize the Crate**: Create the directory under `drivers/<driver\_name>/` and generate a new `Cargo.toml`
2. **Update Workspace**: Add the new crate path to the `members` array in the root `driverspace\_workspace/Cargo.toml`
3. **Define Dependencies**: Add only the necessary `lib/` crates to the driver's `Cargo.toml` (e.g., `kapi-abi`)
4. **Implement Entry Point**: Create `main.rs` (or `lib.rs`), initialize the driver runtime, and establish hardware connections
5. **Implement the Trait**: Implement the required trait from the corresponding `ds-fw-\*` framework.
6. **Hardware Logic**: Implement the actual hardware register manipulation and state machines.

5. Current Status

- [] Workspace structure created and validated.
- [] Legacy `driverspacelib` code analyzed for extraction.
- [] `lib/kapi-abi` definitions finalized.
- [] First driver (`amdgpu\_driver` or `audiodriver`) migrated to the new workspace.

For questions regarding the architecture or the new Kernel API, refer to the main Architecture Document

driverspace_workspace/Tree_readme.md

```
driverspace_workspace/
├── Cargo.toml                # Główny workspace manifest
├── rust-toolchain.toml       # Wymuszenie nightly i specyficznych targetów
├── lib/
│   ├── kapi-abi/             # ■ FUNDAMENT: API, INFRASTRUKTURA I FRAMEWORKI
│   │   ├── Cargo.toml        # Nowe API Jędra: Definicje binarne (Single Source of Truth)
│   │   ├── build.rs          # Generowanie nagłówek C/Odin przez cbindgen
│   │   └── src/
│   │       ├── lib.rs        # Tylko re-eksporty i flagi no_std
│   │       ├── primitives.rs # Podstawowe typy (Handle, CapId, Status)
│   │       ├── opcodes.rs    # Enumy z numerami wywołań (Syscall/IPC opcodes)
│   │       ├── errors.rs     # Wspólne kody błędów (DsError)
│   │       ├── capabilities.rs # Struktury uprawnień (Capability tokens)
│   │       ├── wire/         # Serializacja/Deserializacja wiadomości
│   │       │   ├── mod.rs
│   │       │   ├── encode.rs # Zapis do bufora
│   │       │   ├── decode.rs # Odczyt z bufora
│   │       │   └── endian.rs # Konwersje kolejności bajtów
│   │       ├── payloads/     # Konkretne struktury wiadomości ([repr(C)])
│   │       │   ├── mod.rs
│   │       │   ├── mem.rs    # ■dania mapowania pamięci
│   │       │   ├── ipc.rs    # ■dania tworzenia kanałów
│   │       │   ├── dev.rs    # ■dania rejestracji urządzeń
│   │       │   └── irq.rs    # ■dania obsługi przerwań
│   │       └── kapi-syscall/ # Surowe wywołania systemowe (SVC / INT)
│   │           ├── Cargo.toml
│   │           └── src/
│   │               ├── lib.rs
│   │               ├── arch/ # Implementacje zależne od architektury
│   │               │   ├── mod.rs
│   │               │   ├── x86_64.rs # Makra asm! dla x86_64 (syscall/sysenter)
│   │               │   ├── riscv64.rs # Makra asm! dla RISC-V (ecall)
│   │               └── wrappers.rs # Bezpieczne (lub unsafe) wrappery nad surowym asm
│   └── ds-ipc/               # Infrastruktura IPC (Komunikacja między procesami)
│       ├── Cargo.toml
│       └── src/
│           ├── lib.rs
│           ├── channel.rs    # Abstrakcja kanału (dwukierunkowego)
│           ├── port.rs       # Abstrakcja portu (nasłuchiwanie)
│           ├── router.rs     # Logika routowania wiadomości
│           ├── shared_mem.rs # Zarządzanie buforami pamięci współdzielonej
│           ├── sync/         # Primitives synchronizacji dla IPC
│           │   ├── mod.rs
│           │   └── spinlock.rs # Spinlock (no_std)
```

```

■ ■ ■ ■ semaphore.rs # Semafore dla kolejek wiadomo■ci
■ ■ ■ ■ ds-mem/ # Zarz■dzanie pamie■ci■ w przestrzeni sterownika
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ lib.rs
■ ■ ■ ■ alloc/ # W■asne alokatory (brak std::alloc)
■ ■ ■ ■ mod.rs
■ ■ ■ ■ buddy.rs # Alokator Buddy (dla du■ych bloków)
■ ■ ■ ■ slab.rs # Alokator Slab (dla ma■ych, sta■ych obiektów)
■ ■ ■ ■ heap.rs # Globalny heap sterownika
■ ■ ■ ■ dma/ # Obs■uga DMA (Direct Memory Access)
■ ■ ■ ■ mod.rs
■ ■ ■ ■ buffer.rs # Bufory DMA (pami■ fizyczna/ci■g■a)
■ ■ ■ ■ mapping.rs # Mapowanie pamie■ci dla urz■dze■
■ ■ ■ ■ vmm.rs # Abstrakcje nad Wirtualn■ Pamie■ci■ (stronicowanie)
■ ■ ■ ■ ds-log/ # System logowania
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ lib.rs
■ ■ ■ ■ levels.rs # Enumy poziomów (Trace, Info, Error, Panic)
■ ■ ■ ■ formatter.rs # Formatowanie tekstu (bez std::fmt)
■ ■ ■ ■ transport.rs # Wysy■anie logów przez IPC do managera
■ ■ ■ ■ macros.rs # Makra: ds_log::info!, ds_log::error!
■ ■ ■ ■ ds-fw-block/ # Framework dla urz■dze■ blokowych (Dyski, USB Mass)
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ lib.rs
■ ■ ■ ■ traits.rs # Trait BlockDevice (read, write, flush)
■ ■ ■ ■ request.rs # Struktura ■■dania I/O (LBA, rozmiar, bufor)
■ ■ ■ ■ queue.rs # Kolejka ■■da■ (I/O Scheduler)
■ ■ ■ ■ partition.rs # Abstrakcja tablic partycji (MBR/GPT)
■ ■ ■ ■ ds-fw-audio/ # Framework dla Audio
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ lib.rs
■ ■ ■ ■ traits.rs # Trait AudioDevice (play, record)
■ ■ ■ ■ stream.rs # Abstrakcja strumienia audio
■ ■ ■ ■ formats.rs # Formaty próbek (PCM, Float, Sample rates)
■ ■ ■ ■ ringbuffer.rs # Lock-free ring buffer dla danych audio
■ ■ ■ ■ ds-fw-gpu/ # Framework dla GPU / Display
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ lib.rs
■ ■ ■ ■ traits.rs # Trait DisplayDevice, GpuDevice
■ ■ ■ ■ framebuffer.rs # Abstrakcja bufora ramki (pitch, format)
■ ■ ■ ■ modeset.rs # Ustawianie rozdzielczo■ci i trybu
■ ■ ■ ■ edid.rs # Parser EDID (dane monitora)
■ ■ ■ ■ cmd_buffer.rs # Abstrakcja bufora polece■ GPU (Command Submission)
■ ■ ■ ■ ds-fw-input/ # Framework dla HID (Klawiatura, Mysz, Wacom)
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ lib.rs
■ ■ ■ ■ traits.rs # Trait InputDevice (poll_event)
■ ■ ■ ■ events.rs # Typy zdarze■ (KeyPress, MouseMove, PenPressure)
■ ■ ■ ■ scancodes.rs # Mapowanie scancode'ów na kody klawiszy
■ ■ ■ ■ hid_parser.rs # Parser raportów HID (USB HID descriptors)
■ ■ ■ ■ crates/ # ■ ZARZ■DZANIE: DAEMONY DRIVERSPACE
■ ■ ■ ■ ds-manager/ # G■ówny daemon zarz■dzaj■cy sterownikami
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ main.rs # Entry point, inicjalizacja p■tli zdarze■
■ ■ ■ ■ lifecycle/ # Zarz■dzanie cyklem ■ycia sterowników
■ ■ ■ ■ mod.rs
■ ■ ■ ■ spawn.rs # ■adowanie i startowanie procesów sterowników
■ ■ ■ ■ monitor.rs # Watchdog (restartowanie craszki sterowników)

```

```

■ ■ ■ ■ shutdown.rs # Bezpieczne zamykanie
■ ■ ■ ■ resources/ # Przydzielanie zasobów sprzętowych
■ ■ ■ ■ mod.rs
■ ■ ■ ■ irq.rs # Routing i przydzielanie IRQ
■ ■ ■ ■ mmio.rs # Mapowanie pamięci urządzeń
■ ■ ■ ■ dma.rs # Przydzielanie buforów DMA
■ ■ ■ ■ ipc_handler.rs # Nasłuchiwanie danych od jądra i sterowników
■ ■ ■ ■ config.rs # Wczytywanie konfiguracji (np. z TOML/JSON)
■ ■ ■ ■ ds-registry/ # Rejestr urządzeń i sterowników
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ lib.rs
■ ■ ■ ■ database.rs # Przechowywanie mapy DeviceID -> Driver
■ ■ ■ ■ matching.rs # Logika dopasowywania (PCI ID, USB ID, ACPI)
■ ■ ■ ■ state.rs # Śledzenie stanu urządzeń (Podłączzone, Usunięte, Błęd)
■ ■ ■ ■ drivers/ # ■ STEROWNIKI SPRZĘTOWE
■ ■ ■ ■ amdgpu_driver/ # Sterownik AMD GPU
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ build.rs # (Opcjonalnie) kompilacja C jeśli używasz blobów
■ ■ ■ ■ src/
■ ■ ■ ■ main.rs # Entry point, inicjalizacja ds-runtime
■ ■ ■ ■ probe.rs # Detekcja karty, odczyt BAR-ów
■ ■ ■ ■ mmio.rs # Wrappery do odczytu/zapisu rejestrów MMIO
■ ■ ■ ■ irq.rs # Obsługa przerwa GPU
■ ■ ■ ■ registers/ # Definicje rejestrów (atomowo!)
■ ■ ■ ■ mod.rs
■ ■ ■ ■ gc.rs # Graphics Controller
■ ■ ■ ■ sdma.rs # System DMA
■ ■ ■ ■ nbio.rs # Northbridge IO
■ ■ ■ ■ ring_buffer.rs # Implementacja pierścieni poleceń (Ring Buffers)
■ ■ ■ ■ display/ # Podmoduł wyświetlania
■ ■ ■ ■ mod.rs
■ ■ ■ ■ phy.rs # Inicjalizacja PHY (Physical layer)
■ ■ ■ ■ audiodriver/ # Sterownik Audio (Hybryda Rust/C/Okładka na Odina)
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ build.rs # Kompilacja kodu C i linkowanie z Odinem
■ ■ ■ ■ c/ # Kod C (jeśli potrzebny do specyficznych DSP)
■ ■ ■ ■ dsp.c
■ ■ ■ ■ dsp.h
■ ■ ■ ■ src/
■ ■ ■ ■ main.rs
■ ■ ■ ■ codec.rs # Komunikacja z kodekiem audio (I2C/HDA)
■ ■ ■ ■ dma_stream.rs # Zarządzanie strumieniami DMA dla audio
■ ■ ■ ■ ffi.rs # Funkcje `extern "C"` wywoływane przez C/Okładki
■ ■ ■ ■ intelgpu_driver/ # Sterownik Intel GPU (i915/xe)
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ main.rs
■ ■ ■ ■ probe.rs
■ ■ ■ ■ gt.rs # Graphics Technology (rdzeń GPU)
■ ■ ■ ■ display.rs # Display Engine (DE)
■ ■ ■ ■ guc.rs # GuC (Graphics Microcontroller) firmware loading
■ ■ ■ ■ netcam_driver/ # Sterownik kamer sieciowych (UVC/USB)
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ src/
■ ■ ■ ■ main.rs
■ ■ ■ ■ uvc.rs # USB Video Class protocol parser
■ ■ ■ ■ isoc.rs # Obsługa transferów izochronicznych USB
■ ■ ■ ■ controls.rs # Panel sterowania (jasność, kontrast, zoom)
■ ■ ■ ■ vgpu/ # Wirtualne GPU (Głównie C, z Rust FFI)
■ ■ ■ ■ Cargo.toml
■ ■ ■ ■ build.rs # Kompilacja vgpu_driver.c
■ ■ ■ ■ c/
■ ■ ■ ■ vgpu_driver.c # Główna logika w C
■ ■ ■ ■ vgpu.h
■ ■ ■ ■ vgpu_mmio.c # Operacje na pamięci w C

```

```

■    ■■■ src/
■        ■■■ lib.rs          # Rust wrapper, eksportuj■cy ABI dla ds-manager
■        ■■■ ffi.rs         # Definicje `extern "C"` dla funkcji z C
■
■■■ wacomgraphic_driver/    # Tablet graficzny Wacom (HID)
    ■■■ Cargo.toml
    ■■■ src/
        ■■■ main.rs
        ■■■ hid_report.rs   # Parsowanie specyficznych raportów Wacom
        ■■■ pressure.rs     # Konwersja surowych danych na nacisk
        ■■■ tilt.rs         # Obs■uga nachylenia rysika (Tilt X/Y)

```

driverspace_workspace/Workspace.md

Oto profesjonalna dokumentacja architektury, przygotowana w j■zyku angielskim, zgodnie z Twoimi wytycznymi.

TrangorgeOS Driver Space Architecture and Workspace Documentation

1. Executive Summary

This document outlines the architectural redesign of the Driver Space subsystem within TrangorgeOS. The c

2. Workspace Topology

The `driverspace\_workspace` is divided into three primary pillars. This strict separation of concerns dic

```text

```

driverspace_workspace/
■■■ Cargo.toml # Workspace root configuration
■■■ lib/ # Core libraries, frameworks, and the new Kernel API
■■■ crates/ # Driver space management daemons and services
■■■ drivers/ # Individual hardware driver implementations
```

```

3. Component Breakdown

3.1. `lib/` (Core Libraries and Kernel API)

The `lib/` directory serves as the foundational layer of the driver space. It contains the new Kernel API

```

*   **Kernel API (`kapi-`)**
*   `kapi-abi`: The single source of truth for all data structures, IPC message formats, opcodes, and
*   `kapi-syscall`: Low-level, architecture-specific wrappers for kernel transitions (e.g., SVC calls
*   **Infrastructure (`ds-`)**
*   `ds-ipc`: Inter-process communication primitives, channel management, and message routing logic.
*   `ds-mem`: Memory management utilities, including custom allocators for `no_std` environments, DMA
*   `ds-log`: Standardized logging infrastructure that routes driver logs to the central manager or k
*   **Device Frameworks (`ds-fw-`)**
*   `ds-fw-block`, `ds-fw-audio`, `ds-fw-gpu`, `ds-fw-input`: High-level abstractions for specific de

```

3.2. `crates/` (Driver Space Management)

The `crates/` directory contains the privileged user-space (or kernel-space) daemons responsible for orch

```

*   `ds-manager`: The central daemon responsible for device enumeration, resource allocation (IRQs, MMIO
*   `ds-registry`: A database service that maintains the mapping of hardware device identifiers (e.g., PC

```

3.3. `drivers/` (Hardware Drivers)

The `drivers/` directory contains the actual implementations for specific hardware peripherals (e.g., `am

```

*   Each driver is an independent crate (or an external project for C/Odin implementations).
*   Drivers implement the traits defined in the `ds-fw-` frameworks.
*   Drivers communicate with the `ds-manager` and the kernel exclusively through the APIs provided in `li

```

4. The New Kernel API (`kapi`)

The legacy C headers (`libs/\*.h`) and the old `abi.rs` are deprecated. The new `kapi` (Kernel API) is des

Key characteristics of the new API:

1. **Explicit Message Passing:** All communication between the kernel, the manager, and the drivers is h
2. **Capability-Based Security:** Resource sharing (like memory or IRQs) is handled through explicit cap
3. **Language Agnostic:** The ABI is designed to be easily consumable by Rust, C, and Odin, facilitating

5. Architectural Rules and Dependency Graph

To maintain the integrity of the workspace, the following strict rules apply:

```

1. **Unidirectional Dependencies:**
   * `drivers/` may only depend on `lib/`.
   * `crates/` may only depend on `lib/`.
   * **Crucial Rule:** `drivers/` must **never** depend on `crates/`. Drivers must not have compile-time dependencies on crates.
2. **No Code Duplication:** If a utility, protocol parser, or hardware abstraction is needed by more than one driver, it must be in a crate.
3. **Strict `no_std` Compliance:** All crates within `lib/` and `crates/` must be `#![no_std]`. Standard library crates must be `#![no_std]` compatible.

## 6. Cross-Language Interoperability (FFI)
TrangorgeOS utilizes multiple languages (Rust, C, Odin, Nim). The `lib/kapi-abi` crate acts as the central interface.
* Rust structures must use `#[repr(C)]`.
* A build script (`build.rs`) utilizing `cbindgen` should be implemented in `kapi-abi` to automatically generate C headers.

## 7. Summary
This modular workspace architecture transforms the TrangorgeOS driver space from a tightly coupled monolith into a flexible, maintainable system.

```

driverspace_workspace/crates/ds-detect/README.md

```
# ds-detect
```

The detection layer: find the hardware, decide which drivers to start.

```
## The problem
```

A driver image has to carry every driver the machine *might* need, because at build time nobody knows which machine it will boot on. Starting them all is the obvious answer and the wrong one: each driver probes registers and claims resources, and a driver with no hardware either sits inert or - worse - gets confused by a device that is not its own.

So the image carries everything and the system starts what it finds.

```
## The pipeline
```

```

```text
image (all drivers + manifest)
 ──┬──
 scan read PCI configuration space -> what hardware exists
 ──┬── hw/scan.rs
 match manifest entry vs (vendor, device) -> what could drive it
 ──┬── manifest.rs
 plan one driver per device, honour policy -> what to start
 ──┬── plan.rs
 calibrate read the BARs, check they are real -> what to grant
 ──┬── calibrate.rs
 start load, grant, run
```

```

Each stage is a separate type because each has a different failure mode: a scan can miss a device, a match can be too broad, a plan can drop a driver, and a calibration can find a device that is present in the configuration space and absent on the silicon. Keeping them apart is what makes each diagnosable from a boot log.

```
## Match rules
```

A manifest entry claims a device by ****either**** an exact `(vendor, device)` pair or a PCI class:

| rule | example | meaning |
|-------|-------------------|-------------------------|
| exact | `exact:8086:2415` | only that pair |
| class | `class:04` | every multimedia device |
| class | `class:0401` | every audio device |

An exact match is checked first, because it knows what it is talking to; a class match is the fallback for hardware the build did not enumerate. Both are tried, and the first entry that matches a device wins it - which is how one device never ends up with two drivers.

```
## Load policy
```

| policy | when it starts |
|--------|----------------|
| | |

```
`ondemand`	only when a scanned device matches (the default)
`always`	regardless, for drivers with no enumerable device
`manual`	never automatically; present in the image only
```

The number that matters

With seven drivers in the image and one sound card in the machine, one driver starts. That is the whole point of the layer, and it is what the test ``a_machine_with_one_card_starts_one_driver_out_of_seven`` pins down.

Design notes

****No allocator.**** The layer runs before anything is loaded, so every table is ``heapless::Vec`` over inline storage. Sizes are fixed at build time and bounded (``MAX_DEVICES``, ``Manifest::MAX_ENTRIES``).

****The scan does not probe vendor registers.**** It reads ids, class code and BARs, and stops there. A driver that needs more does it itself after being granted the mapping - probing unknown registers at scan time is how a detection layer turns a harmless machine into a crashing one.

****BAR size is not required.**** Reading a BAR's size means writing all-ones to it, which is a write to hardware this layer does not own. A BAR counts as usable when its `*base*` is non-zero, and the size is left for the driver or the manager to fill in once it holds the resource.

****A failed calibration stops the driver.**** A device that answers in the configuration space but offers no usable BAR is reported in ``Detection::rejected`` and not started, rather than being handed a mapping that does not exist.

Testing

```
```sh
cargo test -p ds-detect
```
```

The whole pipeline is tested against a ``FakeBus`` that is a table of configuration registers, so a whole machine - an audio card, a network card, two identical cards, a bus-bridge stub - can be described in a test with no PCI bus present.

Building the image

```
```sh
just driverimg-list # show the catalogue without building
just driverimg # build dist/TrangorgeOS-drivers-<version>.img
just detect-test # run this crate's tests
```
```

``tools/mkdriverimg.sh`` writes the manifest the loader reads and places it, with the driver binaries, in a FAT16 partition in the image. See the script's header for the layout.

driverspace_workspace/drivers/Alsa-TrangorgeOS/README.md

Alsa-TrangorgeOS

An ALSA-compatible audio stack for TrangorgeOS: the ALSA object model and the ``aplay`` / ``amixer`` tools, ported to Rust and wired into the driver-space architecture.

What it is, and what it is not

It is ****not**** a kernel driver. It runs as a driver-space process, reaches hardware only through resources ``ds-manager`` grants, and serves requests over the same ``kapi-abi`` IPC the rest of the system speaks. There is no ``/dev/snd`` here: the "device files" are IPC endpoints.

What it ***is***, deliberately, is a faithful port of ALSA's concepts, so the mental model survives the port:

| ALSA (C)

| Here (Rust)

|

| | | |
|----------------------------|---|--|
| `snd_card` | [`core::Card`](src/core.rs) | |
| `snd_pcm` | [`core::Pcm`](src/core.rs) | |
| `snd_pcm_hw_params` | [`core::HwParams`](src/core.rs) | |
| `snd_pcm_state` | [`core::Stream`](src/core.rs) | |
| `snd_mixer_selem` | [`core::MixerElem`](src/core.rs) | |
| `snd_ctl_elem_id` | `kapi_abi::payloads::audio::MixerSelem` | |
| `snd_pcm_sw_params` / open | `AudioDevice::open` | |
| `snd_pcm_name_parse` | [`tools::DeviceName`](src/tools.rs) | |

Layering

```

```text
 aplay / amixer userspace tools (src/bin/)
 ■ kapi_abi DsCmd
 ds-fw-audio::AudioClient typed IPC client
 ds-fw-audio::AudioService capability gating, dispatch
 ■
 AlsaDriver handle table, validation
 core ALSA object model
 ac97 register map
 ■ Platform trait
 ds-manager MMIO / DMA / PCI grants
```

```

Each layer is testable on its own, which is why the seams exist at all. The driver carries no IPC code, so the whole ALSA model is exercised against a mock `Platform` with no kernel present.

Layout

| Path | Contents |
|-----------------|---|
| `src/core.rs` | the ALSA object model: cards, PCMs, hw-params, mixer elements, stream state |
| `src/ac97.rs` | the AC97 register map, ported from the legacy Odin driver |
| `src/driver.rs` | `AudioDevice` implementation and the `Platform` trait |
| `src/host.rs` | `SyscallPlatform`, the production `Platform` over IPC |
| `src/tools.rs` | `aplay` / `amixer` argument parsing and arithmetic |
| `src/main.rs` | the service loop `ds-manager` calls |
| `src/bin/` | the ported tools |

Capability model

Two bits, and the split is deliberate:

- * `CapId::ALSA\_ENUMERATE` - list PCMs, mixer elements, jacks and hwdep, and query hardware parameters. Read-only.
- * `CapId::ALSA\_STREAM` - open, prepare, start, transfer, drop, and write mixer values.

The driver is granted **only** `ALSA\_ENUMERATE` at start-up. Playing audio is a decision for `ds-manager`, not something the driver grants itself - a driver that hands itself full rights at boot makes the capability system decorative.

Building

```

```sh
cargo test -p alsa-trangorgeos --lib # the model, driver and tool parsing
cargo check -p alsa-trangorgeos --bins # the three binaries
```

```

The binaries are `no\_std` / `no\_main` and do not link on a host: like every driver-space binary in this workspace they need the target system and its linker script, so `cargo build` is expected to fail at the link step on a development host. `cargo check` is the meaningful host-side verification.

Ported from

The vocabulary and behaviour follow [ALSA](https://www.alsa-project.org/): `alsa-lib` for the model, `alsa-utils` for the tools. The AC97 register map was ported from this repository's own legacy `drivers/audiodriver` (Odin), and the offsets are asserted against it in `ac97.rs`'s tests.

driverspace_workspace/drivers/iommu-TrangorgeOS/README.md

TrangorgeOS IOMMU Driver

IOMMU (DMA remapping) driver for TrangorgeOS. It exposes the machine's DMA remapping units to the rest of the system as a schedulable driver-space device and mediates every IOVA -> PA mapping through `ds-manager`.

> **Status: integration-complete, hardware bring-up partial.**
> The driver is fully wired into the driver space (discovery, capability enforcement, domain/binding/mapping bookkeeping, fault and reservation reporting) and is covered by 76 unit tests. Programming the VT-d / AMD-Vi register files is **not** wired yet - see [Known gaps](#known-gaps).

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What it does

An IOMMU is a hardware unit that intercepts device DMA and rewrites addresses, so a device can only reach memory the OS explicitly granted it. This driver:

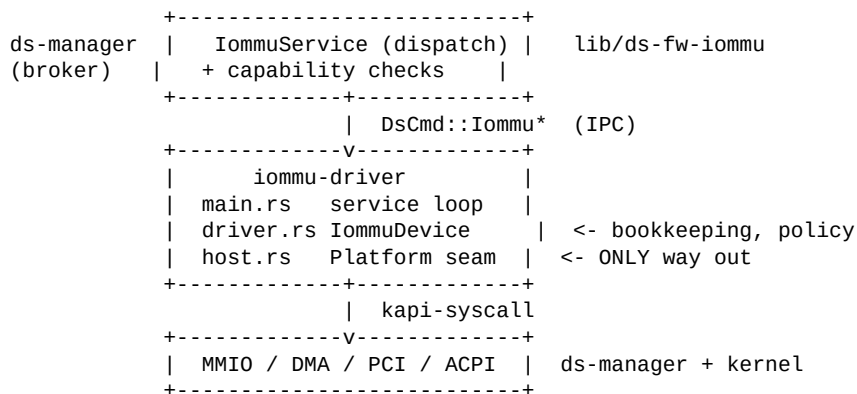
1. **Discovers** remapping units from firmware (on x86_64: the ACPI `DMAR` table, parsed by the hardware core's own parser).
2. **Owns address-space domains** - it allocates the backing page-table memory through the kernel and hands out opaque `DomainId`s.
3. **Binds PCI requesters** to domains and **establishes IOVA mappings** on request, invalidating the relevant TLBs on every state change.
4. **Reports** firmware DMA reservations and IOMMU faults to the system.
5. **Refuses anything it was not granted**: capability checks happen in one place before a request ever reaches the device.

It is a *mediator*, not a bypass. The driver never touches a physical address it has not asked the kernel for.

Layering

The driver obeys the driver-space rules in `../../Workspace.md`:
`drivers/` may depend on `lib/` only, never on `crates/`.

...



...

Two layers are cleanly separated:

- `src/host.rs` - the host resource layer.** A single `Platform`` trait is the `only`` boundary to the system. Its production implementation, `SyscallPlatform``, turns each operation into one `DsCmd``. Because it is a trait, the whole driver is unit-testable with no kernel present.
- `src/driver.rs` - the device layer.** `IommuDriver<P>`` implements the `ds_fw_iommu::IommuDevice`` contract on top of that seam. It holds no hardware knowledge of its own and reuses the hardware core's parsers.

How it talks to the system

Every host resource is requested, never assumed. `Platform`` has exactly seven operations, and each maps to one opcode:

| <code>Platform`</code> operation | Opcode | Purpose |
|--|-----------------------------------|---|
| <code>map_mmio(phys, size)`</code>         | <code>DsCmd::SysMapMmio`</code> | Map a controller register window |
| <code>unmap_mmio(region)`</code>           | <code>DsCmd::SysUnmapMmio`</code> | Release a register window |
| <code>alloc_dma(size, flags)`</code>       | <code>DsCmd::SysAllocDma`</code> | Domain page tables, bounce memory |
| <code>free_dma(region)`</code>             | <code>DsCmd::SysFreeDma`</code> | Release DMA memory |
| <code>read_pci(requester, off)`</code>     | <code>DsCmd::PciRead`</code> | Verify a requester really exists |
| <code>write_pci(requester, off, v)`</code> | <code>DsCmd::PciWrite`</code> | Enable bus mastering, etc. |
| <code>acpi_table(sig, inst)`</code>        | <code>DsCmd::SysAcpiTable`</code> | Fetch the <code>DMAR`</code> / <code>IVRS`</code> table |

`SysAcpiTable`` is important: drivers must **not** walk the RSDT themselves. A driver names the table it needs; the kernel locates it, validates it and maps it into the caller's address space. The reply carries the virtual base in `arg0`` and the length in `arg1``.

Requests are validated **before** they reach the kernel - a misaligned DMA size or a zero-length MMIO window is rejected locally, so a malformed caller can never turn into a kernel round trip.

The device contract

`IommuDriver`` implements `ds_fw_iommu::IommuDevice`` (see `[lib/ds-fw-iommu](../../lib/ds-fw-iommu)`). That trait is the whole surface `ds-manager`` can reach:

| Group | Methods |
|-----------------------|---|
| Discovery | <code>controller_count`</code> , <code>controller_info`</code> |
| Domain lifecycle | <code>domain_create`</code> , <code>domain_destroy`</code> |
| Binding | <code>bind`</code> , <code>unbind`</code> |
| Mapping | <code>map`</code> , <code>unmap`</code> , <code>invalidate`</code> |
| Firmware reservations | <code>reserved_region_count`</code> , <code>reserved_region`</code> |
| Faults | <code>pending_faults`</code> , <code>read_fault`</code> |

Design rules baked into the implementation:

- **A mapping always names a domain.** That is what keeps one device's IOVA space from being shared with another's by accident. Callers wanting pass-through create a domain and map everything, rather than using a domain-less side path.
- **A domain with live mappings cannot be destroyed** (`DsError::DeviceBusy``). Tearing it down would leave a device dereferencing freed frames.
- **Destroying a domain unbinds its requesters** in the same step.
- **Overlapping mappings in one domain are rejected** (`DsError::DeviceBusy``).
- **Bindings are verified against PCI config space**, so a non-existent device is never programmed into a context table.
- **Invalidation scope must reference known state.** Without that check, an invalidate request would be a licence to flush arbitrary hardware state.
- **Every state change invalidates.** Binding, unbinding, mapping, unmapping and domain teardown all drop the relevant IOTLB/context caches, otherwise a device would keep hitting a stale translation.

Capability model

Capability enforcement lives entirely in `IommuService``, before any request reaches `IommuDriver``. The driver implementation contains no permission logic at all, so there is exactly one place to audit.

| Capability | Bit | Grants |
|------------|-----|--------|
|------------|-----|--------|

```
|---|---|---|
| `CapId::IOMMU_ENUMERATE` | `1 << 20` | `IommuEnumerate`, `IommuQueryController`, `IommuReservedRegions`
`CapId::IOMMU_DOMAIN`	`1 << 21`	`IommuDomainCreate`, `IommuDomainDestroy`
`CapId::IOMMU_BIND`	`1 << 22`	`IommuBind`, `IommuUnbind`
`CapId::IOMMU_MAP`	`1 << 23`	`IommuMap`, `IommuUnmap`, `IommuInvalidate`
```

Read-only introspection is grouped under `IOMMU\_ENUMERATE`; each class of state mutation needs its own bit, so a caller that may enumerate controllers gains no ability to reprogram them. On start-up the driver grants **only** `IOMMU\_ENUMERATE` to itself; the mutating capabilities must be granted

Wire protocol

All IOMMU traffic uses the `0x0A` opcode category of `DsCmd`, with `#[repr(C)]` payloads from `kapi-abi` (the single source of truth for the ABI).

```
Opcode	Value	Args / payload	Reply
`IommuEnumerate`	`0x0A00`	`arg0` = index	`arg0` = controller count
`IommuQueryController`	`0x0A01`	`arg0` = index	payload: `IommuControllerInfo`
`IommuDomainCreate`	`0x0A02`	`arg0` = controller, `arg1` = hint	`arg0` = new `DomainId`
`IommuDomainDestroy`	`0x0A03`	`arg0` = domain	-
`IommuBind`	`0x0A04`	payload: `IommuBindPayload`	-
`IommuUnbind`	`0x0A05`	`arg0` = controller, `arg1` = requester	-
`IommuMap`	`0x0A06`	payload: `IommuMapPayload`	`arg0` = physical base
`IommuUnmap`	`0x0A07`	payload: `IommuUnmapPayload`	-
`IommuInvalidate`	`0x0A08`	payload: `IommuInvalidatePayload`	`arg0` = completed scope
`IommuReservedRegions`	`0x0A09`	`arg0` = index	`arg0` = total, payload: `IommuReservedRegionPayload`
`IommuFaultRead`	`0x0A0A`	`arg0` = index	`arg0` = pending total, payload: `IommuFaultPayload`
```

Structured payloads follow the convention already used by `LogPayload`: the request pointer goes in `arg0` and its length in `arg1`. Replies that carry a payload put the element count in `arg0` and the encoded byte count in `arg2`, so a caller can always tell a truncated write (`DsError::BufferTooSmall`) from an empty one.

`ds-manager` also needs one primitive outside the IOMMU category: `DsCmd::SysAcpiTable` (`0x000C`) with `AcpiTableRequest`. Reply `arg0` is the mapped virtual base, `arg1` the length in bytes.

Source layout

```
...
src/
### lib.rs      # crate root; re-exports the hardware core, adds host/driver
### main.rs     # driver_main: probe, wrap in IommuService, serve loop
### host.rs     # Platform trait + SyscallPlatform + test MockPlatform
### driver.rs   # IommuDriver: implements IommuDevice
■
### ctrl.rs     # [upstream] Controller trait, command queues, descriptors
### caps.rs     # [upstream] bindings, permissions, capability flags
### info.rs     # [upstream] controller/domain descriptors
### addr.rs     # [upstream] MMIO/IO/IOVA address types
### error.rs    # [upstream] error taxonomy
### firm/       # [upstream] ACPI + PCIe firmware types
### arch/       # [upstream] per-architecture hardware cores
    ### x86_64/intel/ # VT-d: DMAR parsing, register window, domains
    ### x86_64/amd/   # AMD-Vi: IVRS parsing, second-level page tables
    ### aarch64/     # ARM SMMU v2/v3, IORT parsing
    ### riscv64/     # RISC-V IOMMU, RIMT
...
```

Everything marked `[upstream]` is the `kore` IOMMU core, unchanged. The TrangorgeOS integration is exactly `host.rs`, `driver.rs` and the `main.rs` service loop.

Building and testing

Run from the workspace root (`driverspace\_workspace/`), because the driver is a workspace member:

```
```sh
Build the library and the driver binary
cargo build -p iommu-driver
```

```
Type-check every target (lib, bin)
cargo check -p iommu-driver --all-targets
```

```
Run the unit tests
cargo test -p iommu-driver --lib
```

```
Framework tests
cargo test -p ds-fw-iommu --lib
```
```

The tests run on the host and need no kernel: `MockPlatform`` in `host.rs`` stands in for `SyscallPlatform``, so discovery, domain allocation, mapping bookkeeping, capability gating and the error paths are all genuinely executed.

Note that the binary target sets `test = false` / bench = false`` in `Cargo.toml``: it is `#![no_main]`` with its own `#[panic_handler]``, so Cargo must not build a test harness for it. Unit tests live in the library target.

explicitly by `ds-manager``.

Known gaps

Stated plainly, because a driver that looks finished and is not is worse than one that admits where it stops.

1. **Register programming is not wired yet.** `map``, `bind``, `unbind`` and `invalidate`` currently update bookkeeping and request invalidation, but do not program `VtdUnit`` / the second-level page tables. The remaining work is a thin `arch`` seam: build a `kore_memory::Mapping`` over the MMIO region returned by `Platform::map_mmio``, construct a `VtdUnit``, and route `IommuDevice::map`` into `Controller::remap``. Until that lands, the driver isolates DMA but does not yet translate it.
2. **Only the x86_64 `DMAR` probe is wired.** The IVRS (AMD-Vi), IORT (ARM SMMU) and RIMT (RISC-V) parsers ship in the hardware core and are reachable, but `probe_acpi()`` currently only asks for `DMAR``. Adding AMD is a second `acpi_table("b"IVRS", 0)`` branch.
3. **No end-to-end run yet.** `crates/ds-manager`` is still a skeleton, so the `DsCmd::Iommu*`` path has not been exercised against a live manager. The framework is complete and unit tested; the manager side is what remains.
4. **``kore-memory` is pinned.** It is pinned to rev `e03cdb79`` because later revisions dropped `Mapping::modify_vo32/64``, which the VT-d register window uses. Revisit when the hardware core is rebased.
5. **One flaky upstream test.** `arch::x86_64::intel::ctrl::tests::queued_invalidation_table_address_rejection`` builds its buffer from a heap `Vec`` but requires 32-byte alignment, so it fails depending on the host allocator. It is a pre-existing bug in the untouched upstream core, not in the integration.

Provenance and licence

The hardware core (`src/ctrl.rs``, `src/caps.rs``, `src/info.rs``, `src/addr.rs``, `src/error.rs``, `src/firm/``, `src/arch/``) originates from the `kore`` IOMMU module by Cass Sheng / MicroPerception, vendored under the MIT licence in `[LICENSE](./LICENSE)`. See `[THANKS.md](./THANKS.md)` for the acknowledgement.

The TrangorgeOS integration layer (`src/host.rs``, `src/driver.rs``, the `main.rs`` service loop) and the driver-space support crates (`lib/ds-fw-iommu``, plus the IOMMU additions to `lib/kapi-abi`` and `lib/kapi-syscall``) were written for this project.

The upstream repository notes that the `kore`` module itself has been unmaintained since May 2026 and was not used in `kore``'s production environment. The hardware core is vendored here unchanged; all TrangorgeOS-specific behaviour lives in the integration layer described above.

Further reading

- `[docs/INTEGRATION.md](./docs/INTEGRATION.md)` - message flows, error mapping, extension points.
- `[../lib/ds-fw-iommu](../lib/ds-fw-iommu)` - the device-class contract.
- `[../Workspace.md](../Workspace.md)` - driver-space layering rules.

driverspace_workspace/drivers/iommu-TrangorgeOS/THANKS.md

Acknowledgement

Thank you to Cass Sheng for providing the source code for this driver.

The driver was made available under its original license and will be integrated into the system as part of

The original source and license information are preserved in the corresponding component repository.

driverspace_workspace/drivers/iommu-TrangorgeOS/docs/INTEGRATION.md

IOMMU Driver Integration Guide

How ``iommu-driver`` attaches to TrangorgeOS: the message flows, the error mapping, the seams designed for extension, and what to check when changing anything here.

The [README](../README.md) covers what the driver does and the opcode tables. This document covers the mechanics.

Contents

- [The ``Platform`` seam](#the-platform-seam)
- [Start-up sequence](#start-up-sequence)
- [Service loop](#service-loop)
- [Discovery: the ACPI path](#discovery-the-acpi-path)
- [Domain and mapping bookkeeping](#domain-and-mapping-bookkeeping)
- [Error mapping](#error-mapping)
- [Extension points](#extension-points)
- [Testing strategy](#testing-strategy)
- [Invariants worth preserving](#invariants-worth-preserving)

The ``Platform`` seam

Everything the driver needs from the system is seven methods on one trait (``src/host.rs``). This is the single seam, and it is deliberately narrow:

```
```rust
pub trait Platform {
 fn map_mmio(&self, phys: u64, size: u64) -> Result<MappedRegion, DsError>;
 fn unmap_mmio(&self, region: MappedRegion) -> Result<(), DsError>;
 fn alloc_dma(&self, size: u64, flags: u32) -> Result<MappedRegion, DsError>;
 fn free_dma(&self, region: MappedRegion) -> Result<(), DsError>;
 fn read_pci(&self, requester: u32, offset: u32) -> Result<u32, DsError>;
 fn write_pci(&self, requester: u32, offset: u32, value: u32) -> Result<(), DsError>;
 fn acpi_table(&self, signature: [u8; 4], instance: u32) -> Result<AcpiView, DsError>;
}
```
```

Two properties are worth preserving when this trait changes:

- ****No raw pointers in the interface.**** Regions come back as ``MappedRegion { phys, virt, size }`` values, and turning one into a pointer is an explicit ``unsafe fn`` with a documented contract. The framework and device layers never dereference one.
- ****It is a trait, not a struct.**** ``IommuDriver<P: Platform>`` is generic over it, which is what makes the driver testable without a kernel and lets the transport be swapped (shared-memory ring, ioctl) without touching device logic.

``SyscallPlatform`` is the production implementation; each method is one ``kapi_syscall::sys_ipc_call``. ``MockPlatform`` (test-only) records requests and serves mappings from test-owned buffers.

Start-up sequence

``driver_main()`` in ``src/main.rs``:

1. Construct ``IommuDriver::new(SyscallPlatform::new())``.
2. Call ``probe_acpi()``. A missing ``DMAR`` table is ****not**** an error - it means the machine has no IOMMU, and the driver reports zero controllers.

3. Wrap the driver in ``IommuService`` and grant `**only**` ``CapId::IOMMU_ENUMERATE``. The mutating capabilities stay ungranted.
4. Park the service in a static and enter ``serve()``.

Granting only the read capability at start-up is intentional: a driver that grants itself full rights at boot has made the capability system decorative.

Service loop

```
...
loop {
  match kapi_syscall::sys_try_recv() {
    None => { sys_yield(); continue }           // ring empty, do not spin hot
    Some(msg) => handle(msg),
  }
}

fn handle(msg) {
  copy_in(msg, REQUEST_BUF);                    // msg.arg0/arg1 -> local buffer
  let request = Request::new(msg, &REQUEST_BUF);
  let reply = service.dispatch(&request, REPLY_BUF);
  kapi_syscall::sys_ipc_reply(reply.into_message(&msg), &REPLY_BUF[..len], len);
}
...
```

Points to note:

- `**`sys_try_recv` never blocks.**` It returns ``None`` when the ring is empty and the loop yields. A blocking receive would stall the driver on a request that may never arrive.
- `**The request payload is copied into a local buffer first.**` The framework layer therefore works on byte slices and contains no raw pointers; the copy is length-clamped in ``copy_in``.
- `**Every request gets a reply, including failures.**` A request left unanswered would block the caller on the ring forever. Failures are logged and answered with the error status.
- `**The reply payload is copied into `kapi-syscall`'s own buffer**`, so the driver can reuse its scratch buffer immediately. A payload longer than ``MAX_REPLY_PAYLOAD`` is truncated; the caller checks the length in ``arg2``.

Discovery: the ACPI path

``IommuDriver::probe_acpi()`` (x86_64 only) is the interesting part, because it deliberately does `**not**` re-implement anything:

1. ``platform.acpi_table(*b"DMAR", 0)`` asks the kernel for the table. ``DsError::DeviceNotFound`` means "no IOMMU here" and returns ``Ok(())``.
2. The returned ``AcpiView`` is wrapped in a ``kore_memory::Mapping<VtdSecondLevelPte, VirtAddr, PhysAddr>`` spanning exactly the mapped range.
3. ``DmarTable::from_mapping(&mapping)`` - the hardware core's own parser - walks the table. No second DMAR parser exists in the driver.
4. ``for_each_drhd`` registers one ``IommuControllerInfo`` per remapping unit (segment, register window base and size).
5. ``for_each_rmrr`` records firmware DMA reservations. The payload's ``limit`` is an `**inclusive**` upper bound, so the parser's exclusive ``end`` becomes ``end - 1``.

This is the pattern to follow for AMD-Vi and ARM: wrap the kernel-provided mapping and call the core's parser, instead of adding a second implementation.

Domain and mapping bookkeeping

Fixed-capacity arrays keep the driver ``no_std`` and allocation-free:

| Array | Capacity | Meaning |
|----------------------------|------------------------------------|----------------------------|
| <code>`controllers`</code> | <code>`MAX_CONTROLLERS`</code> = 8 | discovered units |
| <code>`domains`</code> | <code>`MAX_DOMAINS`</code> = 16 | live address-space domains |
| <code>`mappings`</code> | <code>`MAX_MAPPINGS`</code> = 128 | IOVA -> PA entries |
| <code>`bindings`</code> | <code>`MAX_BINDINGS`</code> = 64 | requester -> domain |
| <code>`reserved`</code> | <code>`MAX_RESERVED`</code> = 16 | firmware reservations |

```
| `faults` | `MAX_FAULTS` = 16 | reported faults (ring) |
```

Exhaustion surfaces as `DsError::OutOfMemory` rather than a panic.

Two subtleties that bit during development and are now covered by tests:

- `reserved\_region(index)` checks `index < reserved\_len`, not just `reserved.get(index)`. The array is fixed-size, so `get()` alone happily returns a slot that was never filled.
- `report\_fault` overwrites the **oldest** entry once the ring is full, so the cache always holds the most recent `MAX\_FAULTS` faults.

Error mapping

The driver returns `kapi\_abi::DsError` directly, so no translation layer is needed on the hot path. The choices that matter:

| Situation | Error | Why |
|---|------------------|--|
| Unknown controller / domain / requester | `DeviceNotFound` | The thing being addressed does not exist |
| Malformed request (bad requester id, misaligned IOVA, non-page-multiple size) | `InvalidMessage` | Call |
| Domain with live mappings destroyed | `DeviceBusy` | Would leave a device on freed frames |
| Overlapping mapping in one domain | `DeviceBusy` | Ambiguous ownership of the IOVA range |
| Table or mapping table full | `OutOfMemory` | Fixed-capacity exhaustion |

Extension points

Adding AMD-Vi. In `probe\_acpi`, branch on `acpi\_table(\*b"IVRS", 0)` and register controllers from `IvrsTable`. The `IommuKind` enum already has `AmdVi`, and `IvrsError` already converts into the core `Error`.

Wiring the hardware layer. This is the main remaining work. The shape is:

1. In `probe\_acpi`, `map\_mmio` the register window and keep the `MappedRegion` in the `ControllerEntry`.
2. Build a `kore\_memory::Mapping` over it and construct `VtdRegisterWindow::new(mapping)`.
3. Allocate the root and queued-invalidation tables with `alloc\_dma(.., DMA\_CONTIGUOUS | DMA\_COHERENT)` and build a `VtdUnit`.
4. Route `IommuDevice::map` / `unmap` into `Controller::remap` / `unmap`, and `bind` / `unbind` into `Controller::bind` / `unbind`.

The bookkeeping layer is already shaped for this: domains carry their page table region, and `IommuDevice` already validates everything the hardware layer would otherwise have to re-check.

A second `Platform`. Implement the trait over a different transport and parameterise `IommuDriver<P>`. Nothing else changes.

Manager-side routing. `IommuService::dispatch` is the whole server side; a `ds-manager` only needs to own one and call it. Nothing in the driver changes.

Testing strategy

`MockPlatform` makes the driver testable with no kernel, and the tests exercise real logic rather than just types:

- enumeration and out-of-range `controller\_info`;
- a machine with no `DMAR` reporting zero controllers, not an error;
- domain allocation, duplicate hints rejected, unknown controller rejected;
- a domain with live mappings refusing destruction;
- overlapping mappings rejected, disjoint ones accepted;
- misaligned IOVA, non-page-multiple sizes and unknown domains rejected;
- faults bounded by the ring, out-of-range reads refused;
- reservations reported in insertion order, out-of-range refused;
- invalidate scopes validated against known controller/domain;
- bindings rejected for unknown requesters, malformed ids and unknown domains.

The framework (`lib/ds-fw-iommu`) has its own 17 tests covering payload round-trips for every struct, capability gating, and dispatcher edge cases.

Invariants worth preserving

If you change this driver, keep these. Each exists because removing it breaks a safety property:

1. ****No physical address is used without a `Platform` call behind it.****
2. ****Capability checks live in `IommuService`, not in the driver.**** Duplicating them in the device layer means two places to audit and one to forget.
3. ****A mapping always names a domain**, and a domain with live mappings cannot be destroyed.**
4. ****Every state change invalidates the caches that depend on it.****
5. ****Bindings are verified against PCI config space**** before being recorded.
6. ****Requests are validated locally before a kernel round trip.****
7. ****`IommuDevice` stays object-safe and free of `kore\_memory` types**, so the framework does not inherit the hardware core's dependencies.**

`Platform` errors propagate unchanged. A `PermissionDenied` from the manager stays `PermissionDenied`; the driver does not retry or reinterpret it.

driverspace_workspace/formal/README.md

driverspace_workspace/lib/ds-fw-iommu/README.md

`ds-fw-iommu` - IOMMU Device-Class Framework

The TrangorgeOS driver-space contract for IOMMU devices. This crate holds the ****contract and the wire vocabulary only**** - never hardware register logic.

Hardware lives in `drivers/iommu-driver`; management logic lives in `crates/ds-manager`. Both depend on this crate and nothing else, which is what satisfies the workspace rule that `drivers/` must never depend on `crates/` and the rule against duplicated code.

Layout

| Module | Role |
|-----------|--|
| `traits` | `IommuDevice` - what a driver implements |
| `service` | `IommuService` - server-side dispatch **with capability enforcement** |
| `client` | `IommuClient` - client-side calls encoded as `DsCmd` IPC |
| `codec` | payload encode/decode, shared by both sides |
| `types` | `ControllerId`, `DomainId`, `RequesterId`, `IoRange` |

Every wire type is `#[repr(C)]` and comes from `kapi-abi`. This crate adds ****no**** ABI surface of its own - opcodes and payload layouts live in one place.

The three roles

```
...
      caller                      driver process
      -----
      IommuClient  --DsCmd-->  IommuService --> IommuDevice
      (encodes)          (checks caps)  (hardware)
          ^                      |
          +----- reply payload ---+
...

```

- A ****driver**** implements `IommuDevice` and wraps it in an `IommuService`.
- A ****caller**** (userspace service, or `ds-manager` acting on someone's behalf) uses `IommuClient` to issue requests.
- `IommuService` is the only place capabilities are checked.

`IommuDevice`

```
```rust
pub trait IommuDevice {
 // discovery
 fn controller_count(&self) -> usize;
 fn controller_info(&self, index: usize, out: &mut IommuControllerInfo) -> bool;

 // domain lifecycle
 fn domain_create(&mut self, controller: ControllerId, hint: u32)
 -> Result<DomainId, DsError>;
}
```

```

fn domain_destroy(&mut self, domain: DomainId) -> Result<(), DsError>;

// binding
fn bind(&mut self, request: &IommuBindPayload) -> Result<(), DsError>;
fn unbind(&mut self, controller: ControllerId, requester: RequesterId)
 -> Result<(), DsError>;

// mapping
fn map(&mut self, request: &IommuMapPayload) -> Result<u64, DsError>;
fn unmap(&mut self, request: &IommuUnmapPayload) -> Result<(), DsError>;
fn invalidate(&mut self, request: &IommuInvalidatePayload)
 -> Result<IommuInvalidateScope, DsError>;

// firmware reservations
fn reserved_region_count(&self) -> usize;
fn reserved_region(&self, index: usize, out: &mut IommuReservedRegionPayload) -> bool;

// faults
fn pending_faults(&self) -> usize;
fn read_fault(&mut self, index: usize, out: &mut IommuFaultPayload) -> bool;
}

```

Properties that keep the trait usable as a framework boundary:

- **Object-safe.** No generics, no ``Self`` in return position.
- **No ``kore_memory`` types.** The framework must not inherit the hardware core's dependencies; everything crosses as plain integers.
- **Index-based getters return ``bool``.** Out-of-range returns ``false`` and leaves ``out`` untouched - no sentinel values to guess at.
- **Every fallible method returns ``kapi_abi::DsError``,** which is also the wire status, so no translation layer is needed.

### Domain ownership rule

Domains are allocated by the driver and named by ``DomainId``. **Mapping requests always carry a domain** - that is what keeps one device's IOVA space from being shared with another's by accident. A caller wanting pass-through semantics creates a domain and maps everything into it, rather than reaching for a domain-less side path.

### ``IommuService``

```

`rust
let mut service = IommuService::new(device);
service.grant(CapId::IOMMU_ENUMERATE);

let reply = service.dispatch(&Request::new(msg, &payload_bytes), &mut out);

```

- ``dispatch`` decodes the opcode, checks the required capability, decodes the payload, calls the device, and encodes the reply.
- **Capabilities are checked here and nowhere else.** A driver implementation should contain no permission logic, so there is exactly one place to audit.
- A reply with a payload writes it into ``out`` and reports the byte count in ``Reply::arg2``. A too-small buffer yields ``DsError::BufferTooSmall``, so the caller can always distinguish truncation from an empty reply.
- A non-IOMMU opcode is rejected with ``DsError::InvalidMessage`` - even if the caller holds every capability.

### Capability map

Capability	Bit	Opcodes
<code>`IOMMU_ENUMERATE`</code>	<code>`1 &lt;&lt; 20`</code>	<code>`IommuEnumerate`</code> , <code>`IommuQueryController`</code> , <code>`IommuReservedRegions`</code> , <code>`IommuDomainCreate`</code> , <code>`IommuDomainDestroy`</code>
<code>`IOMMU_DOMAIN`</code>	<code>`1 &lt;&lt; 21`</code>	<code>`IommuDomainCreate`</code> , <code>`IommuDomainDestroy`</code>
<code>`IOMMU_BIND`</code>	<code>`1 &lt;&lt; 22`</code>	<code>`IommuBind`</code> , <code>`IommuUnbind`</code>
<code>`IOMMU_MAP`</code>	<code>`1 &lt;&lt; 23`</code>	<code>`IommuMap`</code> , <code>`IommuUnmap`</code> , <code>`IommuInvalidate`</code>

``required_capability(cmd)`` exposes the mapping, so a manager can implement policy against it without duplicating the table.

```
`IommuClient`
```

Mirrors `ds-ipc::ManagerClient`: it ships a request and reads the reply, and holds no device state. Failures come back as `DsError` and are meant to be propagated with `?`.

```
```rust
let client = IommuClient::new(service_handle);
let count = client.controller_count()?;
let domain = client.create_domain(ControllorId(0))?;
client.bind(ControllorId(0), RequesterId::new(0, 0x2a, 5, 3), domain)?;
```
```

Argument validation that can be done locally is done locally: an invalid `DomainId`, a `RequesterId::NONE`, or a misaligned map size is rejected before any IPC round trip.

```
Payload passing
```

Structured payloads follow the convention already used by `LogPayload`: the pointer goes in `DsMsg::arg0`, the length in `arg1`. `MAX\_PAYLOAD\_LEN` is the worst-case encoded size of any IOMMU payload (48 bytes) and sizes the scratch buffers on both sides.

The framework itself contains **no raw pointers**: the transport copies bytes into a `Request` slice, and the service writes its reply into a caller-provided `&mut [u8]`. That is what lets the tests feed plain byte slices.

```
Types
```

```
Type	Wire form	Notes
`ControllorId`	`u32`	index into `0..controller_count`
`DomainId`	`u32`	`DomainId::INVALID` = `u32::MAX`; `is_valid()` checks it
`RequesterId`	`u32`	`(segment << 16) \| bdf`; `NONE` = unowned; `Display` prints `ssss:bb:dd.f`
`IoRange`	`{ iova: u64, size: u64 }`	`is_valid`, `is_aligned`, `pages`
```

The newtypes exist so a controller index cannot be passed where a domain id belongs. `RequesterId`'s layout matches the driver's `PciDevice`, so conversion between the two is free.

```
Testing
```

17 unit tests, no kernel required:

- payload round-trips for **every** struct (`every\_payload\_round\_trips`), including the padding fields that a naive codec silently misaligns;
- truncated payloads are rejected rather than read past the end;
- capability gating: mutation without the capability is `PermissionDenied`;
- a non-IOMMU opcode is rejected even with `CapId::KERNEL\_ALL`;
- a too-small reply buffer is reported, not silently truncated;
- argument validation in the client that never reaches IPC.

```
```sh
cargo test -p ds-fw-iommu --lib
```
```

```
See also
```

- [../../drivers/iommu-TrangorgeOS/README.md](../../drivers/iommu-TrangorgeOS/README.md) - the driver built on this framework.
- [../../kapi-abi/src/payloads/iommu.rs](../../kapi-abi/src/payloads/iommu.rs) - the payload definitions this codec encodes.

**[kernel\\_Workspace/C\\_base/kc-abi/docs/abi\\_spec.md](#)**

**[kernel\\_Workspace/C\\_base/kc-abi/docs/architecture.md](#)**

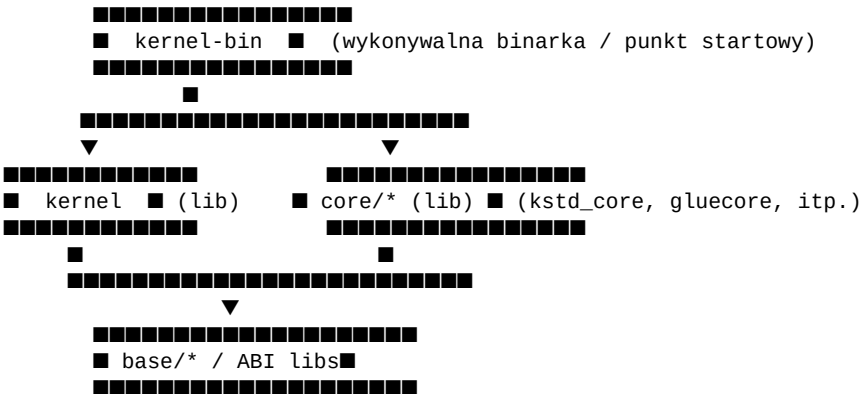
kernel\_Workspace/C\_base/kc-abi/docs/memory\_model.md

kernel\_Workspace/C\_base/kc-abi/kc-abi/docs/abi\_spec.md

kernel\_Workspace/C\_base/kc-abi/kc-abi/docs/architecture.md

kernel\_Workspace/C\_base/kc-abi/kc-abi/docs/memory\_model.md

kernel\_Workspace/README.md



kernel\_Workspace/core/core.md

kernel\_Workspace/kernel/README.md

...

kernel\_Workspace/kernel/src/mm/MM\_PLAN\_ULEPSZENIA.md

```
Plan ulepszenia podsystemu MM w TrangorgeOS

Repozytorium: [CTRL-F-0rg3/TrangorgeOS](https://github.com/CTRL-F-0rg3/TrangorgeOS)
Analizowana gałęź: `stabilizing`
Analizowany commit: `7a0d836f2180874aac3e8c92493c791aec89a8b3`
Autor analizy: ctrl
Data analizy: 20 sierpnia 2026 r.

masz kropke gdzie ostatnio dodałem

1. Zakres i podsumowanie

Przeanalizowany został cały katalog `kernel/src/mm`, w tym: inicjalizacja pamięci architektury, bitmapy.
Architektura jest sensownie rozdzielona na warstwy. Obecny przepływ startowy (`arch_memory_init` → pagin
> **Najważniejsza rekomendacja:** przed dodawaniem nowych funkcji należy ustabilizować kontrakty allocat

2. Ocena stanu obecnego

Obszar	Stan	Ocena
Rozdzielenie PMM/VMM/heap/process	Istnieje i jest czytelne	Dobra baza architektoniczna
Bitmapa ramek	Obsługuje pojedyncze i ciągłe zakresy	Wymaga testów granicznych i lepszej wydajności
Buddy allocator	Działa jako główna pamięć heap	Brak synchronizacji SMP i dokładnego rozmiaru bloków
Slab allocator	Zaimplementowany, ale `HEAP_USE_SLAB` wynosi `0`	Niewykorzystana optymalizacja; ma r
API alokacji	`kmalloc`, `kzalloc`, `kcalloc`, `krealloc`, strony	Kontrakty `aligned/pages` są niepe
```

| Paging/VMM | Słabe mapowanie, translate, protect i address spaces | Brakuje pełnej transakcyjności i walidacji DMA/contiguous | Istnieją osobne moduły | Wymagają bezpiecznego liczenia rozmiarów i modelu cache | Izolacja | SMEP/SMAP/NX i audyt PML4 | Wymaganie mechanizmów nie jest raportowane jako błąd krytyczny | Testy | Jest `kernel/src/testing.rs` i kilka self-testów | Brak automatycznej, szerokiej macierzy testów | Build/CI | W środowisku analizy `cargo` nie było dostępnego | Nie udało się potwierdzić kompilacji; pot

## 3. Problemy wymagające naprawy

### P0 – bezpieczeństwo i poprawność krytyczna

#### P0.1. Blokady wywołujące przerwanie nie są blokadami SMP

PMM, slab i przestrzenie adresowe używają wzorca `pushfq; cli` oraz lokalnego licznika zagnieżdżenia. Wy

**\*\*Pliki:\*\*** `kernel/src/mm/alloc/physical/pmm.c`, `kernel/src/mm/alloc/heap/slab.c`, `kernel/src/mm/process

**\*\*Zadanie:\*\*** zastąpić blokady per-CPU rzeczywistymi spinlockami ticket/MCS albo istniejącym prymitywem S

**\*\*Kryterium akceptacji:\*\*** test wielordzeniowy wykonuje równoległe co najmniej 100 000 alokacji i zwolnień

#### P0.2. Double-free w slabie może uszkodzić listę wolnych obiektów

`slab\_free()` sprawdza zakres i wyrównanie wskaźnika, ale nie przechowuje informacji, czy konkretny obiekt

**\*\*Plik:\*\*** `kernel/src/mm/alloc/heap/slab.c`, szczególnie funkcja `slab\_free()`.

**\*\*Kryterium akceptacji:\*\*** drugie `slab\_free(ptr)` nie modyfikuje allocatorów i generuje błąd diagnostyczny

#### P0.3. Brak walidacji overflow w API przestrzeni adresowych

W wielu miejscach granice są liczone jako `addr + len`, a następnie wyrównywane, np. w `aspace\_map\_at`,

**\*\*Plik:\*\*** `kernel/src/mm/process/address\_space.c`, okolice obliczeń `a`/`b`.

**\*\*Zadanie:\*\*** wprowadzić wspólną funkcję `range\_from\_addr\_len(addr, len, alignment, out\_start, out\_end)`,

1. odrzuca `len == 0`;
2. sprawdza `addr + len` bez overflow;
3. bezpiecznie wyrównuje początek w dół i koniec w górę;
4. sprawdza kanoniczność oraz pełne granice user space;
5. zwraca zakres półotwarty `[start, end)`.

**\*\*Kryterium akceptacji:\*\*** testy dla `UINT64\_MAX`, adresów niekanonicznych, przejścia przez granicę user/

#### P0.4. Nieatomowy rollback w `aspace\_brk()`

`aspace\_brk()` najpierw tworzy lub rozszerza VMA, a dopiero potem wywołuje `map\_anon\_pages()`. Jeżeli al

**\*\*Plik:\*\*** `kernel/src/mm/process/address\_space.c`, funkcja powiększania sterty.

**\*\*Zadanie:\*\*** zastosować transakcję: najpierw przygotować zakres i zmapować strony, a dopiero po pełnym s

**\*\*Kryterium akceptacji:\*\*** wymuszony brak ramek w połowie operacji pozostawia identyczne VMA, `brk` i lic

### P1 – błądy kontraktów allocatorów

#### P1.1. `kmallocc\_aligned()` nie gwarantuje dowolnego wyrównania

`buddy\_alloc\_aligned()` zaokrągla potrzebny rozmiar do potęgi dwójki i zakłada, że baza buddy jest odpow

**\*\*Pliki:\*\*** `kernel/src/mm/alloc/api/alloc.c`, `kernel/src/mm/alloc/heap/buddy.c`.

**\*\*Zadanie:\*\*** jasno zdefiniować API: albo przyjmować wyłącznie potęgę dwójki i to walidować, albo dodać

#### P1.2. `kfree\_pages()` ignoruje parametr `pages`

`kfree\_pages(void \*ptr, size\_t pages)` ignoruje `pages` i deleguje do `heap\_free()`. To tworzy mylący kon

**\*\*Plik:\*\*** `kernel/src/mm/alloc/api/alloc.c`.

#### P1.3. `krealloc()` nie przechowuje rzeczywistego rozmiaru zwrócenia

Poza debug mode `heap\_usable\_size()` zwraca rozmiar zaokrąglonego bloku buddy, a nie liczbę bajtów zwrócon

#### #### P1.4. Overflow w alokacji contiguous i DMA

`contig\_bytes\_to\_frames()` i `dma\_bytes\_to\_frames()` wykonuj` bytes + ARCH\_PAGE\_SIZE - 1` bez jawnej ochrony

**\*\*Pliki:\*\*** `kernel/src/mm/alloc/special/contiguous.c`, `kernel/src/mm/alloc/special/dma.c`.

**\*\*Kryterium akceptacji:\*\*** najwi`ksze warto`ci `size\_t`, `UINT64\_MAX`, d`lugo`ci nieb`d`ce wielokrotno`ci

#### ### P1 – paging, TLB i mapowania

Nale`y przeprowadzi` audyt wszystkich `cie`ek `map`, `unmap` i `protect` pod k`tem transakcyjno`ci. Je`

**\*\*Zadania:\*\***

| Zadanie                              | Oczekiwany rezultat                                                 |
|--------------------------------------|---------------------------------------------------------------------|
| Ujednolici` map_range`/`unmap_range` | Jedna polityka wyrównania, overflow i rollbacku                     |
| Doda` statusy b`dów                  | Rozró`nienie `invalid`, `already mapped`, `no memory`, `permission` |
| Doko`czy batch TLB                   | Jedno `invlpg` na adres lub pe`ny flush przy przepe`nieniu batcha   |
| Doda` shutdown SMP                   | IPI lub bezpieczny mechanizm synchronizacji aktywnych CR3           |
| Walidowa` PTE flags                  | Zakaz USER w mapowaniach kernela i spójne NX/WRITE/DEVICE           |
| Testowa` aliasy                      | Mapowanie tej samej ramki pod ró`ne VA z kontrol` uprawnie`         |

#### ### P1 – semantyka `mmap`/VMA

Obecne `mmap()` rozpoznaje g`ównie `MAP\_FIXED` i kieruje pozosta`e przypadki do anonimowego mapowania. `

**\*\*Zadania:\*\***

- Walidowa` dozwolone kombinacje flag i odrzuca` nieobs`ugiwane zamiast je ignorowa`.
- Rozdzieli` VMA na prefix/zakres/suffix przy cz`ciowym `mprotect`.
- Wykonywa` wyszukanie VMA pod t` sam` blokad` co modyfikacj`.
- Ustali`, czy mapowanie anonimowe ma by` eager czy demand-paged.
- Doda` guard pages dla stosu oraz limit wzrostu stosu.
- Wprowadzi` copy-on-write dla `MAP\_PRIVATE`, gdy pojawi si` fork/procesy potomne.

#### ## 4. Ulepszenia architektury i wydajno`ci

# .

##### ### 4.1. PMM strefowy zamiast jednej bitmapy dla ca`ego maksimum fizycznego

Obecna bitmapa indeksuje PFN-y a` do najwy`szego u`ytecznego adresu, tak`e przez dziury w fizycznej prze

##### ### 4.2. Per-CPU cache dla ma`ych alokacji

Po ustabilizowaniu slaba warto doda` per-CPU magazine/cache dla najcz`stszych klas 16-256 B. Ograniczy t

##### ### 4.3. Lepszy allocator stron dla du`ych bloków

Buddy jest dobrym `ród`em du`ych bloków, ale obecna sterta mapuje i odmapowuje strony przy ka`dej operac

##### ### 4.4. Cache i DMA

`dma\_alloc\_coherent()` potrzebuje jawnego modelu spójno`ci zale`nego od architektury. Nale`y rozró`ni

##### ### 4.5. Bezpieczne flagi ochrony

Nale`y ustali` jedn` reprezentacj` ochrony. Obecnie wyst`puj` osobne zestawy `PROT\_\*` i `virt::\*` flags

#### ## 5. Diagnostyka i obserwowalno`

Debug allocator ma dobre podstawy: magic, poison, canary, leak table i statystyki. Ograniczeniem jest sta

| Ulepszenie                                | Cel                                                        |
|-------------------------------------------|------------------------------------------------------------|
| Rozszerzy` leak table lub u`y` hash table | Brak cichego nie`ledzenia alokacji po przekroczeniu limitu |
| Doda` `alloc_id`                          | Jednoznaczne `ledzenie `ycia obiektu                       |
| Doda` guard pages dla du`ych alokacji     | Wykrywanie overrun/underrun poza canary                    |
| Raportowa` ownera cache i CPU             | Debugowanie SMP                                            |
| Doda` histogram rozmiarów                 | Strojenie klas slab/buddy                                  |

| Wykrywa niezgody par allocator/free | Natychmiastowe raportowanie błądów ownership |  
| Eksportowa snapshot MM | Testy regresyjne: free/used, VMA, mapowania, TLB |

## ## 6. Plan implementacji etapami

| Etap | Zakres                                   | Priorytet | Definicja ukończenia                                       |
|------|------------------------------------------|-----------|------------------------------------------------------------|
| A    | Wspólne helpery overflow/range/alignment | P0        | Wszystkie publiczne API zakresowe używają helperów         |
| B    | Spinlocki SMP + kolejno blokad           | P0        | Test równoległy nie wykazuje race/double allocation        |
| C    | Transakcyjny map/unmap/`aspace_brk`      | P0        | Błąd w połowie operacji nie zmienia stanu                  |
| D    | Bezpieczny slab z bitmap obiektów        | P0        | Double-free i invalid-free są wykrywane                    |
| E    | Kontrakty aligned/pages/realloc          | P1        | API ma dokumentowane i testowane gwarancje                 |
| F    | Overflow DMA/contiguous + strefy PMM     | P1        | Testy graniczne przechodzą, DMA32 działa deterministycznie |
| G    | Pełniejsze VMA/mmap/mprotect             | P1        | Człciowe zakresy są poprawnie dzielone                     |
| H    | TLB shutdown SMP i batchowanie           | P1        | Zmiany mapowania są widoczne na wszystkich CPU             |
| I    | Per-CPU cache i optymalizacja buddy/slab | P2        | Benchmark pokazuje mniejszą kontencję bez regresji         |
| J    | Demand paging/COW/huge pages             | P2        | Funkcje mają testy integracyjne z procesami                |

## ## 7. Minimalny zestaw testów regresyjnych

Należy dodać testy jednostkowe możliwe do uruchamiania poza jądrem dla bitmapy, regionów, alignmentu i o

| Grupa     | Przypadki                                                                             |
|-----------|---------------------------------------------------------------------------------------|
| Bitmapa   | pusta, pełna, jeden bit, granice 63/64/65, zakres przez granic słowa, double-free     |
| PMM       | pojedyncze ramki, contiguous, align 0/1/nie-potęga, max address, brak pamięci         |
| Buddy     | każdy order, koalescencja, fragmentacja, invalid pointer, double-free, overflow align |
| Slab      | każda klasa, pełny/pusty slab, double-free, invalid pointer, równoległość             |
| API heap  | `calloc` overflow, `realloc(NULL)`, `realloc(ptr,0)`, aligned, pages                  |
| Mapowanie | człciowy błąd, rollback, overlap, unmap fragmentu, protect fragmentu                  |
| VMA       | zakresy graniczne, `brk` rollback, guard page, sąsiednie VMA                          |
| DMA       | 32-bit zone, długość overflow, unmap rollback, sync CPU/device                        |
| Ochrona   | NX, SMEP, SMAP, USER/KERNEL PTE, zakaz W^X                                            |

## ## 8. Dokumentacja i CI

Dokumentacja architektury powinna używać gałęzi `stabilizing`, a nie nieaktualnych linków do `unstable`.

Repozytorium powinno mieć CI, który buduje kernel w tym samym cross-toolchainie co Docker/QEMU, uruchamia

## ## 9. Kolejność rekomendowana dla autora

Najpierw należy wdrożyć helpery zakresów i overflow, następnie spinlocki SMP. Dopiero potem warto napraw

> \*\*Nie rekomenduj jeszcze włączania `HEAP\_USE\_SLAB` w produkcyjnej ścieżce.\*\* Najpierw trzeba dodać w

## ## 10. Odwołania do kodu

Najważniejsze ustalenia wynikają bezpośrednio z analizowanej gałęzi:

- [API alokacji](https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/api/alloc.c)
- [Buddy allocator](https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/heap/buddy.c)
- [Slab allocator](https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/heap/slab.c)
- [PMM](https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/physical/pmm.c)
- [Przestrzenie adresowe](https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/process/address\_space.c)
- [DMA](https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/special/dma.c)
- [Dokumentacja architektury](https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/TrangorgeOS%20dokumentacja%20

## ## References

- [1]: <https://github.com/CTRL-F-org3/TrangorgeOS/tree/stabilizing> "TrangorgeOS – gałąź stabilizing"
- [2]: <https://github.com/CTRL-F-org3/TrangorgeOS/commit/7a0d836f2180874aac3e8c92493c791aec89a8b3> "TrangorgeOS: initial commit"
- [3]: <https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/heap/buddy.c> "Buddy allocator"
- [4]: <https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/heap/slab.c> "Slab allocator"
- [5]: [https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/process/address\\_space.c](https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/process/address_space.c) "Address space management"
- [6]: <https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/physical/pmm.c> "Physical memory management"
- [7]: <https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/kernel/src/mm/alloc/special/dma.c> "DMA allocator"
- [8]: <https://github.com/CTRL-F-org3/TrangorgeOS/blob/stabilizing/TrangorgeOS%20dokumentacja%20>

## kernel\_Workspace/lib/README.md

tg-ipc-abi (zero deps, czysty ABI)  
↑

```

tg-cap-core (depends: tg-ipc-abi)
 ↑
tg-cap-table (depends: tg-cap-core)
 ↑
tg-ipc-ring (depends: tg-ipc-abi)
 ↑
tg-ipc-channel (depends: tg-ipc-abi, tg-ipc-ring, tg-cap-core)
 ↑
tg-shmem (depends: tg-cap-core)
 ↑
tg-ipc-dispatch (depends: tg-ipc-abi, tg-cap-core, tg-cap-table)

```

## lib-C/README.md

```

lib-C – C bindings for the TrangorgeOS communication protocol

Thin C client for the Rust core in `../lib`. It mirrors the public
protocol (message layout, layers, opcodes, rights, object types) in
`include/tgcomm.h`, but the capability table, the shared-memory ring and the
authorization gate stay opaque: C only calls the `tgcomm_*` C ABI exported
by `libtg_comm.a`, it never re-implements the logic.

Layout

```
include/tgcomm.h  # protocol types + tgcomm_* declarations
src/tgcomm.c      # thin helper (tgcomm_ring_bytes -> Rust)
test/test.c       # smoke test (authorize, revoke, ring round-trip)
Makefile
```

Build & test

```sh
make          # build libtgcomm_c.a (builds the Rust core first)
make test     # build + link + run the smoke test against libtg_comm.a
make clean
```

The test proves, from C, that:

* a missing capability is denied,
* READ rights do not authorize a CALL,
* a forbidden route (userspace → driverspace) is denied,
* revocation flips a forwarded request back to a deny,
* the ring round-trips a message byte-for-byte.

```

## lib-ada/README.md

```

lib-ada – Ada/SPARK bindings + the mathematical proofs

* `tg_comm.ads` – executable Ada bindings for the `tg_comm` C ABI.
* `tg_comm_security.{ads,adb}` – SPARK proof of the authorization gate.
* `tg_alloc_buddy.{ads,adb}` – SPARK proof of the kernel buddy allocator.

The buddy-allocator proof (`tg_alloc_buddy`)

The kernel buddy allocator
(`kernel_workspace/kernel/src/mm/alloc/heap/buddy.c`) is modelled at the page
level and proven correct with respect to the properties that make it safe.

Modelled state

```ada
ada
Max_Order = 8, Max_Pages = 256
type Buddy_Allocator is record
  State : State_Array; -- Free | Used | Tail per page
  Order : Order_Array; -- block order per head page
  Heads : Heads_Array; -- free-list head per order (-1 = empty)
  Next  : Next_Array;  -- free-list successor (-1 = end)
end record;
```

```

### ### Proved properties

1. **\*\*Invariant (`Valid`)\*\*** – every `Used`/`Free` head block is *aligned* on its own size and *in bounds*, all used blocks are *pairwise disjoint*, and pages are *conserved* (`used + free = Max_Pages`).
2. **\*\*Write correctness (`Used\_Disjoint`)\*\*** – two allocated blocks never overlap, so a write inside one block can never corrupt another:

```
``ada
function Used_Disjoint (A) return Boolean is
 ((for all I => (for all J =>
 (if State(I)=Used and State(J)=Used and I /= J
 then not Overlap(I, Order(I), J, Order(J))))));
 ...
```

3. **\*\*No double-free\*\*** – `Free` has precondition `State(I) = Used`; after one `Free` the block becomes `Free`, so a second `Free` is rejected.
4. **\*\*Size correctness\*\*** – `Allocate` returns a block with `Order_Size(0) >= Size`, aligned and in bounds.
5. **\*\*Conservation\*\*** – `Allocate`/`Free` preserve `used + free = Max_Pages`.

These are expressed as `Pre`/`Post` contracts on `Allocate`/`Free` plus two `Ghost` lemmas (`Lemma_Alloc_Disjoint`, `Lemma_No_Double_Free`).

### ### Verifying

```
* **Compile** (FSF GNAT): `make check` – validates syntax/semantics. Verified
 with `gnatmake -gnat2022`.
* **Prove** (GNATprove): `gnatprove -P tg_comm.gpr --mode=prove` discharges the
 post-conditions and lemmas by unfolding `Valid` and `Used_Disjoint`. GNATprove
 was not installed in the authoring environment; the obligations are provided
 ready-to-discharge and annotated in the sources.
```

### ## Files

```
...
tg_comm.ads / tg_comm.gpr # protocol mirror + C ABI + GNAT project
tg_comm_security.{ads,adb} # authorization-gate proof
tg_alloc_buddy.{ads,adb} # buddy-allocator proof (this document)
Makefile # gnatmake compile-check
...
```

## lib-odin/README.md

# lib-odin – Odin bindings for the TrangorgeOS communication protocol

Thin Odin client for the Rust core in `../lib` (exported as `libtg_comm.a`). It mirrors the public protocol and declares `foreign` procedures that call the `tgcomm_*` C ABI. The capability table, ring and the authorization gate are opaque – Odin never re-implements them.

### ## Files

```
...
tgcomm.odin # constants, enums, TgComm_Msg (#assert size == 64), foreign procs
...
```

### ## Usage (Odin)

```
``odin
import "tgcomm"

msg: tgcomm.TgComm_Msg
tgcomm.build_request(&msg, tgcomm.Layer.Driverspace, tgcomm.Layer.Kernel,
 tgcomm.opcode(tgcomm.Op_Class.Video, 1), /*cap=*/7)

table: tgcomm.TgComm_Cap_Table // opaque storage, 16-byte aligned
tgcomm.tgcomm_cap_table_init(&table)
tgcomm.tgcomm_cap_insert(&table, 7, 42, u8(tgcomm.Object_Type.Device),
 tgcomm.RIGHT_CALL)
```

```

if tgcomm.tgcomm_authorize(&table, &msg) == i32(tgcomm.Result.Forwarded) {
 // forwarded – the kernel may act
}

```

Link with ``-ltg_comm`` (the Rust core built as ``libtg_comm.a``). The Odin compiler was not available in the build environment used to author this binding, so the declarations follow the current ``foreign`/`---`` syntax and are not machine-compiled here.

## lib/PROTOCOL.md

# TrangorgeOS shared-memory communication protocol – specification

Version 1. This document is the normative description of the wire protocol implemented by ``lib/`` (Rust) and mirrored by ``lib-C``, ``lib-odin``, ``lib-ada``.

### ## 1. Constants

| Name                              | Value                     | Meaning                               |
|-----------------------------------|---------------------------|---------------------------------------|
| <code>`COMM_MAGIC`</code>         | <code>`0x5447434D`</code> | <code>`"TGCM"`</code> , little-endian |
| <code>`COMM_VERSION`</code>       | <code>`1`</code>          | wire version                          |
| <code>`COMM_MSG_SIZE`</code>      | <code>`64`</code>         | bytes per message                     |
| <code>`RING_CONTROL_SIZE`</code>  | <code>`16`</code>         | bytes of ring header                  |
| <code>`RING_SLOTS_DEFAULT`</code> | <code>`256`</code>        | default slot count                    |
| <code>`CAP_TABLE_SIZE`</code>     | <code>`512`</code>        | max capability entries                |

### ## 2. Message layout (``CommMsg``, ``#[repr(C)]``, 64 bytes)

| Offset | Size | Field                   | Type                                                            |
|--------|------|-------------------------|-----------------------------------------------------------------|
| 0      | 4    | <code>`magic`</code>    | u32                                                             |
| 4      | 1    | <code>`version`</code>  | u8                                                              |
| 5      | 1    | <code>`kind`</code>     | u8 ( <code>`MsgKind`</code> )                                   |
| 6      | 1    | <code>`layer`</code>    | u8 ( <code>`Layer`</code> , source)                             |
| 7      | 1    | <code>`target`</code>   | u8 ( <code>`Layer`</code> , destination)                        |
| 8      | 4    | <code>`opcode`</code>   | u32                                                             |
| 12     | 4    | <code>`cap`</code>      | u32 ( <code>`CapId`</code> , <code>`u32::MAX`</code> = invalid) |
| 16     | 4    | <code>`seq`</code>      | u32                                                             |
| 20     | 4    | <code>`status`</code>   | i32                                                             |
| 24     | 32   | <code>`a0..a3`</code>   | u64 × 4                                                         |
| 56     | 8    | <code>`reserved`</code> | u8 × 8                                                          |

### ## 3. Layers and routing

...

Kernel = 0, Driverspace = 1, Userspace = 2, Manager = 3, UserDriverSpace = 4

A request is only routed when ``may_target(src, dst)`` holds:

| src \ dst       | Kernel | Driverspace | Userspace | Manager | UDS |
|-----------------|--------|-------------|-----------|---------|-----|
| Kernel          | -      | ✓           | ✓         | ✓       | ✓   |
| Driverspace     | ✓      | -           | -         | ✓       | -   |
| Userspace       | ✓      | -           | -         | ✓       | -   |
| Manager         | ✓      | ✓           | ✓         | -       | ✓   |
| UserDriverSpace | ✓      | -           | -         | ✓       | -   |

### ## 4. Rights bitmask

``READ(0) WRITE(1) EXEC(2) MAP(3) GRANT(4) TRANSFER(5) SEND(6) RECV(7) CALL(8) MANAGE(9)``. ``contains(a,b)`` = ``(a & b) == b``.

### ## 5. Object types

``Memory=0 Endpoint=1 Notification=2 Channel=3 ShmemRegion=4 Device=5 Irq=6``.

### ## 6. Opcodes and requirements

``opcode = (class <= 8) | op``. Classes: ``Sys=0 Mem=1 Cap=2 Ipc=3 Shmem=4 Video=5 Audio=6 Input=7 Block=8 Net=9 Pci=10 Vgpu=11 Fs=12``.

``required(op)`` returns the ``(rights, obj_type)`` a capability must satisfy; an unknown/reserved opcode returns ``None`` and is **never** forwarded.

```
Class	rights	obj_type
Sys	CALL	any
Mem create(1)	MANAGE	Memory
Mem map(3)	MAP\WRITE	Memory
Mem grant(5)	GRANT\MAP	Memory
Cap	MANAGE	any
Ipc send(1)/call(3)	SEND	Endpoint
Ipc recv(2)/reply(4)	RECV	Endpoint
Shmem create(1)	MANAGE	ShmemRegion
Shmem attach/detach	MAP	ShmemRegion
device classes	CALL	Device
```

## 7. Ring (shared-memory SPSC)

```
...
[RingControl: head u32 | tail u32 | capacity u32 | pad u32] (16 bytes)
[slot 0: CommMsg (64 bytes)]
[slot 1: CommMsg (64 bytes)]
...
```

``head`` = producer index, ``tail`` = consumer index, in message units. One slot is always kept free to disambiguate full from empty.

## 8. Capability table

A fixed array of 512 entries; an entry is ``{ obj_id u64, obj_type u8, valid u8, pad u16, rights u32 }``. Absence (or ``valid == 0``)  $\Rightarrow$  deny.

## 9. The authorization gate

```
...
authorize(table, msg):
 if msg.magic != MAGIC or msg.version != VERSION -> Malformed
 if layers undecodable -> Malformed
 if !may_target(src, dst) -> RouteDenied
 if opcode unknown/reserved -> UnknownOp
 if cap absent in table -> NoCapability
 if !rights.contains(required_rights) -> RightsInsufficient
 if required_type present and mismatch -> TypeMismatch
 -> Forwarded
...
```

The kernel handler is invoked **only** on ``Forwarded``. This is the invariant proven in ``lib-ada``.

## lib/README.md

# ``tg-comm`` – strict shared-memory communication protocol for TrangorgeOS

The **single, independent** definition of how the layers of TrangorgeOS talk to each other through shared memory, and of **what they are allowed to ask**.

This crate is **not a member of any Cargo workspace** (the ``[workspace]`` key makes this directory its own isolated workspace). It is the "single source of truth" for the protocol; the language bindings in ``lib-C``, ``lib-odin`` and ``lib-ada`` are thin clients that call into it.

## What it defines

```
Concern	Where
Wire message (64-byte `CommMsg`)	`src/wire.rs`
Layers + routing matrix	`src/layer.rs`
Capability model (`CapId`, `ObjectType`, `CapEntry`, `CapTable`)	`src/caps.rs`
Rights bitmask	`src/rights.rs`
Opcodes + per-opcode requirements	`src/opcodes.rs`
Shared-memory SPSC ring	`src/ring.rs`
Shared-memory regions	`src/region.rs`
```

```
The authorization gate	`src/filter.rs`
Layer channel (wide API)	`src/channel.rs`
Driver manager facade	`src/manager.rs`
Stable C ABI	`src/ffi.rs`
```

## ## Security model

Every request carries a capability handle. The kernel-side `authorize` gate checks, in order:

1. structural validity (magic, version, layers),
2. routing policy (`Layer::may\_target`),
3. that the opcode is known and non-reserved,
4. that the capability exists, carries the required rights, and refers to an object of the required type.

Only `Forwarded` requests reach the kernel. The full argument that this makes unauthorized actions impossible is in `lib-ada` (SPARK) and in `PROTOCOL.md`.

## ## Build & test

```
```sh
cargo build          # no_std rlib + staticlib + cdylib
cargo test           # 11 tests proving "only authorized requests pass"
cargo build --release # -> target/release/libtg_comm.a (for C / Ada / Odin)
```
```

See `PROTOCOL.md` for the complete wire specification, and the sibling folders `lib-C`, `lib-odin`, `lib-ada` for the cross-language bindings.

## math/README.md

### # TrangorgeOS Math Stack

A `#![no\_std]` mathematical library stack for the kernel, drivers and the graphics stack. Everything here is written from scratch, is `forbid(unsafe\_code)`, and is verified against the system's own error vocabulary.

### ## Layout

```
...
math/
├── crates/
│ ├── math-core/ # shared substrate (implemented)
│ │ ├── bits.rs # float classification + field extraction
│ │ ├── consts.rs # constants in f32 and f64
│ │ ├── approx.rs # approximate comparison, lerp, remap
│ │ ├── error.rs # errors mapped onto kapi_abi::Status
│ │ └── feature.rs # compile-time target capability detection
└── ...
```

### ## Design rules

These hold for every crate in the stack, and are enforced rather than merely documented:

1. **\*\*`#![no\_std]` and `#![forbid(unsafe\_code)]`**, the latter enforced at the workspace level through `[workspace.lints]`. Float bit manipulation uses the safe `to\_bits` / `from\_bits` pair, so every result is a pure function of its input with no platform escape hatch hiding a rounding decision.
2. **\*\*No hidden allocation.\*\*** Fixed-capacity arrays only. A math routine that needs memory takes a slice from the caller.
3. **\*\*Errors speak the system's language.\*\*** `[math\_core::error::MathError]` converts into `kapi\_abi::Status`, so a math failure travels through `ds-manager` and IPC like any other error.
4. **\*\*Target capabilities are decided at compile time\*\***, not probed at boot - see `math\_core::feature`.

### ## Building

```
```sh
cd math
cargo test -p math-core
```

```
cargo check -p math-core --target x86_64-unknown-none
````
```

The kernel targets are `x86\_64-unknown-none` and `riscv64gc-unknown-none-elf` (see the repository `rust-toolchain.toml`).

### ## Roadmap

The requested set is 35 libraries. The order below is deliberate: each layer depends only on the ones above it, and nothing below is started until the layer above is green.

```
#	Crate	Replaces	State
1	`math-core`	shared substrate	**done** - 25 tests
2	`math-fp`	`libm`, `micromath`	planned
3	`math-vector`	`glam`, `nalgebra`, `euclid`	planned
4	`math-fixed`	`fixed`, `fpdec`	planned
5	`math-bigint`	`ruint`, `ethnum`, `ibig`	planned
6	`math-complex`	`num-complex`	planned
7	`math-kdtree`	`kdtree`	planned
8	`math-splines`	`splines`, `keyframe`	planned
9	`math-dsp`	`reallfft`, `microfft`, `biquad`	planned
10	`math-hash`	`crc32fast`, `xxhash-rust`, `siphasher`	planned
11	`math-rand`	`rand_core`, `fastrand`	planned
12	`math-crypto`	`curve25519-dalek`, `k256`, `p256`, `subtle`	planned, see warning
13	`math-geo`	`geo-types`, `spade`	planned
```

### ### A note on the security-critical entries

`math-crypto` would cover elliptic curves and constant-time primitives. Writing those from scratch is a genuinely bad idea: timing side channels and subtle field-arithmetic bugs are exactly the failure modes that hand-written crypto produces, and they do not show up in tests. The plan is to implement them as clean, auditable arithmetic (RFC 6234 / FIPS 180-4 style, with constant-time selection at the representation level) and to document the limits plainly - but anything that actually **\*\*signs\*\*** something should still use a vetted crates.io implementation.

### ## Accuracy policy

Each crate states its target ULP and its test tolerances. A routine that cannot meet the stated bound is documented as such rather than quietly rounded up - `math-core` already applies this: the `f32` Cody-Waite split constants were removed because the single-precision remainder falls below one ULP and the split buys nothing, and the documentation says so instead of pretending otherwise.

## panic.md

```
Kernel Hang / Panic Report – 2026-10-09
Kernel Hang / Panic Report – LAPIC base sanity check
```

### Key findings:

- `APIC\_BASE\_MSR` read reports x2APIC when bit 10 set; else LAPIC assumed at `base\_phys` passed in.
- `lapic::init(base\_phys)` uses `base\_phys` straight as virtual if it's `>= 0xFFFFF80000000000`.
- In SMP init, APs are booted with `send\_startup\_ipi` using `trampoline::TRAMPOLINE\_BASE >> 12` as vector

### Open question:

- Where is actual LAPIC physical base loaded from? Current code does not show base-specific usage beyond

### Next active items:

1. Confirm TRAMPOLINE\_BASE value and that identity mapping covers it.
2. Confirm `build.rs` linker path behavior for x86\_64.
3. Add serial console capture in QEMU and reproduce hang location.

### ## Status

Kernel **\*\*builds and boots\*\*** in qemu (GTK display, 4 vCPUs) but **\*\*freezes during runtime\*\***. No console / serial output was captured from the GTK session – the VM stays quiet and appears hung.

Attempted serial capture with qemu (`-serial file:/tmp/qemu\_serial\_out.txt`) did not produce a readable guest log at the time of this report, so the exact

stop point is inferred from the boot path rather than from live kernel prints.

## Boot entry

```
`kernel_main` (boot_info: &'static boot_loader::BootInfo) -> !` (`src/main.rs`)
```

Sequence:

1. ``init()``
2. ``mm::init_from_boot_info` (boot_info)`` → “[mm] allocator initialized OK” expected
3. ``init_permissions()``
4. ``gfx::init()`` → “[gfx] framebuffer initialized OK” expected
5. (optional) ``hdmi::init::init()``
6. ``pci::init()``
7. ``nic::runtime::init()``
8. ``bluetooth::init::init()``
9. ``fs::init()``
10. `**`cpu::init` (boot_info)**` ← SMP bring-up, tight waits per AP
11. ``testing::run_all` (TESTS)``
12. ``println!` (“Welcome in my Galaxy!”)``
13. ``gfx::refresh()``, ``terminal::init()``, ``terminal::run()``

## Where the hang manifests

In the GTK qemu session the kernel reaches ``cpu::init` (boot_info)`` (step 10) and then stops producing visible progress. The SMP initialization polls for each AP to reach its trampoline entry within a tight per-AP timeout (about 100 ms per AP, 3 APs with 4 vCPUs). If an AP does not become ready in time, the boot path blocks waiting for it without emitting a visible panic in the GTK display.

## Observations from the code

- SMP init lives in ``src/cpu/mod.rs`` / ``src/cpu/smp.rs`` and reaches into ``src/cpu/scheduler/smp/mod.rs`` plus the migration / stopper / balancing subtrees. AP boot uses a trampoline at ``TRAMPOLINE_BASE``.
  - The bootloader crate (``boot_loader = 0.9`` with ``map_physical_memory`` +
- ### Changed technical context

- Trampoline base is ``TRAMPOLINE_BASE = 0x8000`` in ``kernel/src/cpu/trampoline.rs`` and ``.set TRAMPOLINE_BASE``
- SMP bringup uses ``send_startup_ipi` (apic_id, (trampoline::TRAMPOLINE_BASE >> 12) as u8)``.
- Paging init in ``kernel/src/mm/arch/x86_64/paging.c`` skips identity work when ``boot_phys_offset == ARCH``
- Kernel MM bridge is compiled by ``kernel/build.rs``, but on x86\_64 it does **not** inject an explicit ``-T``
- Trampoline identity map is done relative to ``TRAMPOLINE_BASE`` before paging is enabled.
- ACPI parsing path uses ``phys_offset`` for RSDP/MADT/FADT reads; MADT can carry ``lapic_base_override`` (ty

### Still blocked

- Cannot reproduce and capture exact panic address or serial log yet in this environment; recommended next

``vga_320x200``) maps physical memory and provides ``BootInfo``. The kernel C bridge (``libmm.a``) is compiled by ``build.rs`` for x86\_64.

- There is **no** explicit x86\_64 kernel linker script in the repo at the
- MADT is already used to read ``lapic_base`` in ``kernel/src/cpu/acpi.rs``; ``lapic_base_override`` is also pa

moment — ``build.rs`` adds ``-T ../riscv64-link.ld`` only for riscv64. On x86\_64 the bootloader’s default layout is used. If that layout does not match kernel assumptions about load address / trampoline placement, early boot can fail silently.

- **566 Rust warnings** remain (Rust 2024 strictness around ``mut static`` / shared references to mutable statics in ``gfx/console.rs``, ``vga_buffer.rs``, ``fs/mbr.rs``, and several scheduler files). These are not compile errors yet, but some are UB-adjacent and can corrupt early-boot state.
- Current ``cargo run`` uses the GTK display variant, so kernel ``println!`` / panic output is not visible from the host side.

## Likely root causes (ranked)

1. **SMP AP bring-up timeout / deadlock** — an AP never reaches the trampoline or does not complete early init within the per-AP timeout, and the BSP waits indefinitely (or until a timeout path that may not panic). This is the most probable hang location with 4 vCPUs / 3 APs.
2. **Incorrect / incomplete physical memory mapping for the kernel image or trampoline region** — if ``BootInfo`` mapping is wrong, AP bootstrap code can fault silently without a visible triple fault in GTK mode.
3. **Undefined behavior from ``mut static`` / shared-ref patterns** in early boot (GFX, scheduler, FS) corrupting state that SMP init / tests depend on.
4. **Missing explicit kernel linker script for x86\_64** — the bootloader default layout may not match kernel expectations for section / trampoline

placement, causing early code to run at wrong addresses.

## Czerwony punkt (hang point)

- ``cpu::init`` in ``src/cpu/mod.rs`` / ``src/cpu/smp.rs`` – specifically the per-AP wait loop after the SMP startup IPI(s). If the AP never signals readiness, the BSP waits here and the system hangs.

## Verification needed

- Run qemu with a `**serial console**` (``-serial stdio`` or ``-serial file:/tmp/qemu_serial_out.txt``) and ``< /dev/null`` so host can read kernel prints and any panic message.
- Determine whether the log stops at ``[mm]``, ``[gfx]``, ``pci``, ``nic``, ``bluetooth``, ``fs``, or at ``cpu::init`` / AP bring-up.
- Inspect the SMP trampoline path and per-AP early stack / GDT / IDT setup.
- Validate the bootloader-provided memory map and kernel load / trampoline addresses against kernel expectations.
- Consider adding an explicit x86\_64 linker script if the bootloader layout is not what the kernel assumes.

## Verification status

- Build: clean (exit 0) for ``driverspacelib``, ``driverspace``, ``kernel``.
- Boot: ``cargo run`` builds the bootimage and launches qemu.
- Runtime: hung; no serial console output captured yet in this session.

## Notes

- The GTK display session shows only qemu theme warnings; no kernel console

### Architecture-specific physical-memory offset notes

- On x86\_64, ``mm::init`` receives ``boot_phys_offset`` from ``boot_info.physical_memory_offset`` and passes it
- Paging code in ``kernel/src/mm/arch/x86_64/paging.c`` has an early exit when ``boot_phys_offset == ARCH_DIRECT_MAP_BASE``
- If the bootloader does not map physical memory at the expected offset, the direct map and/or trampoline
- The LAPIC base can come from ACPI MADT; if that value is used directly as a virtual address somewhere w

### Actionable next steps

1. Boot the kernel with ``-serial stdio`` and capture the exact panic / hang point, not just the GTK display
2. If there is no serial access yet, add an early, unconditional serial write right after paging enable a
3. Compare the bootloader-provided ``physical_memory_offset`` with the value the kernel expects before call
4. If APs are not starting, focus on the trampoline region identity mapping and the startup-IPI path, and

## 2026-09-14 – objdump evidence from debug ELF (``kernel``)

- Static data (confirmed in objdump relocation output):
  - ``boot_phys_offset``: ``0x5c6d88``
  - ``boot_phys_offset_valid``: ``0x5c6d80``
- Disassembly shows explicit checks against ``<boot_phys_offset_valid>`` and ``<boot_phys_offset>`` in main, paging, serial, and CPU bringup paths.
- Paging init path includes early-exit / map logic matching the C/Rust preamble already discussed.

### Observability findings

- The debug kernel ELF (``target/x86_64-kernel/debug/kernel``) exists.
- Serial console capture remains the biggest gap:
  - ``/tmp/kernel_serial_out.txt`` is empty.
  - ``/tmp/kernel_serial_err.txt`` was too short to contain a clear panic line.

### Interpretation

- Because the binary clearly looks at ``boot_phys_offset`` / ``boot_phys_offset_valid`` around paging init ``77b2...a5``, the most likely first failures are:
  1. Paging init fault when bootloader-provided offset is incompatible with ``ARCH_DIRECT_MAP_BASE``, OR
  2. SMP bringup fault when AP trampoline/stack/identity mapping is inconsistent with LAPIC/ACPI MADT info.
- Correlation is now feasible if serial output is captured and mapped back with the debug ELF.

## Branch state

- ``git status``: modified: ``panic.md``; otherwise on ``stabilizing``.
- HEAD: ``183557c`` (``stabilizing``).

output is visible there.

### Current working branch and state

- Branch: `stabilizing` (tracking remote `origin/stabilizing`).
- Working tree has unstaged changes mainly in scheduler / SMP and some target/flycheck artifacts; kernel
  - `kernel/src/cpu/scheduler/collections/mod.rs`
  - `kernel/src/cpu/scheduler/smp/migration/mod.rs`
- The most relevant panic context is in `panic.md`; the key code references are `kernel/src/main.rs`, `ke

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  - `kernel/src/cpu/scheduler/smp/migration/mod.rs`
- The most relevant panic context is in `panic.md`; the key code references are `kernel/src/main.rs`, `ke

## TODO

## References

## References

- [ ] Capture reliable serial console output with `-serial stdio`/`-serial file:`.
- [ ] Add an early unconditional serial write after paging enable, before MM/SMP init.
- [ ] Compare bootloader `physical\_memory\_offset` vs `ARCH\_DIRECT\_MAP\_BASE` and LAPIC direct-map expectations.
- [ ] Confirm trampoline identity mapping covers AP trampoline code/stack.
- [ ] If serial log appears, correlate RIP with `kernel` ELF using addr2line/objdump.
- [ ] If offset mismatch is confirmed, decide whether to support dynamic offset or force early remap before paging enable.
- [ ] Fix `kernel/src/cpu/scheduler/collections/mod.rs` and `kernel/src/cpu/scheduler/smp/migration/mod.rs` (`pub pub mod ...`) so `cargo build --release` can produce the release bootimage again.

- `kernel/src/main.rs`
- `kernel/src/cpu/smp.rs`
- `kernel/src/cpu/lapic.rs`
- `kernel/src/cpu/acpi.rs`
- `kernel/src/mm/mod.rs`
- `kernel/src/mm/arch/x86\_64/paging.c`
- `kernel/src/mm/arch/x86\_64/memory.c`
- `kernel/src/cpu/trampoline.s`
- `kernel/src/cpu/trampoline.rs`
- `panic.md`

This report is intended as a continuation note for kernel hang debugging on x86\_64 and is not yet a root-

- `kernel/src/main.rs`
- `kernel/src/cpu/smp.rs`
- `kernel/src/cpu/lapic.rs`
- `kernel/src/cpu/acpi.rs`
- `kernel/src/mm/mod.rs`
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- `kernel/src/mm/arch/x86\_64/memory.c`
- `kernel/src/cpu/trampoline.s`
- `kernel/src/cpu/trampoline.rs`
- `panic.md`

This report is intended as a continuation note for kernel hang debugging on x86\_64 and is not yet a root-

- [ ] Add a serial-handshake early print right after paging enable, before SMP bringup.
- [ ] Add a small test mode to boot with QEMU and a serial file/stdio in CI or locally, so the hang can b
- [ ] Compare actual bootloader-provided `physical\_memory\_offset` with the offset used by paging init and
- [ ] Confirm trampoline identity map covers at least the trampoline code and stack range used by AP brin

## tools/iso/README.md

# TrangorgeOS – demo ISO images

Build one bootable, publishable `.iso` of the current `stabilizing` (alpha) tree.

```
```sh
```

```

just iso          # build dist/TrangorgeOS-<version>-x86_64-demo.iso
just iso --release # same, kernel built with opt-level 2
just iso-run      # boot the newest ISO in QEMU (window + serial on stdout)
tools/run-iso.sh --headless # automated smoke test, exits with a status code
```

```

The scripts are plain bash; `just` is only a thin wrapper (it runs them through `bash`, so the executable bit is optional - use `chmod +x tools/\*.sh` if you want to call them directly).

Headless runs wait up to 120 s for the kernel to reach the terminal (a cold boot takes ~1-2 minutes in QEMU) and use QEMU's `-snapshot` so repeated/parallel runs neither lock nor modify `dist/demo-root-disk.img`.

| File              | Purpose                                                                                  |
|-------------------|------------------------------------------------------------------------------------------|
| dist/REL          | release notes                                                                            |
| dist/README       | README                                                                                   |
| dist/README.txt   | text file placed at the ISO root (`README.TXT`)                                          |
| dist/Dockerfile   | reproducible builder image (toolchain + xorriso + syslinux)                              |
| dist/mkiso.sh     | build the kernel image, stage the ISO tree, run xorriso, write the checksum + `dist/REL` |
| dist/run-iso.sh   | boot the ISO in QEMU, optionally headless with a serial smoke test                       |
| dist/isolinux.cfg | ISOLINUX menu (ISO `isolinux/isolinux.cfg`)                                              |

## Boot path (and why it is the only one that works today)

The kernel's single working boot path is the **bootloader 0.9 BIOS image** built by `cargo bootimage`:

```

```
kernel_workspace/kernel-bin/src/main.rs -> entry_point!(kernel_main) (the only _start)
kernel_workspace/kernel-bin/build.sh    -> cargo bootimage -Zbuild-std=...
kernel_workspace/target/x86_64-kernel/<profile>/bootimage-kernel-bin.bin
```

```

That artefact is a **raw disk image**: a 512-byte BIOS boot sector at LBA 0 followed by the bootloader stages and the kernel payload, all addressed through the BIOS INT 13h interface with 512-byte sectors.

An El Torito CD is normally booted in **no-emulation** mode, where the BIOS reads and hands over the drive in 2048-byte sectors and does not provide a disk-style INT 13h interface. A raw 512-byte-sector boot image therefore cannot be started that way directly. The ISO is built like this instead:

```

```
El Torito (no emulation) -> ISOLINUX -> MEMDISK -> /boot/trangorge.img
                                     |
                                     the unmodified bootimage, loaded to RAM
```

```

- ISOLINUX** is a no-emulation El Torito bootloader that installs its own INT 13h handlers, which is exactly why the standard syslinux recipe works.
- MEMDISK** claims a chunk of high memory, hooks INT 13h/INT 15h and presents the image as a BIOS hard disk. The raw image is ~3.9 MiB - just under MEMDISK's 4 MiB floppy/hard-disk threshold - so the config passes `harddisk` explicitly instead of relying on size/geometry guessing.

The payload is byte-for-byte the same file `kernel-bin/run.sh` boots, so if the ISO works in QEMU it proves the QEMU boot path (and vice versa).

`comgrub/` (multiboot2) and `comlimine/` (Limine) are **not** usable yet: neither provides the `kernel\_main(magic, info)` symbol the kernel actually exports (`kernel::kernel\_main(&BootInfo)` from bootloader 0.9), and neither hand-off produces a `bootloader::BootInfo`. See `TrangorgeOS - TODO.md`, section 2.6.

## Requirements

```

```sh
sudo apt-get install -y xorriso syslinux-common # Debian/Ubuntu (Ubuntu 24.04 used here)
sudo dnf install -y xorriso syslinux           # Fedora/RHEL
sudo pacman -S --needed libisoburn syslinux    # Arch
```

```

Plus the Rust toolchain from `rust-toolchain.toml` (`nightly-2026-08-01`, `rust-src`, `llvm-tools-preview`) and `cargo-bootimage ^0.10`. `mkiso.sh` installs

`cargo-bootimage` on demand and can be checked with `tools/mkiso.sh --check-deps`.

`mkiso.sh` probes the usual syslinux locations on its own:  
`/usr/lib/ISOLINUX` and `/usr/lib/syslinux/modules/bios` (Debian/Ubuntu),  
`/usr/lib/syslinux/bios` (Arch/Manjaro), `/usr/share/syslinux`. Set  
`SYSLINUX\_DIR=/path/to/flat/dir` to point it at any directory containing  
`isolinux.bin`, `ldlinux.c32`, `memdisk` and (optionally) `isohdpx.bin` - that is  
the escape hatch when the distro does not package them or when root is not  
available:

```
```sh
curl -LO https://www.kernel.org/pub/linux/utils/boot/syslinux/syslinux-6.03.tar.xz
tar -xJf syslinux-6.03.tar.xz
mkdir -p /tmp/syslinux-flat && cp \
  syslinux-6.03/bios/core/isolinux.bin \
  syslinux-6.03/bios/com32/elflink/ldlinux/ldlinux.c32 \
  syslinux-6.03/bios/memdisk/memdisk \
  syslinux-6.03/bios/mbr/isohdpx.bin /tmp/syslinux-flat/
SYSLINUX_DIR=/tmp/syslinux-flat tools/mkiso.sh
```
```

## What the build produces

```
```
dist/
  TrangorgeOS-<version>-x86_64-demo.iso          the image (hybrid MBR: CD or dd to USB)
  TrangorgeOS-<version>-x86_64-demo.iso.sha256    checksum
  RELEASE.txt                                    release manifest (version, git commit, hashes)
  iso-root/                                       staged ISO9660 tree (for inspection)
  demo-root-disk.img                             writable ATA disk for local QEMU runs
  serial-*.log                                   serial capture of the last QEMU run
```
```

Version resolution order: `--version=` → `"\$TRANGORGE\_VERSION"` → `./VERSION` file →  
`git describe --tags` → `dev-<short-sha>`.

## Publishing checklist

1. `just iso --release` on a clean checkout.
2. `tools/run-iso.sh --headless` – the smoke test must print `smoke test PASSED`  
(it greps the serial log for the kernel banner, the allocator, the self-test  
suite and the `SYSTEM STATUS:` line).
3. Verify the manifest: `cat dist/RELEASE.txt`.
4. Attach to the GitHub release: the `.iso`, its `.sha256` and `RELEASE.txt`.
5. Paste the checksum and the `Git commit` line into the release notes so the  
artefact can be tied to a revision.

```
```sh
(cd dist && sha256sum -c TrangorgeOS-*demo.iso.sha256)
```
```

## Verified status

Build and boot verified on 2026-09-25 (`stabilizing` @ `8393847`, Manjaro/Arch host,  
xorriso 1.5.8, syslinux 6.03 assets, cargo bootimage 0.10.5):

```
```
$ tools/mkiso.sh --no-build
[mkiso] ISO size: 5 MiB
dist/TrangorgeOS-dev-8393847-x86_64-demo.iso
sha256 422d2d99965f481afdd837886fc85b6ae183b151cd391c0394f730294665326a
```
```

```
$ tools/run-iso.sh --headless
[OK] kernel banner reached
[OK] memory allocator online
[OK] self-test suite ran
-> SYSTEM STATUS: 19/19 OK
```
```

- * The **memdisk** ISO boots in QEMU: the kernel log on COM1 ends with
`Welcome in my Galaxy!`, `SYSTEM STATUS: 19/19 OK` and the in-kernel terminal
prompt (including the TFS format + write/read roundtrip on the attached ATA disk).
- * A cold boot takes roughly **1-2 minutes** under QEMU (SMP bring-up, PCI/USB/NIC

```

    probing and the 19 module self-tests), hence the 120 s headless timeout.
* `--boot=hdemul` produced **no** boot under SeaBIOS - El Torito hard-disk
  emulation never reached the bootloader's boot sector (empty serial log, QEMU
  exits after the timeout). It is kept only as a syslinux-free fallback; the
  MEMDISK path is the supported one.

## Known limitations

* **Legacy BIOS only.** No UEFI/CSM-free boot: bootloader 0.9 is BIOS-only.
  A UEFI image needs a real Limine/GRUB hand-off (see TODO 2.6).
* **No persistence from the CD.** TFS is read/write only on ATA disks; a CD is
  read-only and ATAPI is not driven by the kernel, so the fs self-test reports
  `FAILED` when no disk is attached (the system keeps running).
  `run-iso.sh` always attaches a writable disk to keep the report green.
* **MEMDISK loads the image into RAM** (~8 MiB of high memory for the current
  image) – irrelevant for QEMU, but it means very large kernels would need a
  different strategy.
* **`--boot=hdemul` fallback**: El Torito hard-disk emulation, no syslinux
  needed. It depends on the BIOS deriving a usable geometry from the image; the
  bootimage's MBR carries no partition table, so `--synthesize-mbr` can write an
  experimental entry at offset 446. Treat this mode as a debugging tool.
* The image is a **pre-release alpha**; `just iso` fetches nothing, so the
  resulting ISO is reproducible from the recorded git commit.

```

uspace/README.md

```

# uspace – TrangorgeOS userspace (nested system + graphical environment)

```

```

Userspace is a **nested system**: it has its own capability-based permissions
(consistent with the kernel's), connects to the kernel only for *executive
actions*, talks to the driver space over IPC, and runs the graphical stack.

```

```

## Layering

```

```

...

```

```

apps (shell, terminal, ...)

```

```

userdrivers (input, gpu, ... specific user-space drivers)

```

- IPC (uspace::ipc) – app <-> userdriver <-> driverspace
- executive (uspace::kernel) – framebuffer / process actions from kernel

```

▼

```

```

uspace::caps nested capability permissions (kernel-consistent)

```

```

▼

```

```

tg-comm shared-memory protocol → driverspace / kernel
`..`

```

```

## Modules

```

```

| module | role |
|---|---|

```

```

| `caps` | nested capability model: `Session` spawns children with **derived** (subset) capabilities; a c

```

```

| `ipc` | `IpcEndpoint` over `tg_comm`; a `Router` (used on host tests; the kernel routes on the real sy

```

```

| `kernel` | `KernelClient` – executive actions (`acquire_framebuffer`) over the shared ring |

```

```

| `gfx` | `compositor` (z-ordered surfaces), `render` (rect/fill/8x8 text), `app` (demo window + termina

```

```

## Permissions (kernel-consistent)

```

```

Userspace uses the *same* capability primitives as the kernel
(`tg_comm::CapTable`, `CapId`, `Rights`, `ObjectType`). The session receives a
capability table from the kernel and derives subsets for apps/userdrivers:

```

```

```rust

```

```

let child = session.spawn_derived(&parent, &[
 (CapId(2), CapEntry::new(20, ObjectType::Memory, Rights::READ)), // ok: subset
]);
// WRITE would be denied: the parent does not hold WRITE over that object.
```

```

```

This is the property proven in `lib-ada`: revocation/derivation can only ever
shrink the authorization set, so the nested user space cannot escalate into a
kernel-damaging capability.

```

```

## Graphical environment

```

``gfx::app::draw_demo_window`` composites a window (with a small terminal) into the kernel-provided framebuffer buffer:

1. fill the background,
2. create a window surface via the compositor,
3. draw border + title bar + terminal text into the surface,
4. ``composite()`` the surface into the framebuffer.

The software renderer stands in for the GPU/Vise-LG pipeline on this reference implementation; the surface/command model matches the driver-space ``gfx-protocol`` (``gfx_protocol_workspace``).

Run the tests

```
```sh
cd uspace && cargo test
```
```

The host tests exercise the nested-capability monotonicity, the IPC router and the demo window renderer end-to-end.